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Compute Project



HIGH BANDWIDTH FLASH (HBF™)

HIGH-LEVEL BASE DIE SPECIFICATION

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List all companies or individuals who may have assisted you with the specification by providing feedback and suggestions but did not provide any IP.

2.Compliance with OCP Tenets

2.1. Openness

The HBF specification is open and available for the industry to use and adopt.

2.2. Impact

Addresses the Memory Wall problem in the data-centric workloads, HBF can augment HBM with the ability to attach terabytes of memory to GPUs.

2.3. Scale

The HBF specification provides a robust interface that could be deployed at scale and enhanced in the future as use cases evolve.

2.4. Efficiency

Higher capacity HBF near GPU reduces off-chip access energy and improves system performance.

2.5. Sustainability

The HBF technology reduces the number of GPUs required to achieve a given system memory capacity.

3.Introduction

The memory wall problem is a well-known issue in the computer architecture domain for decades from the CPU era and continues in the current era of high-volume data-centric processing. The disparity between the speed of GPUs and the response time of memory creates bottlenecks in workload performance. In addition, the LLM (Large Language Model) sizes and context lengths are increasing in every new generation that drives the need for higher memory capacity.

High Bandwidth Flash (HBF), a non-coherent memory-centric flash device, will redefine AI infrastructure by providing extremely high TB/s bandwidth with very high capacity. The HBF sits very close to the GPU/TPU/xPU data processing units. HBF addresses the memory wall problem, where wide-IO NAND-based memory architecture offers 8x to 16x of the capacity of HBM, while delivering the same read bandwidth. In the datacenter, HBF can augment HBM with the ability to attach terabytes of memory to xPUs. This document defines the HBF interface with xPU for datacenter-class deployments, including electrical, packaging, test & debug, reliability, and additional details required to integrate xPU with one or multiple UCle modules. The solution will focus on reducing protocol overhead, enabling more efficient data transfer, and improving scalability across large deployments.

This document describes the High Bandwidth Flash (HBF) High Level Specification. It covers the following sections:

- Product Description
- UCle Host Interface Architecture
- AXI Protocol Layer
- Electrical Requirements, Reset and Initialization
- Test & Debug Interface
- Reliability Specifications
- Packaging Specifications
- Software Userguide
- Security
- HBF Applications

3.1. Acronyms

Acronyms used in the HBF specification:

Table 1: Acronyms

Acronym	Definition
AMBA	Advanced Microcontroller Bus Architecture
BOL	Beginning Of Life
CRC	Cyclic Redundancy Check
CSR	Control Status Register

Acronym	Definition
D2D	Die-to-Die
DDR	Double Data Rate
DWPD	Drive Writes Per Day
DMS	DFx Management Spoke
EOL	End Of Life
ESD	Electrostatic Discharge
FDI	Flit-Aware Die-to-Die Interface
FLITS	A unit for Flow Control across a Link
GiB / GB	GiB = 2^{30} bytes and GB = 10^9 bytes
HBM	High Bandwidth Memory
HBF	High Bandwidth Flash
KGSD	Known Good Stacked Die
KiB / KB	KiB = 2^{10} bytes and KB = 10^3 bytes
LP	Link Layer to PHY
MiB / MB	MiB = 2^{20} bytes and MB = 10^6 bytes
NVM	Non-Volatile Memory
PCB	Printed Circuit Board
PE	Program/Erase
PL	PHY to link Layer
PMIC	Power Management Integrated Circuit
QLC	Quad Level Cell
QoS	Quality of Service
RDI	Raw Die-to-Die Interface
SKH	SK hynix
SNDK	Sandisk Corporation
SSD	Solid State Drive
TBD	To Be Determined
TTP	Time To Performance
TTR	Time To Ready
UBER	Uncorrectable Bit Error Rate
UCle	Universal Chipset Interconnect Express

4. Product Description

The High Bandwidth Flash (HBF) is tightly coupled to the xPU Host Compute die with a distributed interface. The interface is divided into independent channels. Each channel is independent of one another. The host channels are not necessarily synchronous with each other. HBF uses a wide-interface architecture to achieve high-speed and low-power operation. Each host channel interface maintains a full duplex data bus operating based on UCle 3.0 standard guidelines. Each host channel is connected to independent set of NAND dies for that specific host channel through the UCle module.

HBF is optimized for high-bandwidth read operations to a stack of multiple NAND devices across several independent interfaces called channels. Each NAND stack will support up to 16 host channels, and each host channel may have one or more AXI channels. Each HBF stack can support a very high capacity, such as 512 GiB density or higher. Each channel provides access to an independent set of NAND dies. Requests from one UCle channel will not access data attached to a different UCle channel. The UCle channels are independently clocked and synchronous. The Core die refers to the NAND die stack. The Base die refers to the bottom die, where the Core die stack sits on top of the Base die.

4.1. Features

Host Interface Features:

- Burst Length of 4KiB on write access and on a 4KiB aligned address
- Burst Length of 64B to 4KiB in multiple of 64B for read access on 64B aligned address and does not cross 4KiB NAND page boundary
- 64-bit main-band interface data width, each lane running at up to 32Gbps per channel targeting up to 3.072 TiB/s of user bandwidth
- Additional 1-bit sideband interface running at 800 MHz for management and user defined application
- Support UCle Host streaming protocol
- Support 16 host channels – each host channel operating independently of others
- Temperature sensor with 2-bit encoded range output.

NAND Interface Features:

- Supports 16 NAND die stack
- Maintains a read refresh counter
- Internal management of NAND bad blocks
- Supports failure detection and redundancy mapping of TSVs between Base and Core die
- 4KiB NAND page size

Other Features:

- Supports IEEE 1500 Access Port
- Support optional scratchpad memory on all UCle channels for high priority read/write at ultra-low latency, where latency is in the order of L3 cache read/write
- Supports DA Access Port
- Approximate footprint is 10.975mm x 16mm x 775um

4.2. Performance

Table 2: Host Interface Bandwidth Calculation

Host Interface Calculation	Value	Unit
Line Rate	32	GT/s
Data width per Host Channel	64	bit
Bytes data width per Host Channel	8	Byte
Total Bandwidth per Host Channel	256	GB/s
Total Host Channels	16	units
Total Raw BW per UCle Link	4096	GB/s
Interface Efficiency with AXI Link Layer Protocol	75	%
Total Effective BW per Cube	3072	GB/s

Table 3: User Density Calculation

Density Calculation	Value	Unit
Dies per cube	16	units
Banks per channel	16	units
Page size	4096	Bytes
Total Cube Size	512	GiB

Table 4: HBF UCle-A Configuration for Speed Grade

Speed Grade	Maximum User Bandwidth (TB/s)	UCle Configuration			Maximum Stack Height
		Package	UCle Speed (GT/s)	Num AXI Interface Per UCle Channel	
1	0.384	x64	8	1, 2, 4	8
2	1.536	x64	16	1, 2, 4	16
3	3.072	x64	32	1, 2, 4	16

The AXI interfaces within a UCle channel are virtual. The host may interleave addresses across multiple virtual AXI interfaces in the manner best suited to the application. The usable and raw capacity for each UCle channel are fixed, and the capacity of each virtual AXI channel within the UCle channel is divided evenly across the number of AXI channels. The addresses are interleaved at any granularity in multiples of 64B across all the virtual AXI channels.

4.3. System Organization

Figure 1 shows the high-level Base die organization. The HBF is tightly coupled to the Host Compute die with a distributed interface. The interface is divided into independent channels. Each channel is completely independent of one another. The host channels are not necessarily synchronous with each other. HBF uses a wide-interface architecture to achieve high-speed and low-power operation. Each host channel interface maintains a 64-bit full duplex data bus operating based on UCle standard guidelines.

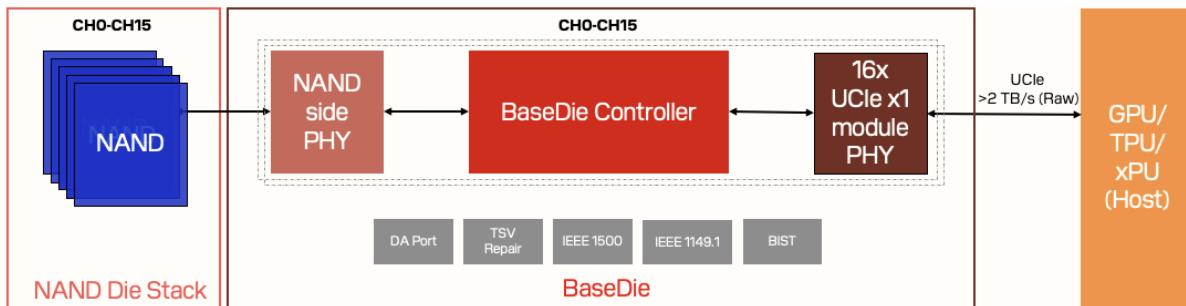


Figure 1: Base Die High-level Organization

Each host channel is connected to an independent NAND die for that specific host channel through UCle PHY module on the host side and NAND side PHY on the NAND side. One such organization structure is shown in the figure above. Common blocks such as DA Port, TSV repair, etc. for all 16 channels are shown in the gray boxes.

HBF is optimized for high-bandwidth read operations to a stack of multiple NAND devices across several independent interfaces (channels). Each NAND stack will support up to 16 host channels. Figure 2 shows an example of four HBF chipsets connected to a host. Each HBF chipset contains 16 host channels, each HBF with 512 GiB density.

Each channel provides access to an independent set of NAND dies. Requests from one channel may not access data attached to a different channel. Channels are independently clocked and need not be synchronous.

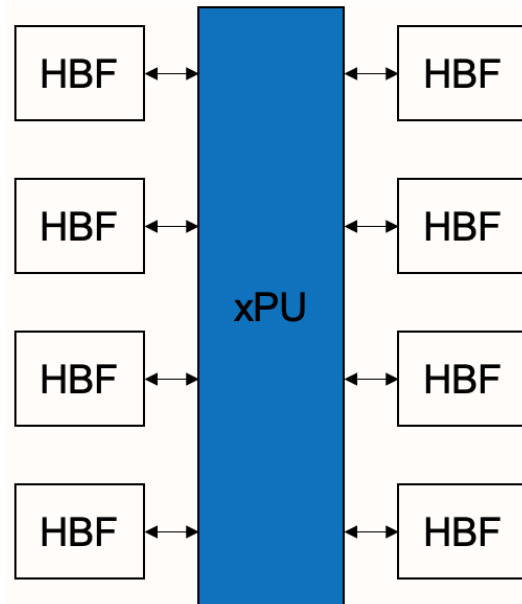


Figure 2: System-level Block Diagram

HBF is made up of three major entities:

1. Base die: The Base die contains four main functionalities,
 - a. The Base die is responsible for managing the host traffic and communicating with the Core die
 - b. The Base die contains the host-side PHY and the host side UCle controller responsible for the UCle protocol management
 - c. The Base die contains the Core die controller, which is responsible for communication and management with the Core die NAND stack
 - d. The Base die also contains the IEEE 1500 and DA port for debugging and out-of-band access to internal registers
2. Core die: The Core die contains 16 NAND dies
3. TSV channels: responsible for routing the signals from the Base die to the NAND dies

4.4. Base Die Organization

The Base die is responsible for managing the UCle protocol and communicating with the Core die through the TSV channels. The above Figure 1 illustrates the main blocks within the Base die for a single channel. The Debug Controller is common to all 16 channels. There are four main subsystems within the Base die: UCle controller, UCle PHY, Base Die controller, and Admin/Debug management.

UCle Controller and UCle PHY

- Responsible for UCle protocol and communication interface with the Host xPU
- Contains the D2D UCle Adapter and UCle Protocol Layer functions

Base Die Controller

- Process commands from the UCle controller
- Responsible for data movement between UCle controller and Core die
- Responsible for both host command management and response
- Responsible for both NAND command management and response
- Provides ECC encoding and decoding
- Error handling and other miscellaneous activities
- Process side-band commands and response
- Responsible for scheduling data transfers to and from the Core dies
- Read/Write/Erase status management
- Responsible for NAND initialization and Power-on-Reset
- Manages TSV redundancy mapping for the core side interface
- Manages Page Read Counter per block for refresh indication to the host

Admin/Debug Management

- Responsible for IEEE 1500 functions
- Responsible for DA Port management
- Can manage TSV repair control

4.5. Host Channel Definition

Each host channel consists of an independent UCle link with 64-bit bus width on the main band and a 1-bit bus width on the sideband. Reset_SB, IEEE 1500 test port, DA port and power supply signals are common to all channels. A channel provides access to a discrete pool of memory; no channel may access the memory storage for a different channel. Each channel interface provides an independent interface to several NAND dies of a defined page size.

4.5.1. Summary of Per-Channel Signals

Table 5 shows the signals required for each channel and adds global signals that are required per HBF device:

Table 5: Single Channel Signal Count

Function	Number of Micro Bumps + Redundancy	Notes
Main Band Signals		
Incoming Main Band Data	64 + 4	IM_DATA [63:0] + redundant
Incoming Main Band Clocks	2+1	IM_CLKn, IM_CLKp + redundant
Incoming Main Band Valid	1+1	IM_VALID + redundant

Function	Number of Micro Bumps + Redundancy	Notes
Incoming Main Band Track	1	IM_TRACK
Outgoing Main Band Data	64 + 4	OM_DATA [63:0] + redundant
Outgoing Main Band Clocks	2+1	OM_CLKn, OM_CLKp + redundant
Outgoing Main Band Valid	1+1	OM_VALID + redundant
Outgoing Main Band Track	1	OM_TRACK
Sideband Signals		
Incoming Sideband Data	1+1	IS_DATA
Incoming Sideband Clock	1+1	IS_CLK
Outgoing Sideband Data	1+1	OS_DATA
Outgoing Sideband Clock	1+1	OS_CLK

4.5.2. Summary of Global Signals

The Table 6 illustrates the common signals among all channels. Each signal with the number of uBump = 2 is duplicated for redundancy protection. Signals are associated with either uBumps, test pads, or both. DA [12] = 0 selects the IEEE 1500 ports and DA [12] = 1 selects the DA ports.

Table 6: Global Signals

Signal Name	I/O Width	IO Pads	Freq (Mhz)	Dir	Num uBumps	Test Pad	Description
RESETSB_n	1	1	NA	In	2	Yes	Reset for sideband. Active Low
RESETBD_n	1	1	NA	In	2	Yes	Reset for Base die (SoC reset). Active Low
REFCLK +/-	1	2	100	In	4	Yes	Reference clock 100MHz. Note1
READY	1	1	NA	Out	2	Yes	HBF is ready to receive UCle CMDS. Active High
DA [39:0]	40	40	30	I/O	40	Yes	See DA Port details. Note2
Vmon	4	4	NA	Out	4	Yes	Core die voltage monitoring signals. Note4
CATTRIP	1	1	NA	Out	2	Yes	Catastrophic temperature sensor. Active High
STRAP[3:0]	4	4	NA	In	8	Yes	Note3
CHIPID[2:0]	3	3	NA	In	6	Yes	JTAG Arc-Num MSB for JTAG IDCODE

Note 1: REFCLK is differential thus width=2. Requires 4 uBumps for redundancy. Wired OR used to drive each pad.

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Note 2: Broken up as DA [39:4] and DA[3:0]. Refer to Debug Organization and Test Access Ports section along with the Test & Debug chapter for additional details. The DA Port MUXing is meant to support the following vendor specific options: IEEE1500, X1 Mode, Scan, ATB, JTAG. Per redundancy needs, 2 uBumps are associated with each signal.

Note 3: Vendor-specific strap pins are used for different boot modes. For example, STRAP = [FW_FDM, FW_DBG, BOOT, !HALT]

Note 4: Vmon is needed to be brought out through Base die for monitoring on a test pad. Vmon are used to monitor voltages. Vmon is used for VPGM/VERA and can currently reach 12V. Core die will provide ESD protection. Base die will pass through only. VMON will be limited to 12V.

For certain signals of width=1, there are 2 uBumps, these are to duplicate the signal on each uBump for link redundancy in-case of damaged uBumps. The signal should connect/short to both uBumps.

CATTRIP: The CATTRIP signal is asserted based on temperature thresholds. If a certain threshold is reached by the temperature sensors, then the CATTRIP signal is asserted. This threshold is programmable.

The Core die temperature sensors reaching a certain threshold may also assert the CATTRIP signal. Firmware reads periodically from the Core die temperature sensors, based on which it can also trigger the CATTRIP signal.

4.5.3. IEEE 1500 Ports Summary

The following Table 7 illustrates the IEEE1500 signals. Each signal with number of uBumps = 2 is duplicated for redundancy protection. DA [12] = 0 selects the IEEE 1500 ports and DA [12] = 1 selects the DA Port.

Table 7: IEEE 1500 Global Signals

Signal Name	I/O Width	IO Pads	Freq (Mhz)	Dir	Num uBumps	Test Pads	Description
H_WRST_n	1	1	na	In	2	N	IEEE 1500 Serial Port Reset from Host
H_WRCK	1	1	100	In	2	N	Host IEEE 1500 Serial Port Clock from Host
H_SelectWIR	1	1	100	In	2	N	Host IEEE 1500 Serial Port Instr Reg Select from Host
H_ShiftWR	1	1	100	In	2	N	Host IEEE 1500 Serial Port Shift from Host
H_CaptureWR	1	1	100	In	2	N	IEEE 1500 Serial Port Capture from Host
H_UpdateWR	1	1	100	In	2	N	IEEE 1500 Serial Port Update from Host
H_WSI	1	1	100	In	2	N	IEEE 1500 Serial Port Data from Host
H_WSO [15:0]	16	16	100	Out	16	N	IEEE 1500 Serial Port Data Out to Host
H_RM[1:0]	2	2	100	Out	2	N	Redundant WSO

The redundant WSOs are used, after detecting damaged WSOs and are to be remapped to the damaged WSOs.

4.6. Address Mapping Scheme

Host software will follow the following address mapping in Figure 3 from the host global address to HBF UCle channel local address. Host software configures the global address and splits the global addresses across the local UCle channel address.

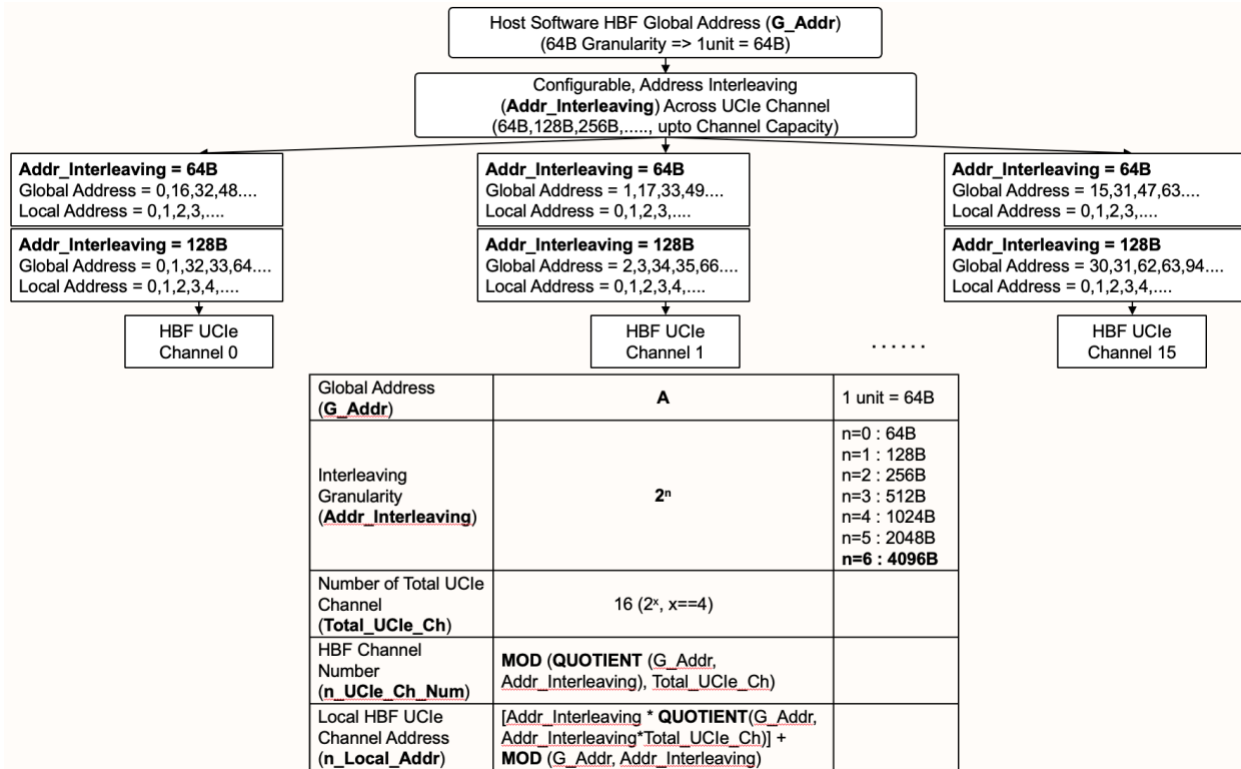


Figure 3: Software Global and Local Address Mapping

Each host HBF UCle channel has a continuous logical space without requiring knowledge of bank or die information; it is up to the HBF Base die architecture to map the HBF UCle channel's local address to the Core die bank, die, and blocks, thereby maximizing Core die channel utilization and performance. However, if the host wants to know the bank and die information from the logical address, the first 4 bits in the local address represent the bank number, and the next 2 bits represent the die number.

The host need not be aware of this, but if the host would like to know, the host interprets it from the configuration, where the host can read the information from the device configuration registers.

Device Logical Unit Size A minimum granularity HBF Base die read/write to Core die (4KiB)

Device Logical Unit Size = 4KiB

Number of banks/die = 16

Number of dies = 4

Device Logical Unit = HBF UCle local channel address >> 6

Figure 4 shows an example of Device Logical Unit data striping for best performance.

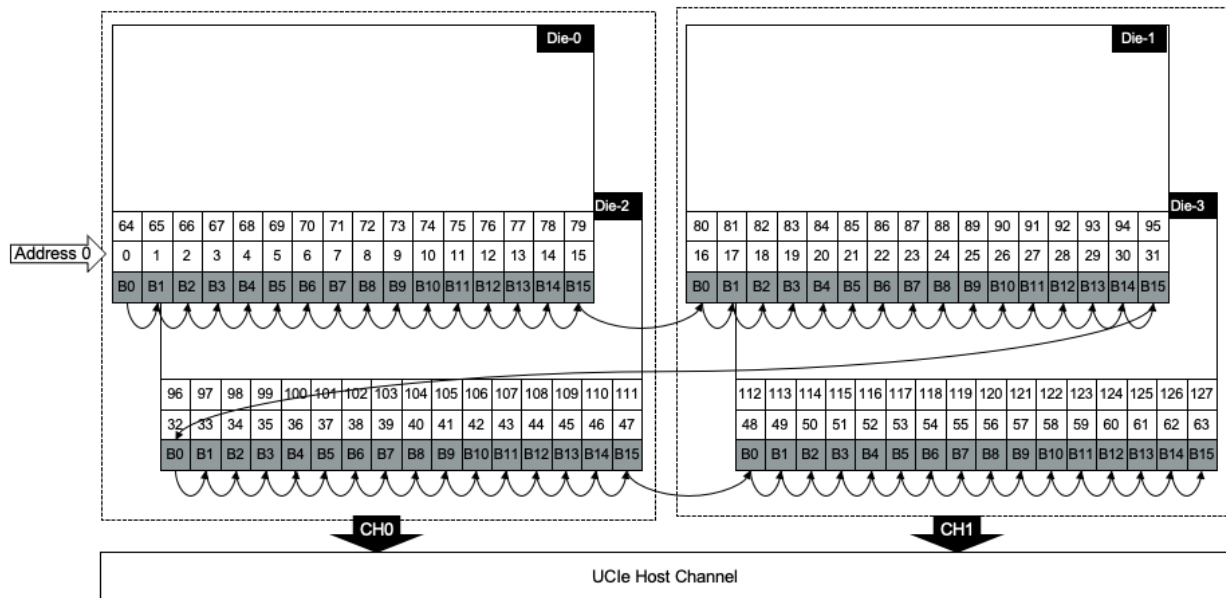


Figure 4: Device Logical Unit Striping

For example, each host channel may be connected to two Core die channels, and each Core die channel may be connected to two dies. The numbers inside the box in Figure 4 represent the device's logical unit address.

4.6.1. Address Mapping with Multiple AXI ports

Address assignment for 4-AXI Port per UCle module is shown in Figure 5. It is flexible for the host to choose one of the three interleaving options as shown in Figure 5, where the host treats each AXI port as a local channel.

In Figure 5, each circled box with number represents local channel ordering from host point of view. The host can choose any of the four options based on the workload requirements. If all four AXI ports are busy responding with full bandwidth to the host at 4KiB Interleaving, the maximum HBF read bandwidth is achieved .

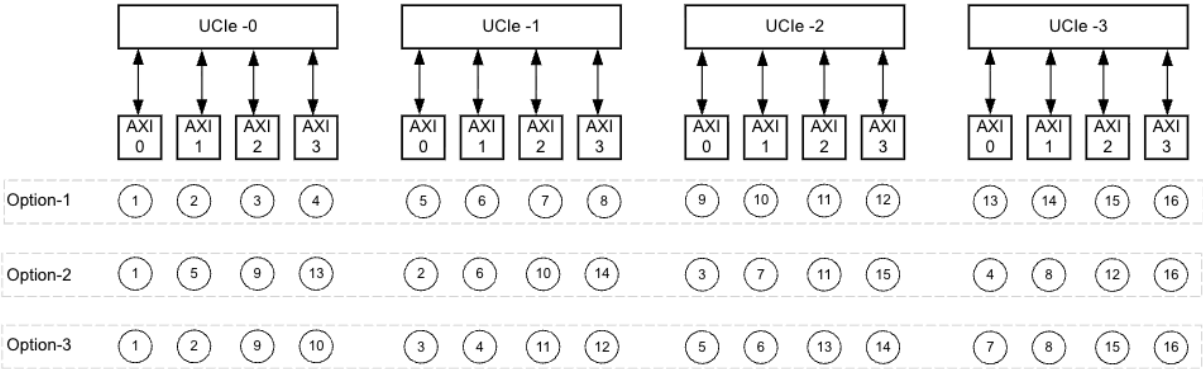


Figure 5: AXI Port Address Interleaving

5.UCLe Host Interface Architecture

5.1. UCLe Host Interface and Requirement

Please note that the following sections discuss a single UCLe interface that belongs to a single host channel. Any feature or functional description is applicable to all 16 host channels.

5.1.1. UCLe Overview

Universal Chipset Interconnect express (UCLe) architecture is an open, multi-protocol capable, on-package interconnect standard for connecting multiple dies on the same package. The chosen advanced packaging technology is used for the performance-optimized application of HBF. Consequently, the channel reach is short (less than 2mm, when measured from a bump on one Die to the connecting bump of the remote Die) and the interconnect is expected to be optimized for high bandwidth and low latency with best performance and power efficiency characteristics.

The advanced package characteristics are summarized as the following for HBF:

- Supported speed per UCLe x64 host module: 256 GB/s for 32 GT/s
- Bump pitch: 45mm
- Channel reach to xPU: 2 mm (maximum-supported for advanced package per UCLe specifications)
- Raw bit error rate: $1e^{-15}$ at greater than or equal to 16 GT/s, $1e^{-27}$ at 12 GT/s (maximum-supported for advanced package per UCLe specifications)

UCLe is a layered protocol, with each layer performing a distinct set of functions. The three main layers are: Protocol layer, Die-to-Die layer, and Physical layer. The HBF specification uses Format 6 flit format. Please refer to the UCLe v3.0 specification for more details. The following Figure 6 shows all three UCLe layers.

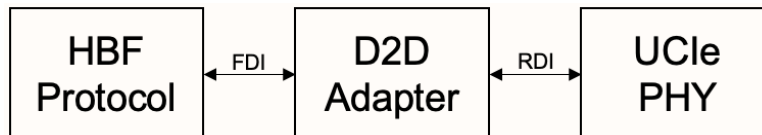


Figure 6: UCLe Layers for a single channel

- Protocol Layer: Streaming Protocol is chosen as the protocol for HBF. HBF enables AXI protocol for die-to-die communication with Streaming protocol.
- Die-to-Die (D2D) Adapter Layer: The D2D Adapter coordinates with the HBF Protocol Layer and the Physical Layer to ensure successful data transfer across the UCLe Link.
- Physical Layer: The physical layer has both mainband and sideband communication paths. The mainband carries the mission mode data, where the bulk of the bandwidth

between the HBF and the xPU is carried. The sideband is mainly used for HBF bring-up, control register access, link training and management related communication.

Figure 7 shows a single module per channel configuration. A single UCle module is self-contained to connect the xPU with the one channel of the base die. The bring-up of each channel happens independently. There is no dependency between the channels in the mission mode and can operate independently.



Figure 7: Single Module Per Channel Configuration

5.1.1.1. Protocol Layer

The Die-to-Die (D2D) Adapter negotiates the protocol with the remote link partner and communicates to the Protocol Layers. The streaming protocol is chosen for HBF. The Latency-Optimized 256B Flit Format with Optional Bytes shall be advertised as the chosen Flit format.

5.1.1.2. Adapter Layer

- The Die-to-Die Adapter ensure reliable data transfer, link state and power management, parameter exchange and negotiation during initialization with remote partner.
- The D2D Adapter interfaces with the HBF Protocol Layer via FDI and with the Physical Layer via RDI. Only one instance of RDI and FDI used.
- Since Streaming protocol is negotiated, there is no notion of Downstream Port and Upstream Port for parameter exchanges, and each side independently advertises its capabilities. Additional vendor-defined sideband messages are permitted to be exchanged to negotiate vendor-specific extensions.

5.1.1.3. Operation Flit Format for Streaming Protocol

AXI Link Layer uses latency optimized 256 bytes with optional bytes for maximum protocol efficiency. Streaming Protocol is the default protocol that shall be advertised and negotiated with the remote link partner. Latency-Optimized 256B Flit Format with Optional Bytes for Streaming Protocol bits are set in the UCle Link Capability register.

Protocol Layer transmits 256B of Flit on FDI and the Adapter fills in the Flit header and CRC information before transmitting on RDI. Adapter forwards the Flit received from the Link to the Protocol Layer to the Protocol Layer. The CRC bytes sent with each 128B of the Flit and optional bytes are used by the Protocol Layer. Figure 8 shows the high-level

Format 6 representation. Refer to the Format 6 Latency-Optimized 256B With Optional Bytes Flit Format in the UCle specification v3.0 for the actual bit definitions.



Figure 8: Format 6 Representation

5.1.1.4. Electrical Layer

A high-level block diagram of UCle-A PHY is shown in Figure 9. The MB is the UCle mainband and SB is the UCle sideband. The UCle physical interface for x64 Advanced Package consists of a clock, 64 single-ended data lanes, and a data valid lane in each direction (Tx and Rx). There is a low-speed sideband bus for initialization, Link training, and configuration reads/writes. HBF specific Electrical requirement captured in Chapter 7.

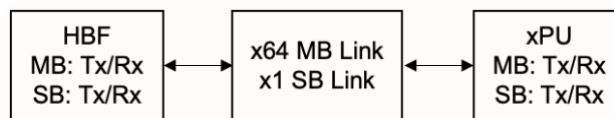


Figure 9: Advanced Package Module for One Channel

5.1.1.5. Sideband Message Support

Three different categories of packets are permitted over UCle sideband: Register accesses, Messages without data, and Messages with data. For more details Review the Sideband protocol specification in the UCle v3.0 specification section 6.1

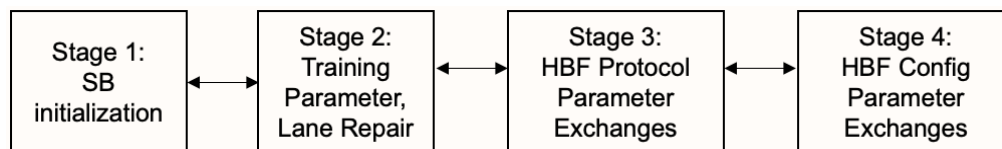
The HBF set/reset the configuration status register sent by the host through the UCle sideband. Figure 10 shows opcode encoding for packet type.

Opcode Encoding	Sideband Packet Type
00000b	32b Memory Read
00001b	32b Memory Write
00010b	32b DMS Register Read
00011b	32b DMS Register Write
00100b	32b Configuration Read
00101b	32b Configuration Write
01000b	64b Memory Read
01001b	64b Memory Write
01010b	64b DMS Register Read
01011b	64b DMS Register Write
01100b	64b Configuration Read
01101b	64b Configuration Write
10000b	Completion without Data
10001b	Completion with 32b Data
10010b	Message without Data
10011b	HBF Vendor Specific
10111b	Management Port Messages without Data
11000b	Management Port Message with Data
11001b	Completion with 64b Data
11011b	Message with 64b Data
Other encodings	Reserved

Figure 10: Packet Type Opcode Encoding

5.1.1.6. Link Initialization

The UCle link initialization goes through four stages as shown in Figure 11. The stage 1 is sideband initialization, and stage 2 is training parameters exchange and lane repair. The stage 3 and stage 4 has HBF protocol specific and HBF configuration parameter exchanges. Review the Link Initialization sequence in the UCle v3.0 specification, Section 3.2.

*Figure 11: High-level UCle Link Initialization*

5.1.2. UCle-A Configurations

As describe in Section 4.2, the HBF UCle module could be arranged for each speed grade and for each speed grade, the AXI channel arrangement to run the interface at different speed that gives the flexibility to integrate at different technology node.

5.1.2.1. UCle-A Speed Grade 1

- UCle-A, X64 8 GT/s – Raw Read/Write Throughput = 64 GB/s
- System Read/Write Throughput Per UCle Module = 48 GB/s
- Maximum number of UCle Modules = 8
- Maximum Stack Height = 8 (Die-0 ~ Die-7)

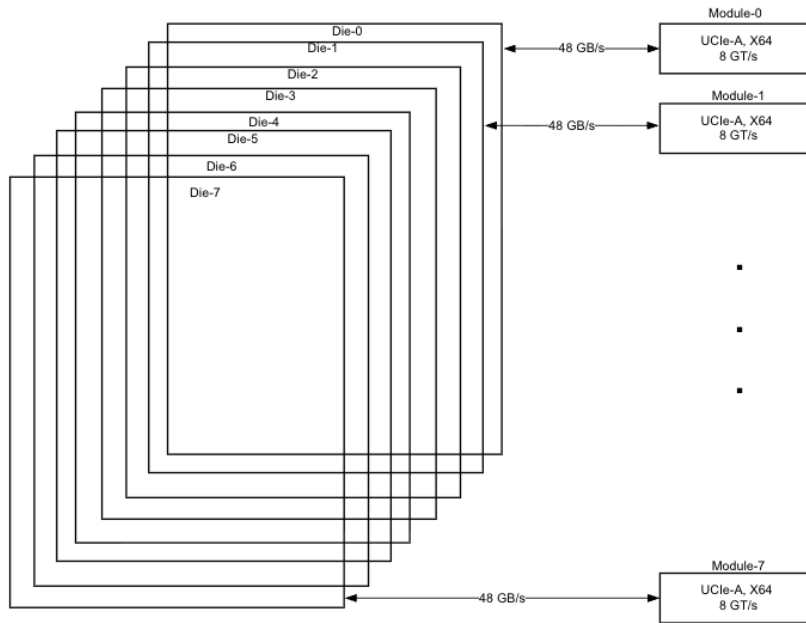


Figure 12: Speed Grade 1 High-Level UCle-A Connection

Option 1

- 1-AXI-Manager/Subordinate per UCle Channel.
- Each AXI Interface running at ≈ 1000 MHz
- Each AXI Channel RAW Throughput = 64 GB/s
- HBF Capacity to each AXI Channel = Capacity of UCle Module
- Scratch pad capacity of each AXI Channel = Scratch pad Capacity of UCle Module

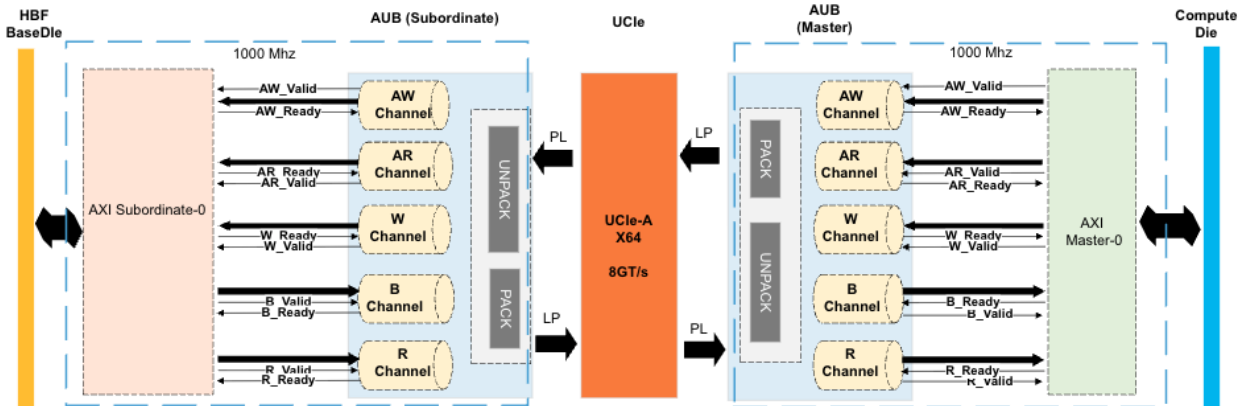


Figure 13: Speed Grade 1, 1-UCle-A Module Connection With 1-AXI Interface

Option 2

- 2-AXI-Manager/Subordinate per UCle Channel.
- Each AXI Interface running at ≈ 500 MHz
- Each AXI Channel RAW Throughput = 32 GB/s
- HBF Capacity to each AXI Channel = Capacity of UCle Module / 2
- Scratch pad capacity of each AXI Channel = Scratch pad Capacity of UCle Module / 2

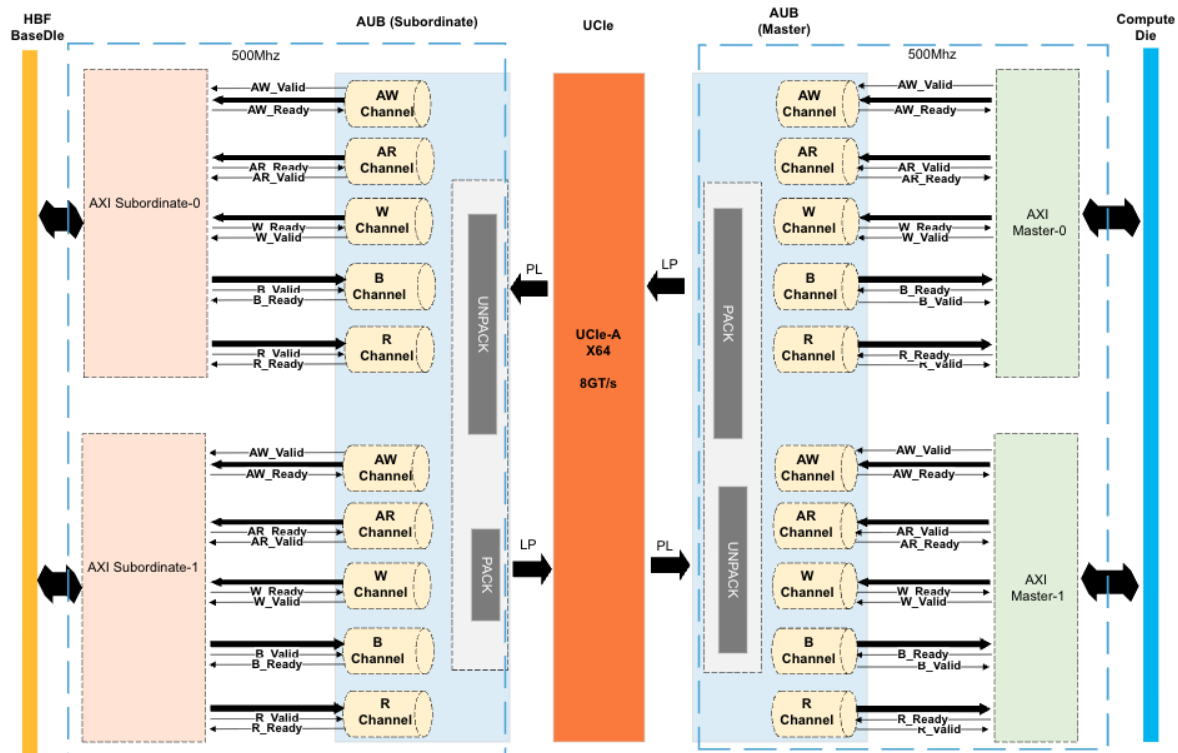


Figure 14: Speed Grade 1, 1-UCle-A Module Connection With 2-AXI Interface

Option 3

- 4-AXI-Manager/Subordinate per UCle Channel.
- Each AXI Interface running at ≈ 250 MHz
- Each AXI Channel RAW Throughput = 16 GB/s
- HBF Capacity to each AXI Channel = Capacity of UCle Module / 4

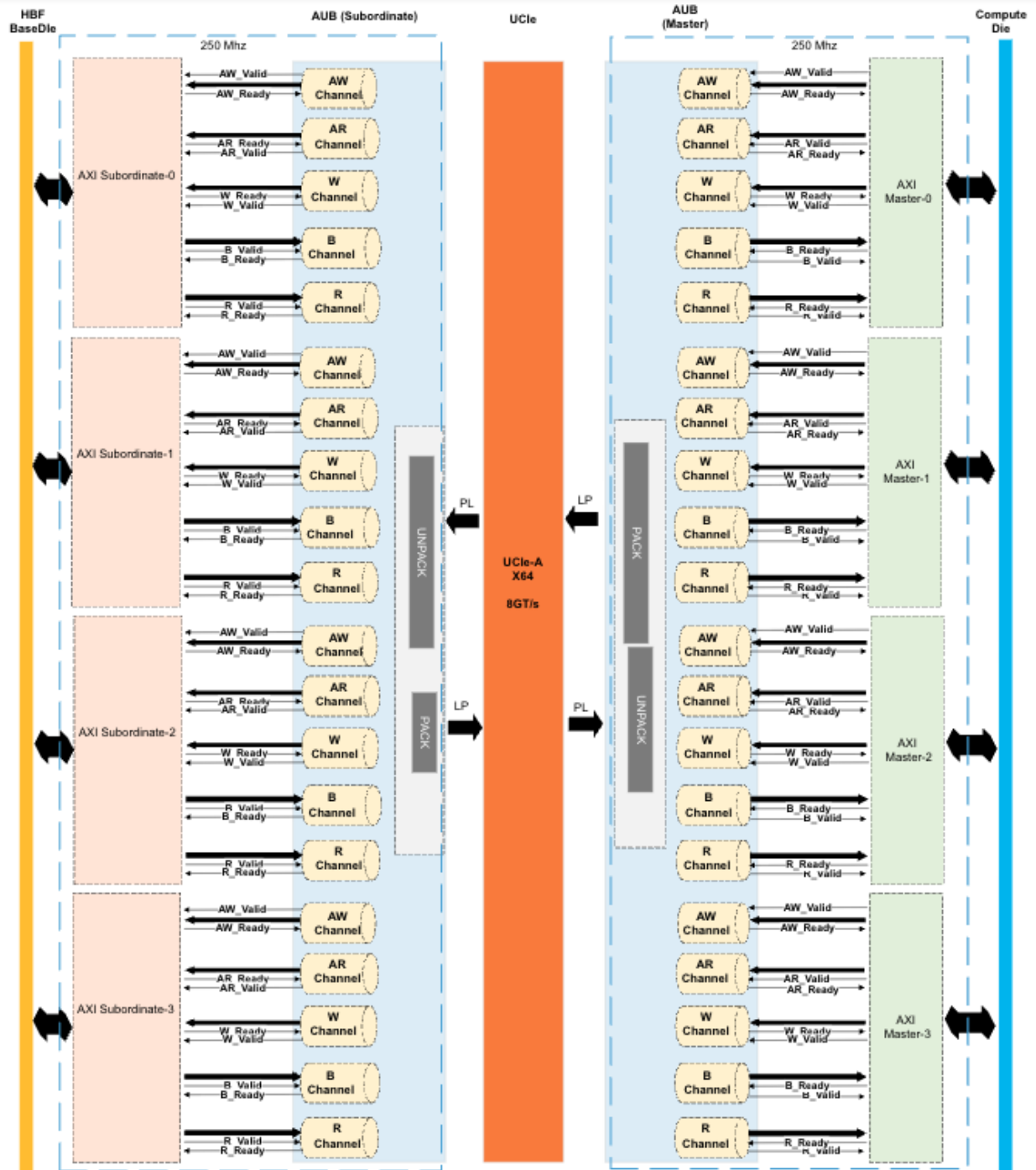


Figure 15: Speed Grade 1, 1-UCle-A Module Connection With 4-AXI Interface

5.1.2.2. UCle-A Speed Grade 2

1. UCle-A, X64 16 GT/s – Raw Read/Write Throughput = 128 GB/s
2. System Read/Write Throughput Per UCle Module = 94 GB/s
3. Maximum number of UCle Modules = 16
4. Maximum Stack Height = 16

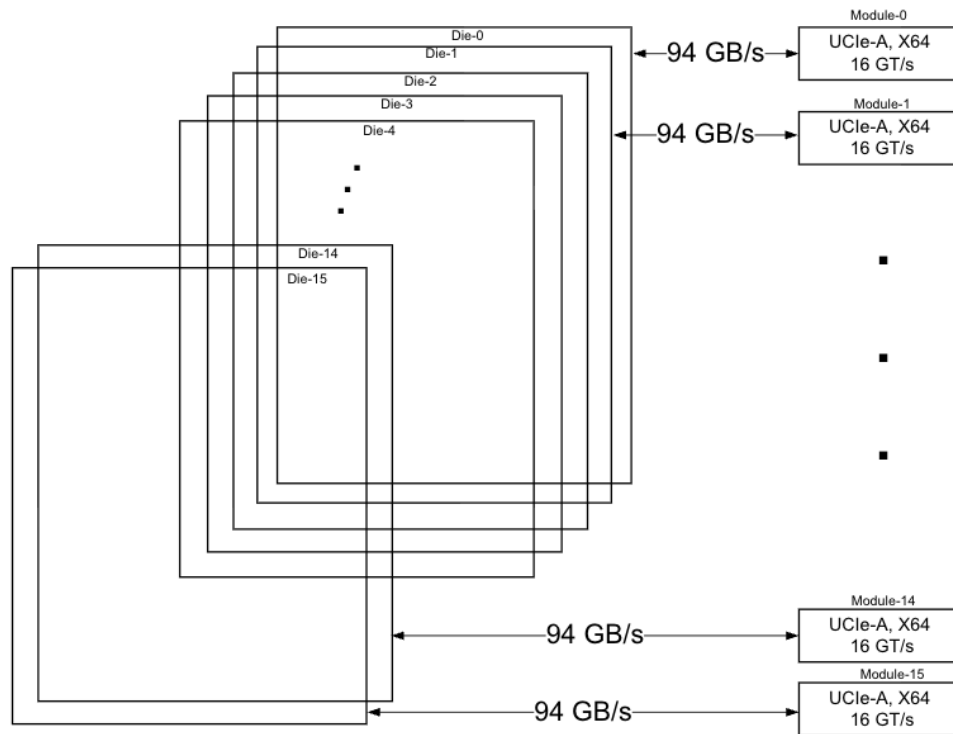


Figure 16: Speed Grade 2 High-Level UCle-A Connection

Option 1

- 1-AXI-Manager/Subordinate per UCle Channel.
- Each AXI Interface running at ≈ 2000 MHz
- Each AXI Channel RAW Throughput = 128 GB/s
- HBF Capacity to each AXI Channel = Capacity of UCle Module"
- Scratch pad capacity of each AXI Channel= Scratch pad Capacity of UCle Module"

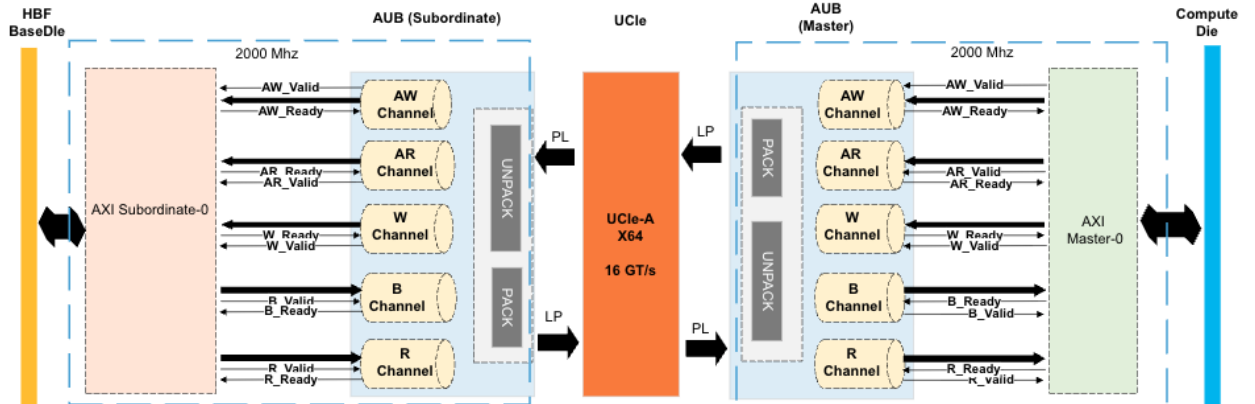


Figure 17: Speed Grade 2, 1-UCle-A Module Connection With 1-AXI Interface

Option 2

- 2-AXI-Manager/Subordinate per UCle Channel.
- Each AXI Interface running at ~ 1000 MHz
- Each AXI Channel RAW Throughput = 64 GB/s
- HBF Capacity to each AXI Channel = Capacity of UCle Module/2
- Scratch pad capacity of each AXI Channel = Scratch pad Capacity of UCle Module/2

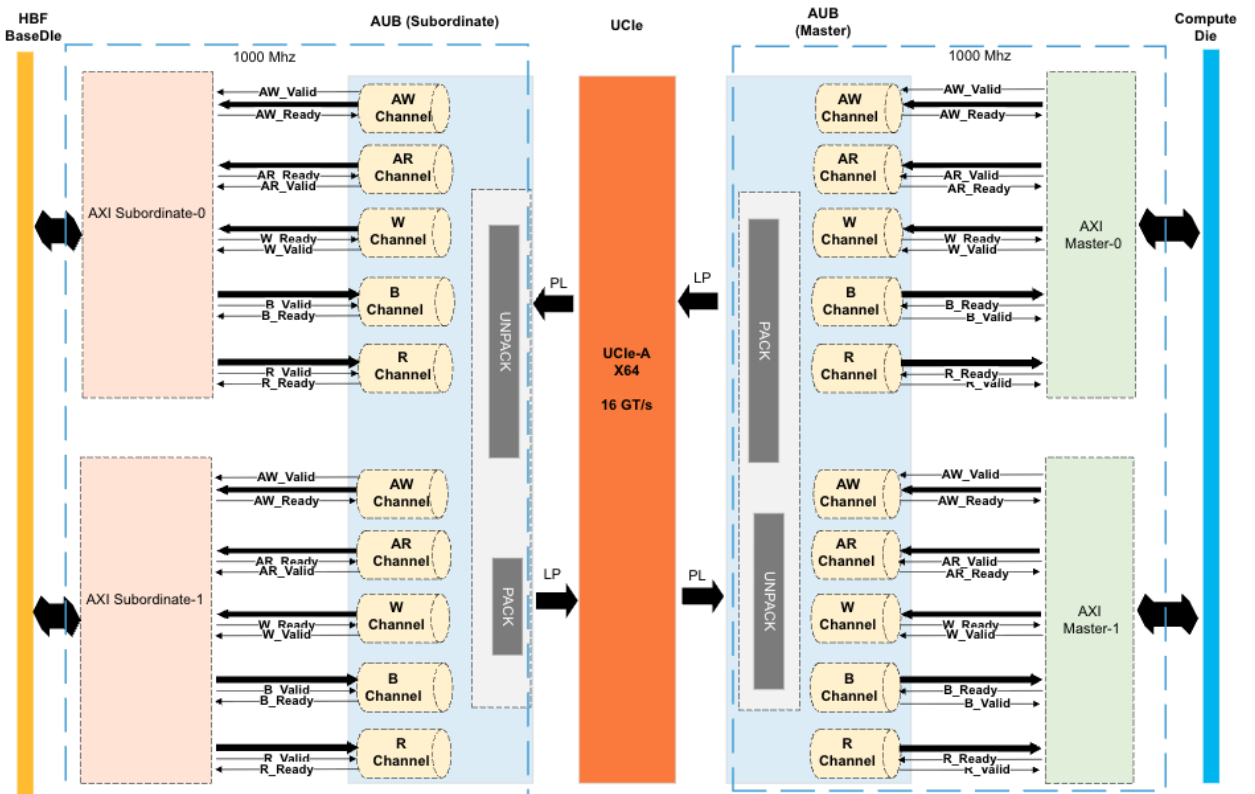


Figure 18: Speed Grade 2, 1-UCle-A Module Connection With 2-AXI Interface

Option 3

- 4-AXI-Manager/Subordinate per UCle Channel.
- Each AXI Interface running at ≈ 500 MHz
- Each AXI Channel RAW Throughput = 32 GB/s
- HBF Capacity to each AXI Channel = Capacity of UCle Module/4
- Scratch pad capacity of each AXI Channel= Scratch pad Capacity of UCle Module/4

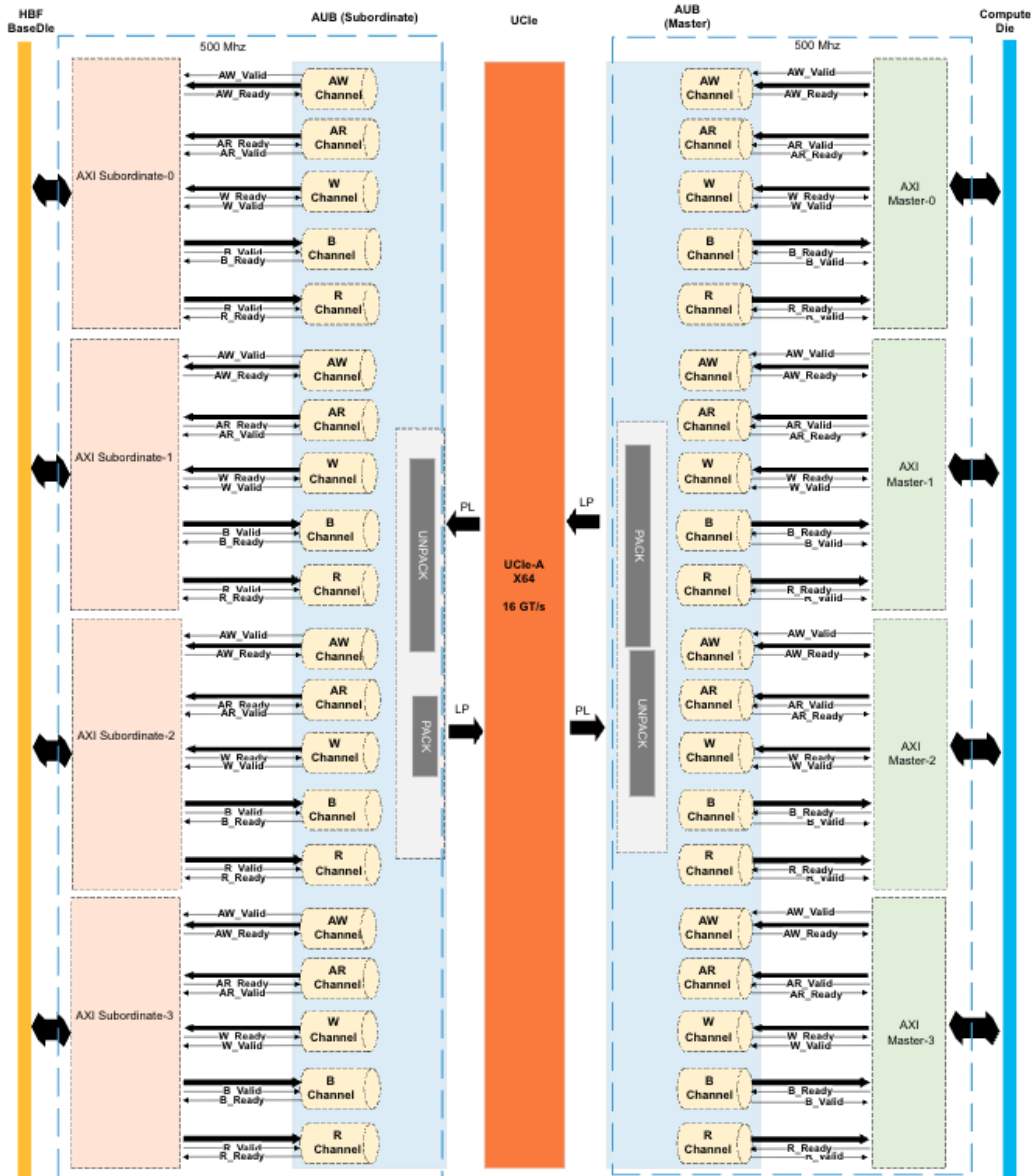


Figure 19: Speed Grade 2, 1-UCle-A Module Connection With 4-AXI Interface

5.1.2.3. UCle-A Speed Grade 3

- UCle-A, X64 32 GT/s – Raw Read/Write Throughput = 256 GB/s
- System Read/Write Throughput Per UCle Module = 188 GB/s
- Maximum number of UCle Modules = 16
- Maximum Stack Height = 16

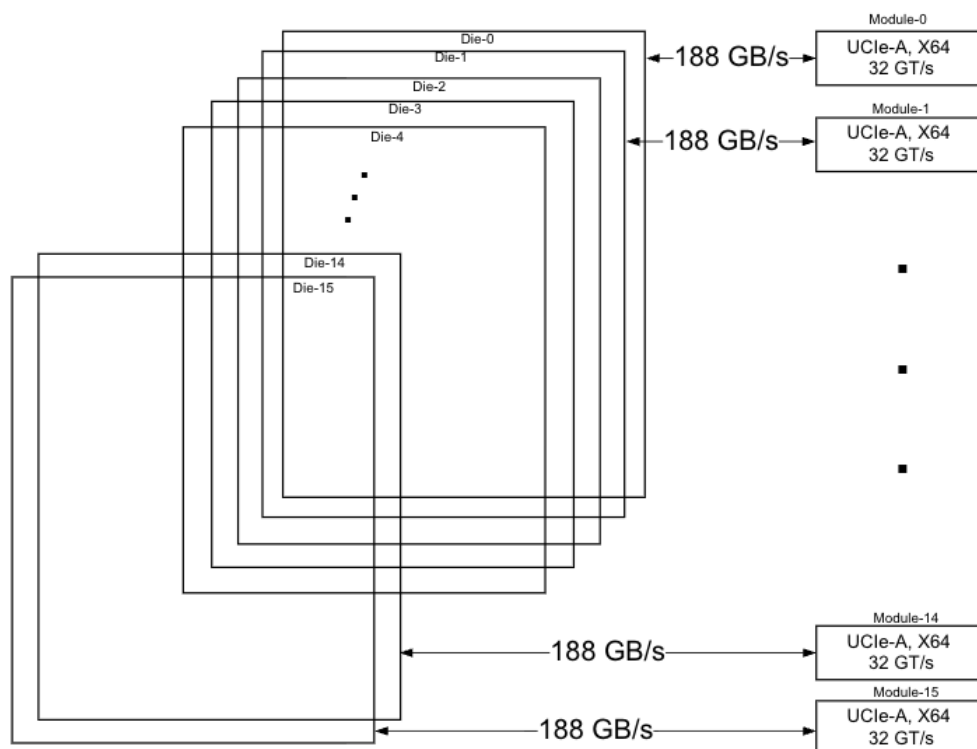


Figure 20: Speed Grade 3 High-Level UCle-A Connection

Option 1

- 1-AXI-Manager/Subordinate per UCle Channel.
- Each AXI Interface running at ≈ 4000 MHz
- Each AXI Channel RAW Throughput = 256 GB/s
- HBF Capacity to each AXI Channel = Capacity of UCle Module"
- Scratch pad capacity of each AXI Channel= Scratch pad Capacity of UCle Module"

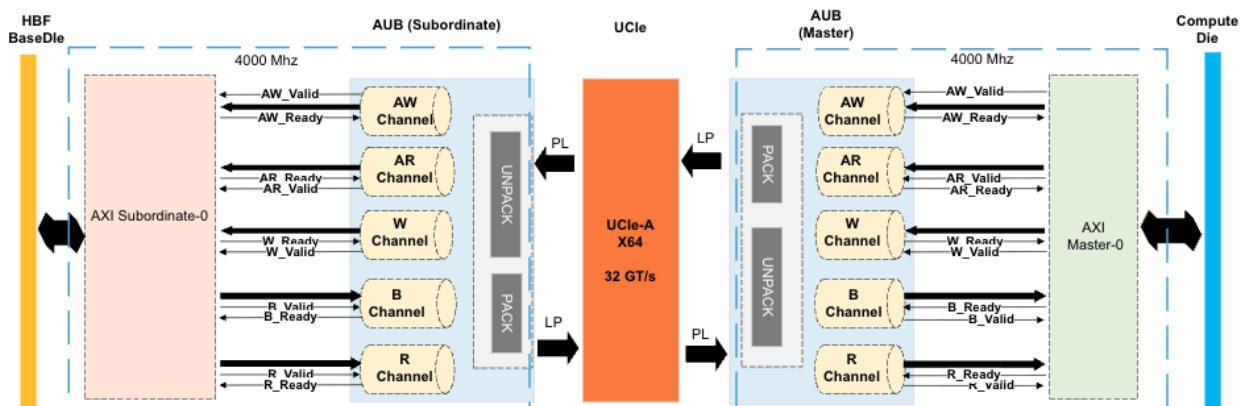


Figure 21: Speed Grade 3, 1-UCle-A Module Connection With 1-AXI Interface

Option 2

- 2-AXI-Manager/Subordinate per UCle Channel.
- Each AXI Interface running at ~ 2000 Mhz
- Each AXI Channel RAW Throughput = 128 GB/s
- HBF Capacity to each AXI Channel = Capacity of UCle Module/2
- Scratch pad capacity of each AXI Channel = Scratch pad Capacity of UCle Module/2

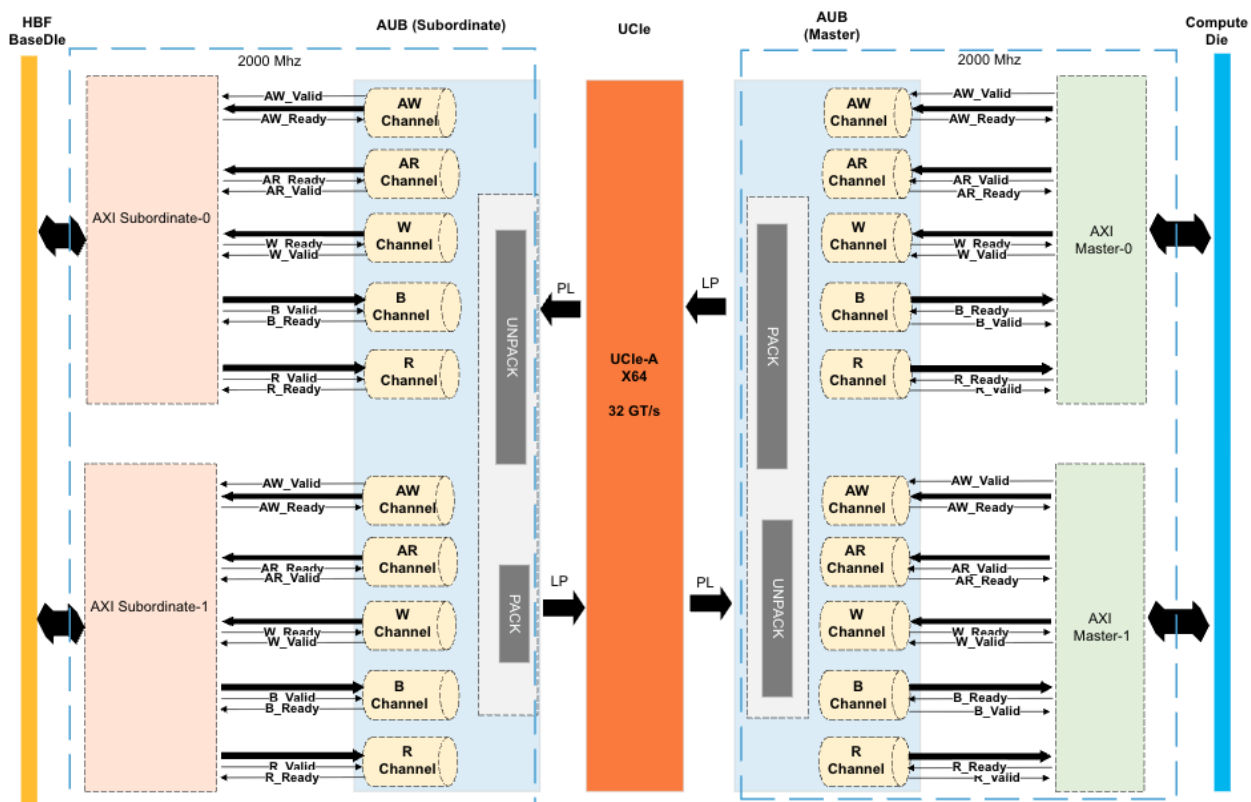


Figure 22: Speed Grade 3, 1-UCle-A Module Connection With 2-AXI Interface

Option 3

- 4-AXI-Manager/Subordinate per UCle Channel.
- Each AXI Interface running at ≈ 1000 MHz
- Each AXI Channel RAW Throughput = 64 GB/s
- HBF Capacity to each AXI Channel = Capacity of UCle Module/4
- Scratch pad capacity of each AXI Channel= Scratch pad Capacity of UCle Module/4

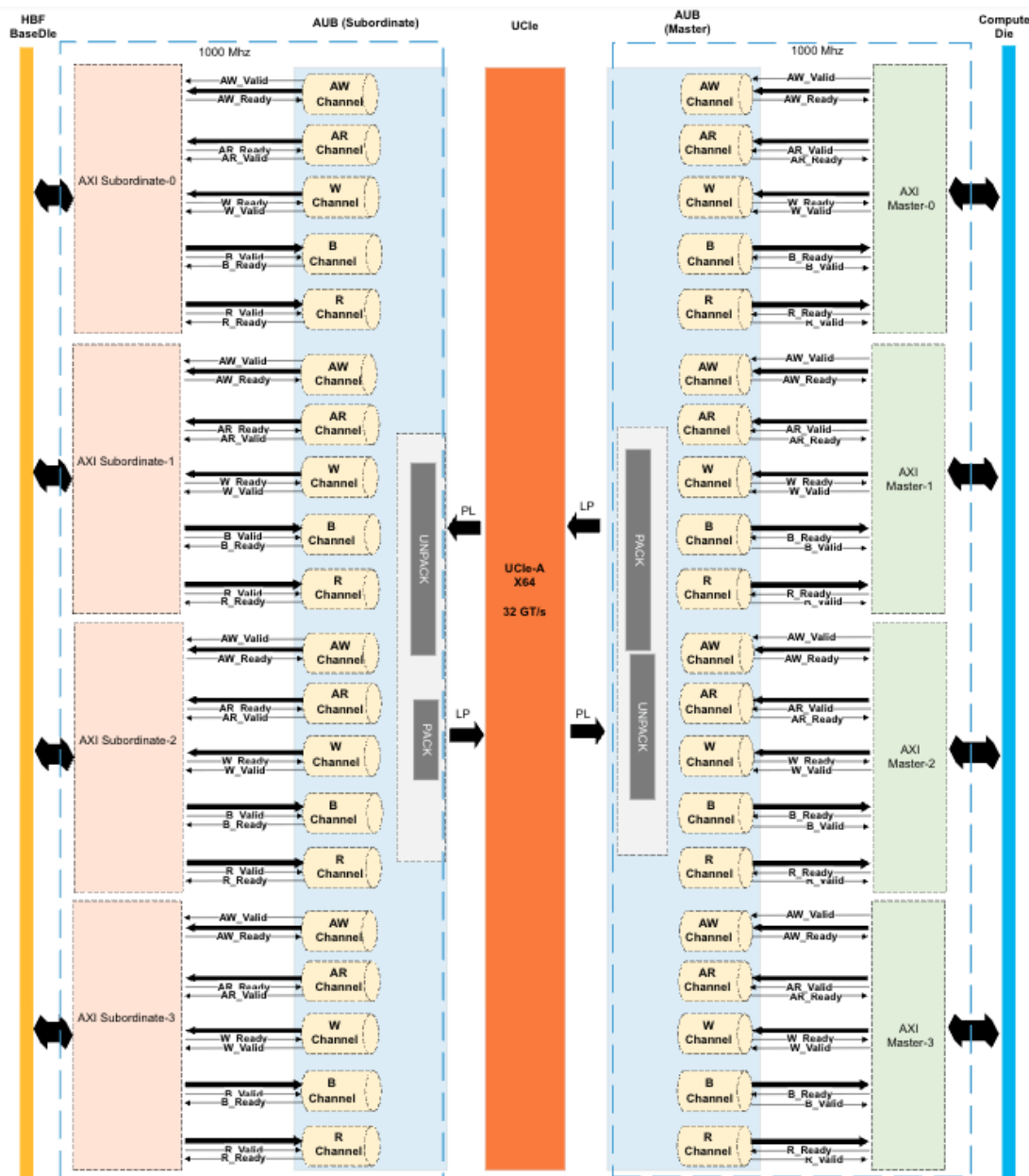


Figure 23: Speed Grade 3, 1-UCle-A Module Connection With 4-AXI Interface

5.2. AXI Host Interface Architecture

5.2.1. AXI over UCle Overview

The AMBA AXI interface is a widely used open specification communication interface that is used to connect one or more initiators to one or more targets. The xPU to HBF communication interface uses AXI as protocol layer running on UCle, which is an on-package interconnect for connecting multiple dies on the same package. The link layer is a wrapper on top of UCle interface for flow control and packing/unpacking AXI channel information in UCle FLIT format.

UCle supports multiple protocols including Streaming, or raw format that are run on UCle if both ends support it. The xPU to HBF link layer protocol is chosen as AXI because the overhead of the protocol is very low. For AXI, only a subset of features is used in the xPU to HBF interface and only several of its features are used, which helps the overhead low and keeps the UCle efficiency high.

The red-dotted box in Figure 24 is the standard UCle module. The two blue-dotted boxes that include the Link Layer are customized by the xPU and HBF vendors. This specification has provided a version of AXI Link Layer protocol specification that runs on UCle.

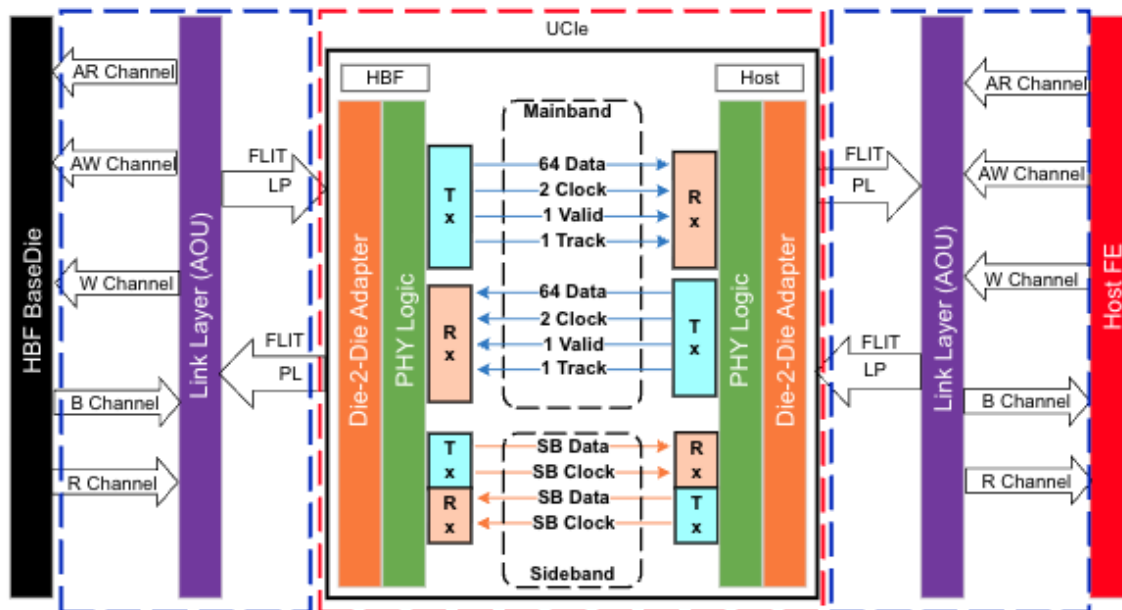


Figure 24: High-level HBF to Host Connection via UCle for AXI

1. Write request channel (AW): Write request channel carries all the required address and control information for a write transaction.
2. Read request channel (AR): Read request channel carries all the required address and control information for a read transaction.
3. Write data channel (W): The write data channel carries the write data from the Manager to the Subordinate.
4. Write response channel (B): A Subordinate uses the write response channel to respond to write transactions. All write transactions require completion signals on the write response channel.
5. Read data channel (R): The read data channel carries both the read data and the read response information from the Subordinate to the Manager.

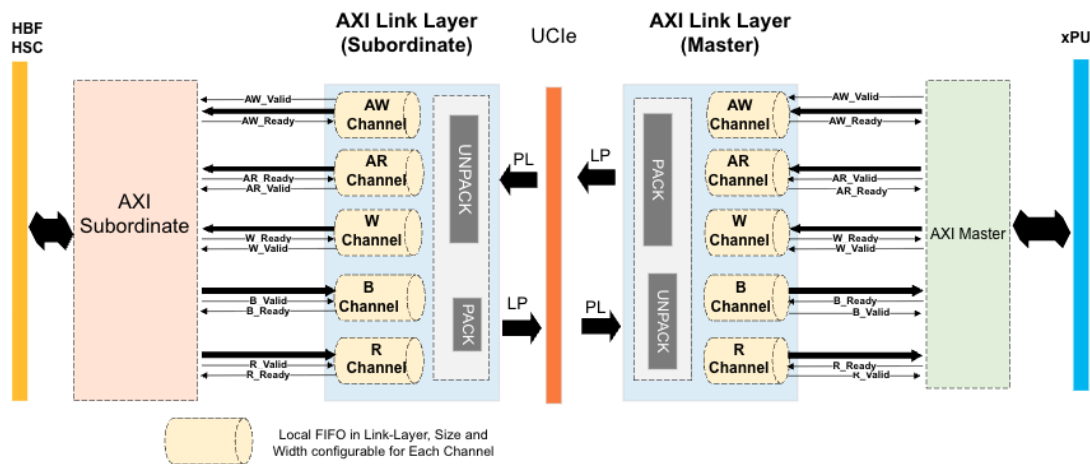


Figure 25: AXI Protocol High-Level Interaction and Flow Control

- Each channel used a valid and ready single for flow control.
- The Link Layer will pack outflow information in a Flit format with flow control information (link-layer FIFO available space) and send it over the UCle interface to the other end of the UCle module.
- The link layer receives inflow information in the Flit format, and the link layer unpacks the information and sends the information to the AXI protocol module.
- In the inflow information Flit, the Link Layer also receives information about available FIFO space at the other end of the UCle interface FIFO. Based on this info, the Link Layer will decide how much channel information the other end of UCle is ready to receive.

5.2.2. AXI Link Layer Adapter

The AXI Link Layer Adapter (ALL Adapter) is situated between the Interface Adapter and AXI Link Layer. It holds a few entries for each AXI channel, controls AXI-ready and valid signals,

and translates descriptor information into AXI signals. As the AXI protocol is standard, the architecture specifications do not include more details related to AXI valid and ready signals.

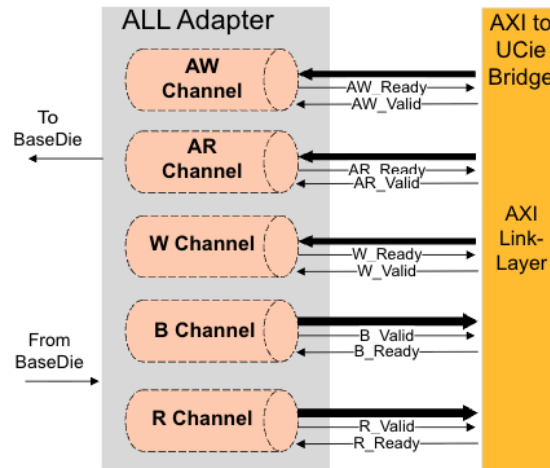


Figure 26: High-level Block Diagram of ALL Adapter

5.2.3. AXI Master Link Layer Adapter

The AXI Master Link Layer Adapter (AMLL Adapter) is an interface between the host traffic software interface and the AXI Link Layer Master. It holds a few entries for each AXI channel, controls AXI-ready and valid signals, and translates descriptor information generated by host software into AXI signals. As the AXI protocol is standard, the architecture specifications do not include more details related to AXI valid and ready signals.

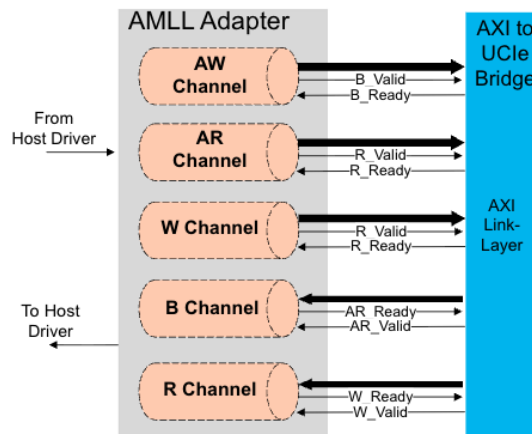


Figure 27: High-level Block Diagram of AMLL Adapter

5.2.4. AXI Protocol for HBF

This section contains the details of the AXI protocol for communicating with HBF. For details of command status and error information, review Sections 5.5 and 5.6.

Table 8: AXI Protocol

AXI Channel	Signals	Description
Read Command (AR Channel)	ARADDR[AR_ADDR_WIDTH-1:0]	Cmd_packet_type: IO CMD [AR_ADDR_WIDTH-1:0] : Read Address Cmd_packet_type: CSR/ADMIN [7:0] : Opcode [AR_ADDR_WIDTH-1:8] : Address/reserved
	ARUSER[AR_USER_WIDTH-1:0]	ARUSER [1:0]: Cmd_packet_type 2b'00: Flash IO CMD 2b'01: Scratchpad IO CMD 2b'10: CSR/ADMIN ¹ 2b'11: Reserve ARUSER [2]: Read Type 1b'0: Regular Read 1b'1: Batch Read ARUSER [2] is used only for Flash IO Cmd or else Reserved ARUSER [5:2]: Reserved ARUSER [AR_USER_WIDTH-1:6]: Read Retry Stage (product specific)
	ARID[AR_ID_WIDTH-1:0]	Read ID (For out of order command execution, host software will assign a unique read ID for each command)
	ARLEN[AR_LEN_WIDTH-1:0]	Read length (if HBF product support burst read/write, this defines length/number of beats in burst command)
Write Command (AW Channel)	AWADDR[AW_ADDR_WIDTH-1:0]	Cmd_packet_type: IO CMD [AW_ADDR_WIDTH-1:0] : Write Address Cmd_packet_type: CSR/ADMIN [7:0] : Opcode [AW_ADDR_WIDTH-1:8] : Address/reserved
	AWUSER[AW_USER_WIDTH-1:0]	AWUSER [1:0]: Cmd_packet_type 2b'00: Flash IO CMD 2b'01: Scratchpad IO CMD 2b'10: CSR/ADMIN 2b'11: Reserved AWUSER [2] is used only for Flash IO Cmd or else Reserved AWUSER [AW_USER_WIDTH-1:2]: Reserved

AXI Channel	Signals	Description
	AWID[AW_ID_WIDTH-1:0]	Write ID (For out of order command execution, host software will assign a unique write ID for each command)
	AWLEN[AW_LEN_WIDTH-1:0]	Write length (if HBF product support burst read/write, this defines length/number of beats in burst command)
Write Data (W Channel)	WDATA[W_DATA_WIDTH-1:0]	Write Data
	WLAST	Last transaction of Write
Read Response (R Channel)	RRESP[R_RESP_WIDTH-1:0]	Read Response (AXI protocol specific read response, detail read response flash specific will be part of user field)
	RUSER[R_USER_WIDTH-1:0]	RUSER [1:0]: RResp_packet_type 2b'00: Flash IO CMD 2b'01: Scratchpad IO CMD 2b'10: CSR/ADMIN 2b'11: Reserved RUSER [5:2]: CMD Status ² RUSER [R_USER_WIDTH-1:6]: ErrorInfo
	RID[R_ID_WIDTH-1:0]	Read ID
	RDATA[R_DATA_WIDTH-1:0]	Read data
	RLAST	Last Read data transaction
Write Response (B Channel)	BRESP[B_RESP_WIDTH-1:0]	Write Response (AXI protocol specific write response, detail write response flash specific will be part of user field)
	BUSER[B_USER_WIDTH-1:0]	BUSER [1:0]: BResp_packet_type 2b'00: Flash IO CMD 2b'01: Scratchpad IO CMD 2b'10: CSR/ADMIN 2b'11: Reserved BUSER [5:2]: CMD Status ³ BUSER [B_USER_WIDTH-1:6]: ErrorInfo
	BID[B_ID_WIDTH-1:0]	Write ID

Note 1: The CSR (control status register) provides access to all control register read/write operations and the protocol interface for admin commands. The same AXI ID will return in-order response and different AXI ID may return out of order response. It is recommended to use the same AXI ID for all CSR commands.

Note 2: The command status and error details are in Table 12

Note 3: The command status and error details are in Table 13

Table 9: Opcodes for CSR/ADMIN Commands

CMD TYPE	OPCODE	AXI-CHANNEL	Description
Set Feature	0x01	AW	Set product-specific feature

CMD TYPE	OPCODE	AXI-CHANNEL	Description
Get Feature	0x02	AR	Get product-specific feature enable/disable
Secure Erase	0x03	AW	Erase part of UCle Channel or complete channel capacity
Get Log Page	0x04	AR	Read smart /Error/Telemetry log
RSVD	0x05-0x6	AR	Reserved
BIST	0x07	AW	Run product-specific BIST test.
Zone Remapping	0x08	AW	Remapped a zone UCle local channel address to less wear blocks
Read UCle Errors	0x09	AR	Read UCle errors for that specific channel
Reduced Capacity	0x0A	AR	Read reduced capacity information
RSVD	0x0B-0x1F		Reserved
Register Read/Write	0x20	AR/AW	Register read/writes

5.2.4.1. Set Feature command

AXI Channel	Signals	Description
Write Command (AW Channel)	AWADDR[AW_ADDR_WIDTH-1:0]	[7:0] : 0x01 [AW_ADDR_WIDTH-1:8] : Set-Feature Address (Product Specific)
	AWUSER[AW_USER_WIDTH-1:0]	AWUSER [1:0]: 2b'10 AWUSER [AW_USER_WIDTH-1:2]: Reserved
	AWID[AW_ID_WIDTH-1:0]	Write ID (we recommended for all CSR/ADMIN command use same ID to avoid out of order execution)
	AWLEN[AW_LEN_WIDTH-1:0]	0x0
Write Data (W Channel)	WDATA[W_DATA_WIDTH-1:0]	[W_DATA_WIDTH-1:0] : Value
	WLAST	1b'1
Write Response (B Channel)	BRESP[B_RESP_WIDTH-1:0]	Write Response (AXI protocol specific write response, detail write response flash specific will be part of user field)
	BUSER[B_USER_WIDTH-1:0]	BUSER [1:0]: 2b'10 BUSER [5:2]: CMD Status 0x0: Success 0x1: Invalid Set-Feature Address 0x2: Invalid Value 0x3: Access Denied 0x4: HBF Internal Error (vendor specific)

AXI Channel	Signals	Description
		BUSER [B_USER_WIDTH-1:6]: Vendor specific Error Details
	BID[B_ID_WIDTH-1:0]	Write ID

5.2.4.2. Get Feature command

AXI Channel	Signals	Description
Read Command (AR Channel)	ARADDR[AR_ADDR_WIDTH-1:0]	[7:0] : 0x02 [AR_ADDR_WIDTH-1:8] : Address/reserved
	ARUSER[AR_USER_WIDTH-1:0]	ARUSER [1:0]: 2b'10 ARUSER [AR_USER_WIDTH-1:2]: Reserved
	ARID[AR_ID_WIDTH-1:0]	Read ID (we recommend to used same ID for all CSR/ADMIN Command for in order execution)
	ARLEN[AR_LEN_WIDTH-1:0]	0x0
Read Response (R Channel)	RRESP[R_RESP_WIDTH-1:0]	Read Response (AXI protocol specific read response, detail read response flash specific will be part of user field)
	RUSER[R_USER_WIDTH-1:0]	RUSER [1:0]: 2b'10 RUSER [5:2]: CMD Status 0x0: Success 0x1: Access Denied 0x2-0x3: Reserved 0x4: HBF Internal Error (vendor specific) RUSER [R_USER_WIDTH-1:6]: Vendor specific Error Details
	RID[R_ID_WIDTH-1:0]	Read ID
	RDATA[R_DATA_WIDTH-1:0]	[R_DATA_WIDTH-1:0] : Value
	RLAST	1b'1

5.2.4.3. Secure Erase command

AXI Channel	Signals	Description
Write Command (AW Channel)	AWADDR[AW_ADDR_WIDTH-1:0]	[7:0] : 0x03 [9:8] : Capacity Type (CT) 00b: Complete AXI Port Capacity 01b: Only HBF Capacity 10b: Only HBF Zone Specific 11b: Only Scratchpad RAM [11:10] : Secure Erase Setting (SES) 00b: No secure Erase 01b: HBF secure Erase 10b: Cryptographic Erase 11b: Reserved [AW_ADDR_WIDTH-1:12] : Reserved
	AWUSER[AW_USER_WIDTH-1:0]	AWUSER [1:0]: 2b'10 AWUSER [AW_USER_WIDTH-1:2]: Reserved
	AWID[AW_ID_WIDTH-1:0]	Write ID (we recommended for all CSR/ADMIN command use same ID to avoid out of order execution)
	AWLEN[AW_LEN_WIDTH-1:0]	0x0
Write Data (W Channel)	WDATA[W_DATA_WIDTH-1:0]	If Capacity Type (CT) == 10b [31:0] : ZoneBitMap else [31:0] : Reserved [W_DATA_WIDTH-1:32] : Reserved
	WLAST	1b'1
Write Response (B Channel)	BRESP[B_RESP_WIDTH-1:0]	Write Response (AXI protocol specific write response, detail write response flash specific will be part of user field)
	BUSER[B_USER_WIDTH-1:0]	BUSER [1:0]: 2b'10 BUSER [5:2]: CMD Status 0x0: Success 0x1: Invalid Field (SES Not supported) 0x2: Invalid Zone Bitmap 0x3: Command Sequence Error (Another Secure Erase already in process) 0x4: HBF Internal Error (vendor specific)

AXI Channel	Signals	Description
		BUSER [B_USER_WIDTH-1:6]: Vendor specific Error Details
	BID[B_ID_WIDTH-1:0]	Write ID

5.2.4.4. Get Log Page Command

AXI Channel	Signals	Description
Read Command (AR Channel)	ARADDR[AR_ADDR_WIDTH-1:0]	[7:0] : 0x04 [11:8] : Log Page Type (LPT) 0x0: Smart Log 0x1: HBF Error Log 0x2: UCle Error Log 0x3: Vendor Specific Telemetry Log 0x4-0xF: Reserved [15:12]: Reserved [31:16] : Start Log Offset (unit-64B) [AR_ADDR_WIDTH-1:32] : reserved
	ARUSER[AR_USER_WIDTH-1:0]	ARUSER [1:0]: 2b'10 ARUSER [AR_USER_WIDTH-1:2]: Reserved
	ARID[AR_ID_WIDTH-1:0]	Read ID (we recommend to used same ID for all CSR/ADMIN Command for in order execution)
	ARLEN[AR_LEN_WIDTH-1:0]	Transaction Length (Number of Beats)
Read Response (R Channel)	RRESP[R_RESP_WIDTH-1:0]	Read Response (AXI protocol specific read response, detail read response flash specific will be part of user field)
	RUSER[R_USER_WIDTH-1:0]	RUSER [1:0]: 2b'10 RUSER [5:2]: CMD Status 0x0: Success 0x1: Invalid LPT (Not supported) 0x2: Invalid offset and Length 0x3: No Error log Entry Present 0x4: HBF Internal Error (vendor specific) 0x5: Media Access denied

AXI Channel	Signals	Description
		RUSER [R_USER_WIDTH-1:6]: Vendor specific Error Details
	RID[R_ID_WIDTH-1:0]	Read ID
	RDATA[R_DATA_WIDTH-1:0]	[R_DATA_WIDTH-1:0]. : Data
	RLAST	Last Read data transaction

5.2.4.5. BIST(Built-In-Self-Test) command

The BIST command (opcode=0x7) shall be used by the Host to initiate product-specific BIST test operations within HBF Device. Since the BIST operations may be destructive to the data stored or may disrupt normal IO operations, the Host should only issue this command.

AXI Channel	Signals	Description
Write Command (AW Channel)	AWADDR[AW_ADDR_WIDTH-1:0]	Cmd_packet_type: CSR/ADMIN [AW_ADDR_WIDTH-1:16] : Reserved [15:8] : BIST Type; This field specifies the type of BIST to run. (product-specific) [7:0] : Opcode; 0x07 (BIST)
	AWUSER[AW_USER_WIDTH-1:0]	AWUSER [AW_USER_WIDTH-1:2]: Reserved AWUSER [1:0]: Cmd_packet_type 2b'10: CSR/ADMIN
	AWID[AW_ID_WIDTH-1:0]	Write ID (For all CSR/ADMIN command, using same ID to avoid out of order execution is recommended)
	AWLEN[AW_LEN_WIDTH-1:0]	Write length (if HBF product support burst read/write, this defines length/number of beats in burst command)
Write Data (W Channel)	WDATA[W_DATA_WIDTH-1:0]	Write Data
	WLAST	Last transaction of Write
Write Response (B Channel)	BRESP[B_RESP_WIDTH-1:0]	Write Response (AXI protocol specific write response, detail write response flash specific will be part of user field)

	BUSER[B_USER_WIDTH-1:0]	BUSER [B_USER_WIDTH-1:6]: ErrorInfo BUSER [5:2]: CMD Status 0x00: Success 0x01: Invalid BIST Type 0x02: Device Busy; already another BIST is in-processing or BIST cannot be initiated (e.g., normal IO traffic is in-processing) 0x03: Reserved 0x04: HBF Internal Error (vendor specific) 0x05 – 0x0F: product-specific field BUSER [1:0]: BResp_packet_type 2'b10: CSR/ADMIN
	BID[B_ID_WIDTH-1:0]	Write ID

5.2.4.6. Zone Remapping command

The Zone Remapping command (Opcode=0x8), only valid when BUCC.WLS is set to ‘10b: Host-controlled Wear Leveling’, shall be used by the Host to swap the Hot Zone(High-PEC) with Cold Zone(Low PEC) as remapping a zone to less-worn physical NAND blocks. This command only updates the logical-to-physical mapping within HBF device. It does not migrate or copy the user data between physical NAND blocks. Therefore, upon completion of the remapping, Host is responsible for rewriting the data which was written in the Source Zone to Destination Zone.

AXI Channel	Signals	Description
Write Command (AW Channel)	AWADDR[AW_ADDR_WIDTH-1:0]	Cmd_packet_type: CSR/ADMIN [AW_ADDR_WIDTH-1:16] : Reserved [15:8] : Remap Type; 0x00: Swap – Swap the physical mapping of the Source Zone and Destination Zone. 0x01 – 0xFF: Reserved (product-specific) [7:0] : Opcode; 0x08 (Zone Remapping)
	AWUSER[AW_USER_WIDTH-1:0]	AWUSER [AW_USER_WIDTH-1:2]: Reserved AWUSER [1:0]: Cmd_packet_type 2b'10: CSR/ADMIN
	AWID[AW_ID_WIDTH-1:0]	Write ID

		(For all CSR/ADMIN command, using same ID to avoid out of order execution is recommended)
	AWLEN[AW_LEN_WIDTH-1:0]	Write length (if HBF product support burst read/write, this defines length/number of beats in burst command)
Write Data (W Channel)	WDATA[W_DATA_WIDTH-1:0]	Write Data; Within a single WDATA payload, each Zone ID shall appear only once. WDATA[511:496]: 16 th Destination Zone ID WDATA[495:480]: 16 th Source Zone ID ... WDATA[63:48]: 2 nd Destination Zone ID WDATA[47:32]: 2 nd Source Zone ID WDATA[31:16]: 1 st Destination Zone ID (Cold Zone) WDATA[15:00]: 1 st Source Zone ID (Hot Zone)
	WLAST	Last transaction of Write
Write Response (B Channel)	BRESP[B_RESP_WIDTH-1:0]	Write Response (AXI protocol specific write response, detail write response flash specific will be part of user field)
	BUSER[B_USER_WIDTH-1:0]	BUSER [B_USER_WIDTH-1:6]: ErrorInfo; Additional Error Information BUSER [5:2]: CMD Status 0x00: Success 0x01: Invalid ZoneID 0x02: Invalid Remap Type 0x03: Reserved 0x04: HBF Internal Error (vendor specific) 0x05 – 0x0F: product-specific field BUSER [1:0]: BResp_packet_type 2b'10: CSR/ADMIN
	BID[B_ID_WIDTH-1:0]	Write ID

5.2.4.7. Reduced Capacity command

The Reduced Capacity command (Opcode=0xA) shall be used by the Host to retrieve the unusable logical capacity information of the HBF device, Refer to section 11.5.1.8. The HBF device maintains a bitmap representing the health status of each NAND block. The Host shall read this bitmap, after a power cycle or device initialization, to calculate the usable capacity address space and prevent issuing IO requests to unusable NAND blocks.

AXI Channel	Signals	Description
Read Command (AR Channel)	ARADDR[AR_ADDR_WIDTH-1:0]	Cmd_packet_type: CSR/ADMIN [AR_ADDR_WIDTH-1:32] : Reserved [31:16] : Start Info Offset (unit- 64B) [15:8] : Report Info Type; 0x00: Per-Block Bitmap 0x01 – 0xFF: product-specific [7:0] : Opcode; 0x0A (Reduced Capacity)
	ARUSER[AR_USER_WIDTH-1:0]	ARUSER [AR_USER_WIDTH-1:2]: Reserved ARUSER [1:0]: Cmd_packet_type 2b'10: CSR/ADMIN
	ARID[AR_ID_WIDTH-1:0]	Read ID (For all CSR/ADMIN command, using same ID to avoid out of order execution is recommended)
	ARLEN[AR_LEN_WIDTH-1:0]	Read length (if HBF product support burst read/write, this defines length/number of beats in burst command)
Read Response (R Channel)	RRESP[R_RESP_WIDTH-1:0]	Read Response (AXI protocol specific read response, detail read response flash specific will be part of user field)
	RUSER[R_USER_WIDTH-1:0]	RUSER [R_USER_WIDTH-1:6]: ErrorInfo RUSER [5:2]: CMD Status 0x00: Success 0x01: Invalid Report Info Type 0x02: Invalid Start Info Offset or Length 0x03: Reserved 0x04: HBF Internal Error (vendor)

		specific) 0x05 – 0x0F: product-specific field RUSER [1:0]: RResp_packet_type 2b'10: CSR/ADMIN
	RID[R_ID_WIDTH-1:0]	Read ID
	RDATA[R_DATA_WIDTH-1:0]	Read data; Reduced Capacity Information When Report Type(RT) is 0x00, Device shall return 'Per-Block Bitmap'. The LSB of the first returned bitmap corresponds to NAND block 0(the lowest NAND block address). Each bit represents 1-block. If it is set to '1', the corresponding NAND block is unusable, and if it is cleared to 0, the corresponding NAND block is usable.
	RLAST	Last Read data transaction

5.2.4.8. Register Read command

The Register Read command (Opcode=0x20) shall be used by the Host to directly access the Always-On Configuration registers and MMIO-mapped registers through UCle mainband.

AXI Channel	Signals	Description
Read Command (AR Channel)	ARADDR[AR_ADDR_WIDTH-1:0]	Cmd_packet_type: CSR/ADMIN [AR_ADDR_WIDTH-1:32] : Reserved [31:8] : Register Offset; This field specifies the target register's byte offset. The offset shall be Dword-aligned (i.e., bits[9:8] shall be 00b). (Refer to Table 15, and Table 16) [7:0] : Opcode; 0x20 (Register Read/Write)
	ARUSER[AR_USER_WIDTH-1:0]	ARUSER [AR_USER_WIDTH-1:3]: Reserved ARUSER [2]: Register Type; This field indicates the accessing Register Type. 0b: Always-On Configuration Registers 1b: MMIO-mapped Registers ARUSER [1:0]: Cmd_packet_type 2b'10: CSR/ADMIN

	ARID[AR_ID_WIDTH-1:0]	Read ID (For all CSR/ADMIN command, using same ID to avoid out of order execution is recommended)
	ARLEN[AR_LEN_WIDTH-1:0]	Read length (if HBF product support burst read/write, this defines length/number of beats in burst command)
Read Response (R Channel)	RRESP[R_RESP_WIDTH-1:0]	Read Response (AXI protocol specific read response, detail read response flash specific will be part of user field)
	RUSER[R_USER_WIDTH-1:0]	RUSER [R_USER_WIDTH-1:6]: ErrorInfo RUSER [5:2]: CMD Status 0x00: Success 0x01: Invalid Register Offset 0x02: Reserved 0x03: Reserved 0x04: HBF Internal Error (vendor specific) 0x05 – 0x0F: product-specific field RUSER [1:0]: RResp_packet_type 2b'10: CSR/ADMIN
	RID[R_ID_WIDTH-1:0]	Read ID
	RDATA[R_DATA_WIDTH-1:0]	Read data
	RLAST	Last Read data transaction

5.2.4.9. Register Write command

The Register Write command (Opcode=0x20) shall be used by the Host to directly access the Always-On Configuration registers and MMIO-mapped registers through UCle mainband.

AXI Channel	Signals	Description
Write Command (AW Channel)	AWADDR[AW_ADDR_WIDTH-1:0]	Cmd_packet_type: CSR/ADMIN [AR_ADDR_WIDTH-1:32] : Reserved [31:8] : Register Offset; This field specifies the target register's byte offset. The offset shall be Dword-aligned (i.e., bits[9:8] shall be 00b). (Refer to Table 15, and Table 16) [7:0] : Opcode; 0x20 (Register Read/Write)

	AWUSER[AW_USER_WIDTH-1:0]	AWUSER [AW_USER_WIDTH-1:3]: Reserved AWUSER [2]: Register Type; This field indicates the accessing Register Type. 0b: Always-On Configuration Registers 1b: MMIO-mapped Registers AWUSER [1:0]: Cmd_packet_type 2b'10: CSR/ADMIN
	AWID[AW_ID_WIDTH-1:0]	Write ID (For all CSR/ADMIN command, using same ID to avoid out of order execution is recommended)
	AWLEN[AW_LEN_WIDTH-1:0]	Write length (if HBF product support burst read/write, this defines length/number of beats in burst command)
Write Data (W Channel)	WDATA[W_DATA_WIDTH-1:0]	Write Data
	WLAST	Last transaction of Write
Write Response (B Channel)	BRESP[B_RESP_WIDTH-1:0]	Write Response (AXI protocol specific write response, detail write response flash specific will be part of user field)
	BUSER[B_USER_WIDTH-1:0]	BUSER [B_USER_WIDTH-1:6]: ErrorInfo BUSER [5:2]: CMD Status 0x00: Success 0x01: Invalid Register Offset 0x02: Reserved 0x03: Reserved 0x04: HBF Internal Error (vendor specific) 0x05 – 0x0F: product-specific field BUSER [1:0]: BResp_packet_type 2b'10: CSR/ADMIN
	BID[B_ID_WIDTH-1:0]	Write ID

5.2.4.10. AXI Config for HBF

This section contains the details of the AXI configuration used to configure AXI master and subordinate port signals.

Table 10: AXI Interface Configuration, UCle (Host-side)

#	Port	Signal Width	Signal width
1	oaxi_irq	1	1
2	iaxi_awid	AW_ID_WIDTH	14
3	iaxi_awaddr	AW_ADDR_WIDTH	36
4	iaxi_awregion	AW_REGION_WIDTH	0
5	iaxi_awlen	AW_LEN_WIDTH	8
6	iaxi_awsz	AW_SIZE_WIDTH	0
7	iaxi_awburst	AW_BURST_WIDTH	0
8	iaxi_awlock	AW_LOCK_WIDTH	0
9	iaxi_awcache	AW_CACHE_WIDTH	0
10	iaxi_awprot	AW_PROT_WIDTH	0
11	iaxi_awqos	AW_QOS_WIDTH	0
12	iaxi_awuser	AW_USER_WIDTH	9
13	iaxi_awvalid	1	1
14	oaxi_awready	1	1
15	iaxi_wid	W_ID_WIDTH	0
16	iaxi_wdata	W_DATA_WIDTH	512
17	iaxi_wstrb	W_STRB_WIDTH	0
18	iaxi_wlast	1	1
19	iaxi_wuser	W_USER_WIDTH	0
20	iaxi_wvalid	1	1
21	oaxi_wready	1	1
22	iaxi_arid	AR_ID_WIDTH	14
23	iaxi_araddr	AR_ADDR_WIDTH	36
24	iaxi_arregion	AR_REGION_WIDTH	0
25	iaxi_arlen	AR_LEN_WIDTH	8
26	iaxi_arsz	AR_SIZE_WIDTH	0
27	iaxi_arburst	AR_BURST_WIDTH	0
28	iaxi_arlock	AR_LOCK_WIDTH	0
29	iaxi_arcache	AR_CACHE_WIDTH	0
30	iaxi_arprot	AR_PROT_WIDTH	0
31	iaxi_arqos	AR_QOS_WIDTH	0
32	iaxi_aruser	AR_USER_WIDTH	9
33	iaxi_arvalid	1	1
34	oaxi_arready	1	1
35	oaxi_bid	B_ID_WIDTH	14
36	oaxi_bresp	B_RESP_WIDTH	3
37	oaxi_buser	B_USER_WIDTH	9

#	Port	Signal Width	Signal width
38	oaxi_bvalid	1	1
39	iaxi_bready	1	1
40	oaxi_rid	R_ID_WIDTH	14
41	oaxi_rdata	R_DATA_WIDTH	512
42	oaxi_rresp	R_RESP_WIDTH	3
43	oaxi_rlast	1	1
44	oaxi_ruser	R_USER_WIDTH	9
45	oaxi_rvalid	1	1
46	iaxi_rready	1	1

Table 11: AXI Interface Configuration, HBF Device-side

#	Port	Signal Width	Signal width
1	oaxi_irq	1	1
2	oaxi_awid	AW_ID_WIDTH	14
3	oaxi_awaddr	AW_ADDR_WIDTH	36
4	oaxi_awregion	AW_REGION_WIDTH	0
5	oaxi_awlen	AW_LEN_WIDTH	8
6	oaxi_awsz	AW_SIZE_WIDTH	0
7	oaxi_awburst	AW_BURST_WIDTH	0
8	oaxi_awlock	AW_LOCK_WIDTH	0
9	oaxi_awcache	AW_CACHE_WIDTH	0
10	oaxi_awprot	AW_PROT_WIDTH	0
11	oaxi_awqos	AW_QOS_WIDTH	0
12	oaxi_awuser	AW_USER_WIDTH	9
13	oaxi_awvalid	1	1
14	iaxi_awready	1	1
15	oaxi_wid	W_ID_WIDTH	0
16	oaxi_wdata	W_DATA_WIDTH	512
17	oaxi_wstrb	W_STRB_WIDTH	0
18	oaxi_wlast	1	1
19	oaxi_wuser	W_USER_WIDTH	0
20	oaxi_wvalid	1	1
21	iaxi_wready	1	1
22	oaxi_arid	AR_ID_WIDTH	14
23	oaxi_araddr	AR_ADDR_WIDTH	36
24	oaxi_arregion	AR_REGION_WIDTH	0
25	oaxi_arlen	AR_LEN_WIDTH	8

#	Port	Signal Width	Signal width
26	oaxi_arsize	AR_SIZE_WIDTH	0
27	oaxi_arburst	AR_BURST_WIDTH	0
28	oaxi_arlock	AR_LOCK_WIDTH	0
29	oaxi_arcache	AR_CACHE_WIDTH	0
30	oaxi_arprot	AR_PROT_WIDTH	0
31	oaxi_arqos	AR_QOS_WIDTH	0
32	oaxi_aruser	AR_USER_WIDTH	9
33	oaxi_arvalid	1	1
34	iaxi_arready	1	1
35	iaxi_bid	B_ID_WIDTH	14
36	iaxi_bresp	B_RESP_WIDTH	3
37	iaxi_buser	B_USER_WIDTH	9
38	iaxi_bvalid	1	1
39	oaxi_bready	1	1
40	iaxi_rid	R_ID_WIDTH	14
41	iaxi_rdata	R_DATA_WIDTH	512
42	iaxi_rresp	R_RESP_WIDTH	3
43	iaxi_rlast	1	1
44	iaxi_ruser	R_USER_WIDTH	9
45	iaxi_rvalid	1	1
46	oaxi_rready	1	1

Note: Width-0 signifies that the following signal is not present for the AXI interface.

5.3. Host Read

5.3.1. Host Read Requirements

The host will communicate all read requests in AXI format as defined in the AXI Protocol for HBF section 5.2.4.

1. All Read requests are 64-byte granularity; however, the host shall send 64 read commands to retrieve the complete 4KiB data with different AXI IDs. Or if product support burst read then send burst read with burst length (0-63).
2. AXI ID width is fixed, however if xPU would like to use lesser number of AXI-ID width , MSB will remain reserved, where the host can define it based on the maximum number of outstanding 64B transactions the host would like to manage.
3. The host can choose to allocate either the least significant or the most significant 6 bits to manage the same 4KiB transaction in AXI IDs.
4. The read request address is at 64-byte granularity.

5. Read command completion may be out of order for different AXI IDs, while read order is maintained with the same AXI IDs.
 - a. Cache hit reads are served immediately if there are no ordering violations.
 - b. For cache miss reads, Core die sense is triggered when prior read for that specific bank is done.
6. HBF follows strict ordering for sense requests to Core die within the same bank requests.
7. HBF supports two cache buffers for each bank to hold up to at least two pages of data. It is recommended to read the full first 4KiB page before issuing a third 4KiB address that will evict the first 4KiB data.
8. The Base die shall immediately serve the cache hit read from the cache buffer if ordering is not violated.
9. In the case of read failure like UECC or high correctable errors, the host configures the Base die to retry, or the host can retry the read.

5.3.2. Regular Read and Batch Reads

In the case of read failure like UECC or high correctable errors, the host can configure the Base die to retry, or host can retry the read.

5.3.2.1. Regular Reads

The reads that are issued from the host without the batch read hint are called regular reads. The regular reads are accumulated together and scheduled to the Core die optimally for best performance.

5.3.2.2. Batch Reads

Batch reads issued by the host will indicate that the read is batch read by setting AXI user bit 0 to 1 in the command fields. The Base die will not wait for the timer to expire and will schedule all the reads that were received until a regular read command is issued. The Base die will assume the end of batch reads whenever a regular read after batch reads.

R R R R R R R – No batch read hint set in any of the read commands

R R R R R (Batch Read bit set) R R R R – The first five read commands are scheduled together and will not wait for the read commands behind it.

5.3.3. Scratchpad Reads

Scratchpad reads are reads sent by the host to the scratchpad memory region in the HBF. The host can use the scratchpad IO command to read temporarily stored data from scratchpad memory if the HBF product supports scratchpad. Scratchpad reads are high-priority operations, and a response from HBF is expected within a few cycles, similar to that of the host cache memory. HBF is unaware of the contents and validity of scratchpad

memory, and the host software is responsible for managing the scratchpad region. All reads to the scratchpad region are regular reads and the batch read hint is ignored. Scratchpad support by HBF is optional and is product specific.

5.4. Host Write

5.4.1. Host Write Requirement

1. All writes are non-posted writes. A write command received with any length receives only one response sent to the host irrespective of length size, which follows AXI protocol.
2. The write command is completed to the host only after the data is written to the Core die.
3. The write granularity to the Core die is 4KiB. The Base die controller first accumulates the complete 4KiB write commands and data is sent to the Core die.
4. No skipping or jumping of 4KiB local address is allowed within a host channel's allocated global address range.
5. HBF will internally auto-erase the NAND block from the host point of view, when receiving a page-0 (WL-0, STR-0, NAND block-X) write request of a NAND block.
6. When HBF gets the first 64Byte write request for a local address for that channel, HBF shall receive all the 4KiB writes within a time limit. The time limit is set by the host in the HBF config register during initialization. If the HBF does not receive all the 4KiB within the time limit, the HBF will error out and send the error response to the host.
7. Maximum of product specific outstanding device logical unit write requests per host channel. For example, it is 64 or 128 or any other number of write requests supported by the product.
8. Write ID width is defined by the number of outstanding writes as defined in the product specification.
9. The host shall write the 4KiB local address consecutively within the NAND block. The NAND block size is defined in the product specification.

5.4.2. Scratchpad Write

The host can use the scratchpad write IO command to temporarily store host content in the scratchpad region, if HBF supports the scratchpad feature. During the scratchpad write, IO command writes the data to the scratchpad address. Scratchpad write IO is a high-priority command, and HBF response completion occurs within a few cycles. The content of the scratchpad remains valid until the HBF power cycle, and the host can read it back at any time at high priority. The scratchpad write does not have to be 4KiB page size for the SRAM write to take place. All scratchpad writes have 64B granularity and are written to the SRAM with much less latency. HBF is unaware of the contents and validity of scratchpad memory, and the host software is responsible for managing them. The scratchpad functionality is optional and is product specific.

5.4.3. Mixed Read/Write Workload

During mixed workloads, the Base die maintains the order between writes and reads at 64B granularity with the same address even with different AXI IDs. If a read follows a write and only some parts of 4KiB-write have been accumulated in the Base die and is still waiting for the remaining of the 4KiB, the read that hits the accumulated part returns the data from the accumulated part. If the read hits the pending part of the write, a read error is returned to the host with read command status error code 0xA.

5.5. Read Error Types

High-level read error description and the actions taken by the Base die controller.

Table 12: Read Error Type to Host

Error-Type	Description	Base Die Action	Command Status ruser[5:2]	User Info ruser[8:6]
Invalid Address	Address > UCle Channel Capacity or Logical Address Already marked bad.	Complete CMD with Error, Read Data=Invalid	0x1	0
Temporarily restricted read (Product specific)	Address belongs to a temporarily restricted address	Complete CMD with Error, Read Data=Invalid	0x2	Vendor-specific info
Invalid user field in command	User field not defined	Complete CMD with Error, Read Data=Invalid	0x3	0
UECC (Block Data Refresh required)	ECC Engine reported UECC and data refresh required	Complete CMD with Error, Read Data=Invalid	0x4	Additional Info 0x1: Read Count > Max_Read_Threshold 0x2: Block Data Health (Read Count <= Max_Read_Threshold) 0x3-0x7: Reserved
CECC (Block Data Refresh required)	ECC Engine reported a high correctable error. It is recommended to refresh NAND block data.	Complete CMD with Warning, Read Data = Valid	0x5	Additional Info 0x1: Read Count > Max_Read_Threshold

Error-Type	Description	Base Die Action	Command Status ruser[5:2]	User Info ruser[8:6]
				0x2: Block Data Health (Read Count <= Max_Read_Threshold) 0x3-0x7: Reserved
UECC - Read Retry with Copying userinfo[8:6] in next retry-read command	ECC Engine reported UECC, and Host Need to retry the same address read so Base die could perform the next step of read-retry flow.	Complete CMD with retry-required, Read Data=Invalid set ruser[8:6]	0x6	Retry Stage Info
Read from ErasePage	ECC Engine Detected All 1 data on this page.	Complete CMD with Error, Read Data= Invalid	0x7	0
Mark UCle channel Local Capacity-unusable	Base die detects that the following NAND block/Bank/Die/Channel is unrecoverable and need to be retired.	Complete CMD with Error, Read Data= Invalid Followed command on these logical addresses will be completed with the error type -0x01	0x8	Retired UCle Channel Capacity Granularity 0x1: NAND block 0x2: Bank 0x3: Die 0x4: Channel
Die temporary block for recovery	Base die mark following Die temporarily blocked to perform recovery flow	Complete CMD with Error, Read Data= Invalid	0x9	0x1: Severe Throttling Error 0x2-0x7: Reserved
Read from a pending Write	Partial write data from host is received and Base die is waiting for the remaining part of the write	Complete Read with Error	0xA	0
Reserved	Reserved	Reserved	0xB-0xC	Reserved
Vendor-Specific Errors	Vendor-defined product specific read errors	Vendor will define for each error behavior	0xD-0xF	Vendor-specific

5.6. Write Error Types

High-level write error description and the action taken by the Base die controller.

Table 13: Write Error Type to Host

Error Type	Description	Base die Action	Command Status buser[5:2]	User Info buser[8:6]
Invalid Address	Address > UCle Channel Capacity or Logical Address Already marked bad.	Complete CMD with Error	0x1	0
Overlap Address	Already received write request for this address and pending because complete write unit 4KiB data not received	Complete this CMD with Error	0x2	0
Invalid user field in command	User field not defined	Complete this CMD with Error	0x3	0
Maximum pending device logical unit (4KiB) limit reached	UCle channel already processing Maximum allow device logical unit. Host software needs to wait to get completion of pending writes. If any data is pending for prior-issued device logical unit, prioritize it.	Complete this CMD with Error	0x4	0
Complete Device Logical Unit (4KiB) data accumulation timeout, retry	UCle channel did not receive complete device logical unit (4KiB) data with-in time-limit	Complete all CMD with error associated with this device logical unit.	0x5	0
Write order violation	Host software didn't follow programming order	Complete all CMD with error associated with this device logical unit.	0x6	Additional Info 0x1: No skipping or jumping between page allow 0x2: Next page couldn't program as program failure on prior page

Error Type	Description	Base die Action	Command Status buser[5:2]	User Info buser[8:6]
Program fail, replay NAND block data	Core die encountered program failure on this page and could not continue programming this NAND block. Replay this NAND block data from page-0.	Complete All CMD with Error associated with this device logical unit. Ensure all pending/follow up pages for given bank-block also error out with status 0x06 to ensure host start same NAND block write with page-0 (WL-0 Str-0) For page-0 local address calculation please review the address calculation section.	0x7	0
Mark UCle channel local capacity unusable	Base die detects that the following NAND block/Bank/Die/Channel is unrecoverable or All spare NAND blocks used for this bank need to be retired.	Complete all CMD with error associated with this device logical unit. Followed command on these logical addresses will be completed with the error type -0x01	0x8	Retired UCle channel capacity granularity 0x1: NAND block 0x2: Bank 0x3: Die 0x4: Channel
Die temporary blocked for recovery	Base die mark following die temporarily blocked to perform recovery flow	Complete all CMD with error associated with this device logical unit.	0x9	0x1: Severe Throttling Error 0x2-0x7: Reserved
Reserved	Reserved	Reserved	0xA-0xC	Reserved
Vendor-Specific Errors	Vendor-defined product specific read errors	Vendor will define for each error behavior	0xD-0xF	Vendor-specific

5.7. UCle Local Channel NAND block Address Calculation

In the case of a program failure and NAND block data refresh, host software shall calculate the page-0 address of the failed logical address and replay NAND block data write to refresh the data. The host shall read the following information from the device config register at the time of initialization.

R1 = Number of dies

R2 = Number of banks in die

R3 = Number of pages in bank NAND block

R4 = Number of 64 Bytes in a page.

R5 = Data stripe width in device logical units

A1 = UCle channel local address of failed command

Device logical unit (L1) = $\text{QUOTIENT}(A1, R4)$

Block Number (B1) = $\text{QUOTIENT}(L1, R1 \cdot R2 \cdot R3)$

Fail Page Number in NAND block (P1) = $\text{Remainder}(L1, R1 \cdot R2 \cdot R3)$

Bank Number (Bank_Num) = $\text{Remainder}(\text{Remainder}(L1, R1 \cdot R2 \cdot R3), R1 \cdot R2)$

Start device logical unit (L2) = $B1 \cdot R1 \cdot R2 \cdot R3 + \text{Bank_Num}$

UCle Local Channel Address = $L2 \cdot R4, L2 \cdot R4 + 1, \dots, L2 \cdot R4 + R4 - 1$

To complete program failure or NAND block data refresh, the host needs to send writes for device logical units L2, L2 + R5, L2 + 2*R5, ..., L2 + (R3-1)*R5.

It may be possible that the Base die reports failure for multiple reads or writes of same NAND block, so if for a given NAND block data refresh is in progress, host should ignore multiple requests on same NAND block unless HBF completes a refresh write command with an error.

5.8. Sideband Message Support

The purpose of the sideband link is to provide an out-of-band channel for link training and an interface for accessing the sideband registers of the link partner. It is also used for Link Management packets and parameter exchanges with a remote link partner.

The same protocol is also used for local die sideband accesses over FDI and RDI. When relevant, FDI-specific rules are pointed out using “FDI sideband:”. When relevant, RDI-specific rules are pointed out using “RDI sideband:”. When relevant, UCle link-specific rules are communicated using the UCle link sideband. If no prefix is mentioned, it is a common rule across FDI, RDI, and UCle link.

The physical layer is responsible for framing and transporting sideband packets over the UCle link. Direct sideband access to remote die can originate from the adapter or the physical layer. The adapter forwards remote die sideband access over RDI to the physical layer for framing and transport. These include register access requests, completions, or messages.

The protocol layer has indirect access to remote die registers using the sideband mailbox mechanism. The mailbox registers reside in the adapter, and it is the adapter's responsibility to initiate remote die register access requests when it receives the corresponding access trigger for the mailbox register over FDI

Sideband Packet Types: Three different categories of packets are permitted:

- **Register Accesses:** Configuration (CFG) or Memory-Mapped access for both reads and writes are supported. These are 32-bit or 64-bit in length.
- **Messages without data:** These are Link Management (LM), or Vendor-defined packets. These don't carry additional data payloads.
- **Messages with data:** These are Parameter Exchange (PE), link training-related, or vendor-defined messages. All three carry 64b of data.

Every packet carries a 5-bit opcode, 3-bit source identifier (srcid), and a 3-bit destination identifier (dstid). The 5-bit opcode indicates the packet type, as well as whether the packet carries 32 bits of data or 64 bits of data. For details of each opcode request and completion format, please refer to the UCle specification documents.

Table 14: Opcode Encoding Mapping for Packet Type

Opcode Encoding	Sideband Packet Type
00000b	32b Memory Read
00001b	32b Memory Write
00010b	32b DMS Register Read
00011b	32b DMS Register Write
00100b	32b Configuration Read
00101b	32b Configuration Write
01000b	64b Memory Read
01001b	64b Memory Write
01010b	64b DMS Register Read
01011b	64b DMS Register Write
01100b	64b Configuration Read
01101b	64b Configuration Write
10000b	Completion without Data
10001b	Completion with 32b Data
10010b	Message without Data
10011b	HBF Vendor Specific
10111b	Management Port Messages without Data
11000b	Management Port Message with Data
11001b	Completion with 64b Data
11011b	Message with 64b Data
Other encodings	Reserved

5.9. Device Configuration and Protocol Register

HBF defines all device configurations and protocol registers in the MMIO-mapped config space and register space that are shadowed in the adapter, where host can access it. Each UCle channel has its own always-on register and MMIO register sets.

5.9.1. Protocol-Specific Always-On Configuration Registers

All device capabilities using the configuration read/write register sideband packet type with selection addr[23:0] in request format as

- [11:0] = Byte offset (implementation specific)
- [19:12] = Reserved
- [23:20] = 0xF (access protocol-specific configuration registers)

Table 15: Protocol Specific Always-On Config Registers

Start	End	Size (Bytes)	Symbol	Name
0x0000	0x0003	4	ID	Identifiers
0x0004	0x0007	4	STS	Device status
0x0008	0x000B	4	RID	Revision ID
0x000C	0x000F	4	RSVD	Reserved
0x0010	0x0017	8	MADDR	MMIO MAP register address
0x0018	0x001F	8	PWR	UCle channel power supply
0x0058	0x0097	64	UCONFIG	UCle channel configuration, product-specific
0x0098	0x0197	256	CAP	Device capabilities
0x0198	0x01FF	104	RSVD	Reserved
0x0200	0x0203	4	VPR	Voltage Pointer register
0x0204	0x03FF	508	VCONFIG	Vendor-specific configuration

5.9.1.1. ID

Bits	Type	Reset	Description
31:16	RO	Implementation specific	Device ID Register: Device ID register is Hwlnit and the value in this register. Device ID shall be allocated by Vendor. The Device ID, in conjunction with the Vendor ID and Revision ID, are used as one mechanism for software to determine which driver should be loaded
15:00	RO	Implementation specific	Vendor ID Register: the Vendor ID register is Hwlnit and the value in this register identifies the manufacturer. Value 0xFFFF indicate invalid.

5.9.1.2. Device Status

Bits	Type	Reset	Description
31:08	RO	0x0	Reserved
07	RO	Implementation specific	Scratchpad Ready (SPR) : when this is set indicate device is ready to accept scratchpad commands
06	RO	Implementation specific	CSR Space Ready (CSR) : when this is set indicate device CSR space ready to accept host commands
05	RO	Implementation specific	IO Space Ready (ISR) : when this is set indicate device is ready to accept IO commands
04	RO	Implementation specific	Capability list : indicate presence of extended capability list item.
03:01	RO	0x0	Reserved
00	RO	Implementation specific	Immediate Readiness : when this is set, indicates the device is guaranteed to be ready for successful completion of a valid configuration request at any time.

5.9.1.3. Revision ID

Bits	Type	Reset	Description
31:00	RO	Implementation specific	Revision ID Register : The Revision ID Register is HwInit and the value in this register specifies a Function specific revision identifier. The value is chosen by the vendor. Zero is an acceptable value while 0xFFFFFFFF is invalid.

5.9.1.4. MMIO MAP Register Address (MADDR)

Bits	Type	Reset	Description
31:00	RW	Implementation specific	MMIO MAP Register Address (MADDR) : Host can program this offset if host willing to map Multiple device MMIO space continuous and Device will offset MMIO space base on this register value.

5.9.1.5. UCle Channel Power Capability (PCAP)

Bits	Type	Reset	Description
63:32	RO	0xFFFFFFFF	Extension offset (0xFFFFFFFF is invalid or no additional info)
31:16	RW	TBD	Power management control/status (PMCSR)
15:00	RW	TBD	Power management capability (PMC)

5.9.1.6. UCle Channel Configuration (UCONFIG)

Bits	Type	Reset	Description
31:16	RO	0x0	Reserved
15:08	RO	Implementation specific	Maximum Supported UCle Speed 0x0: 4 GT/s 0x1: 8 GT/s 0x2: 16 GT/s 0x3: 24 GT/s 0x4: 32 GT/s 0x5: 40 GT/s 0x6: 64 GT/s 0x7: Reserved
07:02	RO	0x0	Reserved
01:00	RO	Implementation specific	Number of AXI Port 00b: 1 AXI Port 01b: 2 AXI Port 10b: 4 AXI Port 11b: Reserved

5.9.1.7. Device Capabilities (CAP)

Bits	Type	Reset	Description
31:24	RO	0x0	Reserved
23:20	RO	Implementation specific	Slot Power Limit
19:16	RO	Implementation specific	Reset Capabilities
15:14	RO	Implementation specific	Maximum ScratchPad Read Transaction Size 00b: 64 Bytes 01b: 256 Bytes 10b: 512 Bytes 11b: 4096 Bytes
13:12	RO	Implementation specific	Maximum ScratchPad Write Transaction Size 00b: 64 Bytes 01b: 256 Bytes 10b: 512 Bytes 11b: 4096 Bytes
11:10	RO	Implementation specific	Minimum ScratchPad Read Transaction Size 00b: 64 Bytes 01b: 256 Bytes 10b: 512 Bytes 11b: 4096 Bytes
09:08	RO	Implementation specific	Minimum ScratchPad Write Transaction Size 00b: 64 Bytes 01b: 256 Bytes

			10b: 512 Bytes 11b: 4096 Bytes
07:06	RO	Implementation specific	Maximum Read IO Transaction Size 00b: 64 Bytes 01b: 256 Bytes 10b: 512 Bytes 11b: 4096 Bytes
05:04	RO	Implementation specific	Maximum Write IO Transaction Size 00b: 64 Bytes 01b: 256 Bytes 10b: 512 Bytes 11b: 4096 Bytes
03:02	RO	Implementation specific	Minimum Read IO Transaction Size 00b: 64 Bytes 01b: 256 Bytes 10b: 512 Bytes 11b: 4096 Bytes
01:00	RO	Implementation specific	Minimum Write IO Transaction Size 00b: 64 Bytes 01b: 256 Bytes 10b: 512 Bytes 11b: 4096 Bytes

5.9.1.8. Voltage Pointer register (VPR)

Bits	Type	Reset	Description
31:16	RO	Implementation specific	Offset pointer for voltage (VPR)
15:00	RO	Implementation specific	Number of voltage's requirement

5.9.2. Protocol-Specific MMIO Registers

All device protocol MMIO-mapped registers to read/configure the device status using the register read/write sideband packet type with selecting addr[23:0] in request format as

- [19:0] = Byte offset (implementation-specific)
- [23:20] = 0xF (access protocol-specific configuration registers)

Table 16: Protocol Specific MMIO Map Registers

Start	End	Size (Bytes)	Host R/W	Symbol	Name
0x0000	0x0007	8	R	BUCCAP	Base die UCle Channel

					Capabilities
0x0008	0x000B	4	R	VS	Protocol version number
0x000C	0x000F	4	R/W	BUCC	Base die UCle Channel Configuration
0x0010	0x0013	4	R	BUCSTS	Base die UCle Channel status
0x0014	0x0017	4	R	BUCCR	Base die UCle Channel reset
0x0018	0x001B	4	R	TMON	Temperature Monitoring Register
0x001C	0x0023	8	R	SMEMSA	Scratchpad Memory Start Address
0x0024	0x0033	16	R	SMEMSZ	Scratchpad Memory Size
0x0034	0x003B	8	R	FIOSA	Flash IO start address
0x003C	0x004B	16	R	FIOSZ	Flash IO Size
0x004C	0x0053	8	R	SMSZ	Smart log size (in Bytes)
0x0054	0x005B	8	R	VSTSZ	Vendor-specific telemetry size
0x005C	0x0063	8	R	ESZ	Error log entry size
0x0064	0x0067	4	R	ENC	Maximum error log count
0x0068	0x00FF	152	R	RSVD	Reserved
0x0100	0x013F	64	R/W	TMR	Product-specific timers
0x0140	0x0143	4	R	LOCKOUT	IEEE 1500 lockout
0x0144	0x0147	4	R	MAXPEC	Indicate the NAND block's Maximum PEC
0x0148	0x014B	4	R	AVGPEC	Indicate the channel-average PEC from all NAND blocks
0x014C	0x014F	4	R	REDCAP	Reduced capacity information bit 0: If set to 1, it indicates reduced capacity. If set to 0, it indicates full capacity available
0x0150	0x0153	4	R/W	TTTEMP	Thermal Throttling TEMP
0x0154	0x02FF	428	R	RSVD	Reserved

All the always-on config registers and MMIO registers are accessed by both the UCle mainband and sideband.

5.9.2.1. Offset 0x00: BUCCAP – Base Die UCle Channel Capabilities

This register indicates basic capabilities of the Base Die UCle Channel to a Host.

Bits	Type	Reset	Description
63:44	RO	0x0	Reserved
43:40	RO	Implementation specific	Number of Core Dies per UCle Channel (NCDU): The number of Core Dies per UCle Channel. The value is expressed as a power of 2: 0x00: 1 Core Die (20)

			0x01: 2 Core Die (21) 0x02: 4 Core Die (22) 0x03: 8 Core Die (23) 0x04: 16 Core Die (24) 0x05 – 0x0F: Reserved
39:36	RO	0x01	Number of Cache Buffers per Bank (NCBB): Indicates the number of cache buffers per NAND Bank in the Base die. Each Cache Buffer Size is same as NAND Page Size(4KiB). 0x00: Reserved 0x01: 2 Cache Buffers per Bank (default) 0x02 – 0x0F: Reserved
35:32	RO	Implementation specific	Maximum Outstanding Commands Supported (MOCS): Indicates the maximum number of outstanding commands supported by the Base Die command Queue. The value is expressed as a power of 2: (0-based, except 0x00) 0x00: 1 outstanding command 0x07: 256 outstanding commands (27+1) 0x08: 512 outstanding commands (28+1) 0x09: 1,024 outstanding commands (29+1) 0x0A: 2,048 outstanding commands (210+1) 0x0B: 4,096 outstanding commands (211+1) 0x0C: 8,192 outstanding commands (212+1) 0x0D: 16,384 outstanding commands (213+1) - All other values are reserved.
31:24	RO	0x0	Timeout (TO): This is the worst-case time that a host should wait for the BUCSTS.RDY bit to transition from: '0' to '1'; or '1' to '0'. This field is in 500 millisecond units. 0x00: Reserved. 0x01 – 0xFF: Represents 1 to 255 units (e.g., 0x0A indicates a timeout limit of 5.0 seconds)
23:16	RO	Implementation specific	Supported Speed Grades (SSG): Each bit location, when set to '1', indicates the UCle transfer speeds supported by the physical interface layer of this channel. (refer to Table 4) - Bit [16]: UCle-A Speed Grade 1 (8 GT/s) supported - Bit [17]: UCle-A Speed Grade 2 (16 GT/s) supported - Bit [18]: UCle-A Speed Grade 3 (32 GT/s) supported - Bit [19-23]: Reserved
15:10	RO	0x0	Reserved
09:08	RO	Implementation specific	Burst Command Supported (BCS): 00b: Both Read and Write do not support Burst commands. 01b: Only Write supports Burst command. 10b: Only Read supports Burst command. 11b: Both Read and Write support the Burst command.

07:06	RO	Implementation specific	Wear Leveling Supported (WLS): 00b: Reserved 01b: Only Base Die-driven automatic wear leveling is supported. 10b: Only Host-controlled wear leveling is supported. 11b: Both Base Die-driven and Host-controlled wear leveling are supported.
05	RO	Implementation specific	Security Supported (SECS): 0: Security functions are not supported. 1: Security functions (Power-on Erase, Host Key and Random Seed Authentication, etc) are supported.
04	RO	Implementation specific	Scratchpad Memory Supported (SMEMS): 0: Scratchpad SRAM is not supported. 1: Scratchpad SRAM is supported. Its address range is defined in SMEMSA and BCAP.SMEMSZ.
03:00	RO	Implementation specific	Capability Version (VER): Indicates the version of the capability structure.

5.9.2.2. Offset 0x08: VS – Version

This register indicates the version of this specification that the Base die supports.

Bits	Type	Reset	Description
31:16	RO	Implementation specific	Major Version Number (MJR): An integer value indicating the major version number of this specification which is supported by the Base Die.
15:00	RO	Implementation specific	Minor Version Number (MNR): An integer value indicating the minor version number of this specification which is supported by the Base Die.

5.9.2.3. Offset 0x0C: BUCC – Base Die UCle Channel Configuration

This register modifies settings for the Base Die UCle Channel.

Bits	Type	Reset	Description
31:28	RO	Implementation specific	Reserved
27:24	RW	Implementation specific	Maximum Outstanding Commands Selected (MOCS): Indicates the maximum number of outstanding commands selected by a Host. The value is expressed as a power of 2: (0-based, except 0x00) 0x00: 1 outstanding command 0x07: 256 outstanding commands (27+1) 0x08: 512 outstanding commands (28+1) 0x09: 1,024 outstanding commands (29+1) 0x0A: 2,048 outstanding commands (210+1) 0x0B: 4,096 outstanding commands (211+1) 0x0C: 8,192 outstanding commands (212+1)

			0x0D: 16,384 outstanding commands (213+1) - All other values are reserved.
23:16	RW	Implementation specific	Selected Speed Grade (SSG): - Bit [16]: This bit is set to '1' means, UCle-A Speed Grade 1 (8 GT/s) is selected. - Bit [17]: This bit is set to '1' means, UCle-A Speed Grade 2 (16 GT/s) is selected. - Bit [18]: This bit is set to '1' means, UCle-A Speed Grade 3 (32 GT/s) is selected. (32 GT/s) selected. - All other values are reserved.
15:08	RO	Implementation specific	Reserved
07:06	RW	Implementation specific	Wear Leveling Selected (WLS): 00b: Reserved 01b: Base Die-driven automatic wear leveling is selected. 10b: Host-controlled wear leveling is selected. 11b: Reserved
03:01	RO	0x0	Reserved
00	RW	0x0	Enable (EN): This bit is set to '1' when the Base Die UCle Channel shall accept IO commands. This bit is set to '0' when Base Die UCle Channel shall not accept IO commands.

5.9.2.4. Offset 0x10: BUCSTS – Base Die UCle Channel Status

Bits	Type	Reset	Description
31:04	RO	0x0	Reserved
03:02	RO	0x0	Shutdown Status (SHST): Indicates the status of shutdown processing. 0x00: Normal operation (no shutdown) 0x01: Shutdown in processing 0x02: Shutdown processing complete 0x03: Reserved
01	RO	Implementation specific	Base Die UCle Channel Fatal Status (BUCFS): This bit is set to '1' when a fatal Base die UCle channel error occurred that could not be communicated. This bit is cleared to '0' when a fatal Base die UCle channel error has not occurred. The reset value of this bit is set to '1' when a fatal Base die UCle channel error is detected during Base die initialization.
00	RO	0x0	Ready (RDY): This bit is set to '1' when the Base Die UCle Channel is ready to accept the commands through mainband AXI Channel. This bit shall be cleared to '0' once the Base Die UCle Channel is not ready to accept the commands.

5.9.2.5. Offset 0x14: BUCR – Base Die UCle Channel Reset

Bits	Type	Reset	Description
31:08	RW	0x0	Base Die UCle Channel Reset Control (BUCRC): A write of the value 0x484246 (“HBF”) to this field initiates a Base Die UCle Channel Reset. A write of any other value has no functional effect on the operation of the Base Die UCle Channel. This field shall return the value 0x0 when read.
07:02	RO	0x0	Reserved
01:00	RW	0x0	UCle Channel Reset Type (UCRST): 00b: Only UCle PHY + D2D Adapter reset 01b: UCle PHY + D2D Adapter + AOU reset 10b: Channel Warm Reset 11b: Reserved

5.9.2.6. Offset 0x18: TMON – Temperature Monitoring

Bits	Type	Reset	Description
31:02	RO	0x0	Reserved
01	RW	0x0	Light Throttling Mode Status (LTMS): If this bit is set to ‘1’, the device is in Light Throttling Mode. When the device goes back to Normal mode, this bit will be cleared to ‘0’.
00	RW	0x0	CATTRIP Event Status (CES): Indicates the HBF Catastrophic Temperature Report, which is following the CATTRIP signal, refer to section 4.5.2 and 9.2, CATTRIP pin drives a static HIGH. Note that this is sticky and not cleared by a HBF_RESET.

5.9.2.7. Offset 0x1C: SMEMSA – Scratchpad Memory Start Address

Bits	Type	Reset	Description
63:06	RO	0x0	Scratchpad Memory Start Address (SMEMSA): This field specifies the 58 most significant bits of the 64-bit start address for the Scratchpad Memory address range.
05:00	RO	0x0	Reserved

5.9.2.8. Offset 0x24: SMEMSZ – Scratchpad Memory Size

Scratchpad Memory Address is a contiguous range. If BUCC.SNAU=0x00 (1-AXI Interface), SMEMSZA2~SMEMSZA4 Fields are reserved.

Bits	Type	Reset	Description
127:96	RO	0x0	Scratchpad Memory Size for 4th AXI Channel (SMEMSZA4):

			Indicates the size of the 4th AXI Channel's scratchpad memory range in multiples of 64B.
95:64	RO	0x0	Scratchpad Memory Size for 3rd AXI Channel (SMEMSZA3): Indicates the size of the 3rd AXI Channel's scratchpad memory range in multiples of 64B.
63:32	RO	0x0	Scratchpad Memory Size for 2nd AXI Channel (SMEMSZA2): Indicates the size of the 2nd AXI Channel's scratchpad memory range in multiples of 64B.
31:00	RO	0x0	Scratchpad Memory Size for 1st AXI Channel (SMEMSZA1): Indicates the size of the 1st AXI Channel's scratchpad memory range in multiples of 64B.

5.9.2.9. Offset 0x34: FIOSA – Flash IO Start Address

Bits	Type	Reset	Description
63:12	RO	Implementation specific	Flash IO Start Address (FIOSA): This field specifies the 52 most significant bits of the 64-bit start address for the Flash IO address range. This address shall be 4KiB NAND Page size aligned
11:00	RO	0x0	Reserved (NAND Page Size(4KiB) Aligned)

5.9.2.10. Offset 0x3C: FIOSZ – Flash IO Size

Flash IO Address is a contiguous range. If BUCC.SNAU is set to 0x00 (1-AXI Interface), FIOSZA2~FIOSZA4 Fields are reserved.

Bits	Type	Reset	Description
127:96	RO	0x0	Flash IO Size for 4th AXI Channel (FIOSZA4): Indicates the size of the 4th AXI Channel's flash io range in multiples of 4KiB.
95:64	RO	0x0	Flash IO Size for 3rd AXI Channel (FIOSZA3): Indicates the size of the 3rd AXI Channel's flash io range in multiples of 4KiB.
63:32	RO	0x0	Flash IO Size for 2nd AXI Channel (FIOSZA2): Indicates the size of the 2nd AXI Channel's flash io range in multiples of 4KiB.
31:00	RO	0x0	Flash IO Size for 1st AXI Channel (FIOSZA1): Indicates the size of the 1st AXI Channel's flash io range in multiples of 4KiB.

5.9.2.11. Offset 0x4C: SMSZ – Smart Log Size (in Bytes)

Bits	Type	Reset	Description
63:32	RO	Implementation specific	Smart log Size in Byte for Each AXI Port
31:16	RO	0x0	Reserved
15:00	RO	Implementation specific	Smart log Revision Format

5.9.2.12. Offset 0x54: VSTSZ – Vendor Specific Telemetry Info

Bits	Type	Reset	Description
63:40	RO	Implementation specific	Telemetry snapshot size in Bytes
39:32	RO	Implementation specific	Number of Telemetry snapshot retained
31:16	RW	Implementation specific	Collection Interval value
17:16	RW	Implementation specific	Collection Interval for telemetry 00b: in seconds 01b: in minutes 10b: in hours 11b: reserved
15:00	RO	Implementation specific	Telemetry log Revision Format

5.9.2.13. Offset 0x5C: ESZ – Error Log Entry Size

Bits	Type	Reset	Description
63:48	RO	Implementation specific	UCle Error Log Size
47:32	RO	Implementation specific	HBF Error Log Size
31:16	RO	Implementation specific	UCle Error log Revision
15:00	RO	Implementation specific	HBF Error log Revision

5.9.2.14. Offset 0x64: ENC – Maximum Error Log Count

Bits	Type	Reset	Description
63:32	RO	0x0	Reserved
31:16	RO	Implementation specific	Maximum UCle Error log Count
15:00	RO	Implementation specific	Maximum HBF Error log Count

5.9.2.15. Offset 0x100: TMR – Product-specific Timers

Following is each timer format, number of timers required will be product specific. Total supported timer 64.

Bits	Type	Reset	Description
31:08	RW	Implementation specific	Timer Value

07:00	RW	Implementation specific	Timer resolution 0x0: Nano Sec. 0x1: Micro Sec. 0x2: Milli Sec. 0x3: Sec. 0x4: Minute 0x5: Hours 0x6-0x7: Reserve
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5.9.2.16. Offset 0x140: LOCKOUT – IEEE 1500 Lockout

Bits	Type	Reset	Description
31:00	RO	Implementation specific	IEEE 1500 Lockout

5.9.2.17. Offset 0x144: MAXPEC – Maximum Program/Erase Cycle

This register indicates the NAND block's Maximum PEC value.

Bits	Type	Reset	Description
31:00	RO	Implementation specific	Current Maximum PE value of the NAND block for that AXI port capacity

5.9.2.18. Offset 0x148: AVGPEC – Average Program/Erase Cycle

This register indicates the channel average PEC value from all NAND blocks

Bits	Type	Reset	Description
31:00	RO	Implementation specific	Current average PE value for AXI port capacity

5.9.2.19. Offset 0x14C: REDCAP – Reduced Capacity Information

Bits	Type	Reset	Description
31:01	RO	0x0	Reserved
00	RO	Implementation specific	AXI Port Capacity 1 : AXI Port capacity is reduced 0 : AXI Port operating at full capacity

5.9.2.20. Offset 0x150: TTTEMP – Thermal Throttling Temperature

This register indicates the Thermal throttling mode and temperatures.

Bits	Type	Reset	Description
31:24	RO	0x0	Reserved
23	RW	Implementation specific	Severe Throttling Mode Enable/Disable. 0x1 : Enable, 0x0: Disable
22:16	RW	Implementation	STT: Severe Throttling Temperature

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		specific	60 °C + Value*0.5 °C, Range is 60 °C, ~124 °C
15	RW	Implementation specific	Light Throttling Mode Enable/Disable. 0x1 : Enable,0x0:Disable
14:08	RW	Implementation specific	LTT: Light Throttling Temperature 60 °C + Value*0.5 °C, Range is 60 °C, ~124 °C
07	RO	0x0	Reserved
06:00	RW	Implementation specific	RTT: Release Throttling Temperature 60 °C + Value*0.5 °C, Range is 60 °C, ~124 °C

6.AXI Protocol Layer – Link Layer

The UCle flit format is Format 6: Latency-Optimized 256B With Optional Bytes Flit Format. The AXI protocol layer received flit (in selected flit format) from Die-to-Die adapter, flit is constructed with many packets, each packet is AXI protocol message follow common structure of various size, link layer job is to accept the flit from D2D Adapter or issue flit to D2D Adapter.

Link-layer will unpack the flit to AXI message and drive it to AXI interface or receive message from AXI interface and pack it to flit. Link-layer should be compliant to D2D Adapter FDI interface.

The D2D interface shall support these high-level features to achieve interoperability between HBF and hosts:

1. A defined mechanism to pack AXI transactions into the FDI flit
2. Independent flow control for each AXI channel to avoid inter-channel blocking
3. Provision for multiple independent AXI ports per UCle instance to enable interleaving and bandwidth optimization
4. Flexibility to interface with hosts with differing AXI data widths

To transport AXI across UCle flits, HBF adopts the AXI over UCle Protocol Specification (AoU), revision 0.8, available at openchipletatlas.org. To address HBF specific use cases not covered with the AoU Basic Profile, the HBF Profile defined in Appendix D is adopted.

The HBF Profile shall support the following:

1. Sync message
2. 8b SYNC field added to most transactions
3. 9 USER bits
4. 14 bits ID
5. Error message

7. Electrical Requirements

7.1. DC Operating Conditions

The following diagram depicts the power sources for HBF.

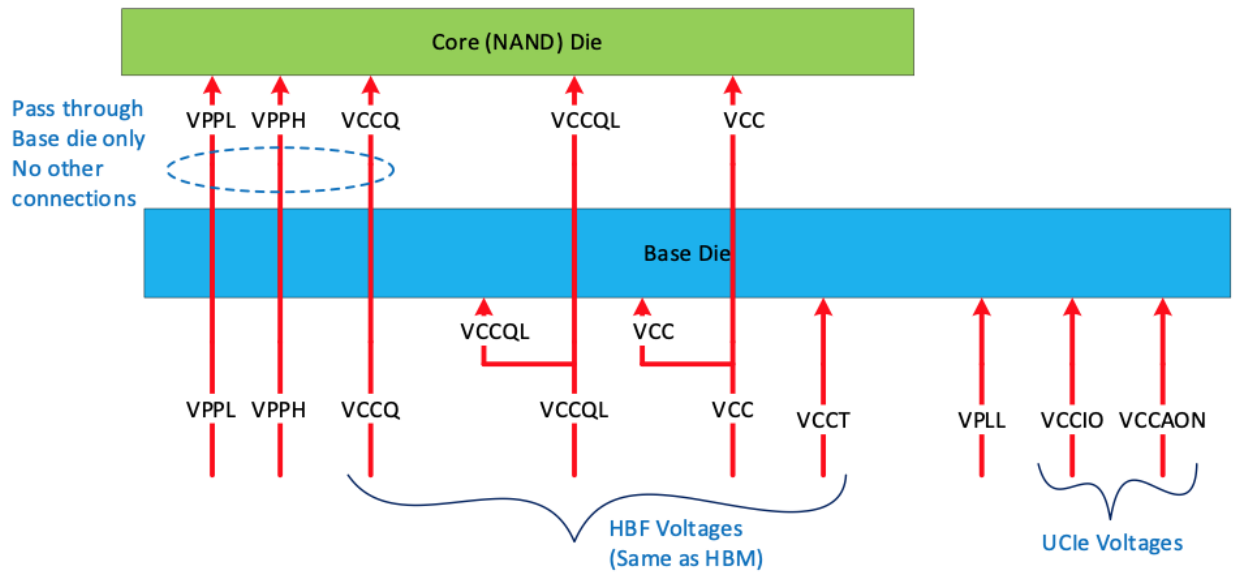


Figure 28: HBF Power Rails

Table 17: UCle DC Operating Conditions

UCle Voltage Supplies						
Description	Symbol	Minimum	Typical	Maximum	Unit	Notes
Ground Reference	VSS	-	0	-	V	
I/O Supply	VCCIO	-	0.5 or 0.7	-	V	
Forwarded power supply from remote Transmitter supply to local Receiver AFE	VCCFWDIO	-	0.5 or 0.7	-	V	
Always on Aux supply (sideband)	VCCAON	0.65	0.75	-	V	
Voltage for PLL to provide 800MHz SB clock from Refclk	VPLL	1.12	1.2	1.32	V	

Table 18: HBF DC Operating Conditions – Case 1

HBF Voltage Supplies						
Description	Symbol	Minimum	Typical	Maximum	Unit	Notes
I/O Supply Voltage	VCCT	0.675	0.75	0.825	V	
Core Supply	VCCQ	1.12	1.2	1.32	V	
Pump Supply	VCCE	-	2.5	-	V	
NAND Supply	VPPL	4.5	5.0	5.5	V	
NAND Supply	VPPH	10.8	12.0	13.2	V	

Table 19: HBF DC Operating Conditions – Case 2

HBF Voltage Supplies						
Description	Symbol	Minimum	Typical	Maximum	Unit	Notes
I/O Supply Voltage	VCCT	0.776	0.8	0.856	V	
Core Supply	VCCQ	1.018	1.05	1.124	V	
Pump Supply	VCCE	1.746	1.80	1.95	V	
NAND Supply	VPPL	3.80	3.90	4.0	V	
NAND Supply	VPPH	10.5	11.0	11.5	V	
Core Supply Low	VCCQL	0.38	0.40	0.44	V	Sandisk product only ¹

¹The Sandisk HBF product requires additional voltage power supply, Core Supply Low (VCCQL) and the host shall support this voltage rail.

The host reads 0x200 the Voltage Pointer register from the protocol-specific configuration register, which points to another address that is vendor-specific. All vendor-related parameters are in this address region. The host is required to provide the above range of voltage supplies to HBF. The actual voltage supplied is set in the configuration register settings during initialization setup and boot process. Based on the register settings, the host supplies the correct voltage for different rails. This is required as the host can use the HBF from different vendors.

7.2. Clock Requirements

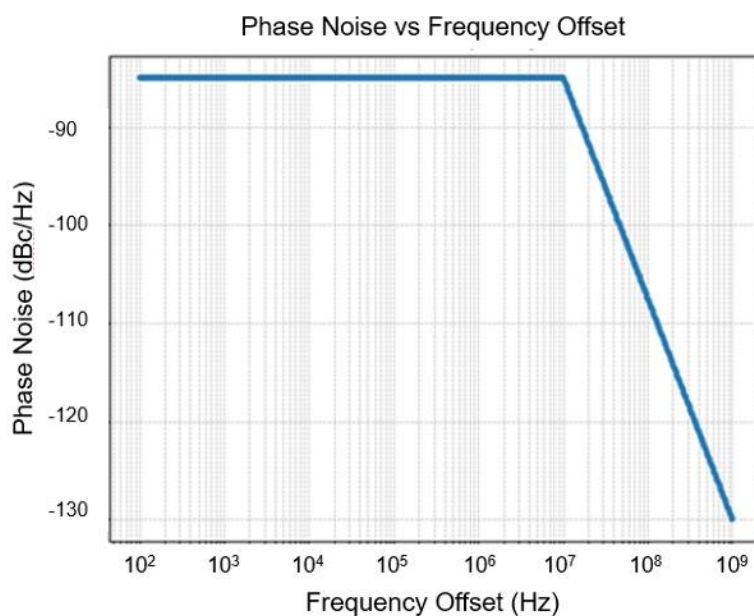
The host is required provide a 100MHz reference clock to the HBF. The cycle-to-cycle jitter requirement is 150ps for this reference clock.

Table 20: Reference clock frequency PPMs and SSC PPMs

Symbol	Description	Limits			Unit	Notes
		Min	Rec	Max		
F_{REFCLK}	REFCLK Frequency	100	100	200	MHz	
$F_{\text{REFCLKDEVIATION}}$	REFCLK Frequency Deviation	-300		300	ppm	Maximum deviation allowed from ideal target frequency
F_{SSC}	SSC Modulation Frequency	30		33	kHz	
$T_{\text{SSC-FREQ-DEVIATION}}$	SSC Deviation	-0.5		0	%	Tracks for different frequencies
$T_{\text{TRANSPORT-DELAY}}$	Tx-to-Rx Transport Delay			5	ns	
$T_{\text{SSC-MAX-FREQ-SLEW}}$	SSC df/dt			1250	ppm/us	

Table 21: Reference clock requirements

Parameter	Minimum	Maximum	Units
Refclk frequency	100	200	MHz
Duty cycle	40	60	%

**Figure 29: Reference clock phase noise**

7.3. Reset Requirements

The host is required to make sure all the ongoing transactions are complete before issuing a reset to the HBF. If a write to the Core die is pending and reset comes from host, the write state is non-deterministic, and the subsequent read may or may not have the written data. The UCle reset triggers a Base die reset then sets the ready bit in the UCle interface.

Table 22: Reset Signal Electrical Parameters

Parameter	Symbol	Min	Max	Unit
Input HIGH Voltage	V_{IH}	$0.65 \cdot V_{CCAON}$	$V_{CCAON} + 0.3$	V
Input LOW Voltage	V_{IL}	$V_{SS} - 0.3$	$0.35 \cdot V_{CCAON}$	V

Table 23: Reset Timing Parameters

Symbol	Min	Max	Unit	Description
tRSTW	1	-	us	RST_n/RESETSB_n/ RESETBD_n pulse width
tRSTH	1	-	us	RST_n/RESETSB_n/ RESETBD_n high period
tRSTF	100	-	ns	RST_n/RESETSB_n/ RESETBD_n filter



Figure 30: Reset Timing

7.4. Power Up Sequence

1. Base die requirement is that VCCE is fully up before VPPH/L starts to ramp
2. HBF requirement is VPPH/L shall be greater than VCCE, VCCQ, VCCT
3. During both power ramp-up and down sequence, the voltage level shall be maintained $VCCE > VCCQ > VCCT$, $VPPH > VPPL$
4. VPPH and VPPL shall initiate their ramp-up 100us after VCCT reaches its minimum DC level.
5. During the VCCE power-down sequence, both VPPH and VPPL shall be maintained below VCCE level until they are fully discharged.
6. The ramp slope for all power supplies during both ramp-up and down transitions shall be from 20us/V to 20ms/V.
7. Ramp time to be less than 200 ms
8. After power rails ramp up, the Base die reset is de-asserted and the Base die goes through boot, resetting the Core die and configuring the interfaces

- Then the ready signal to the host is raised to indicate ready for host commands via UCle

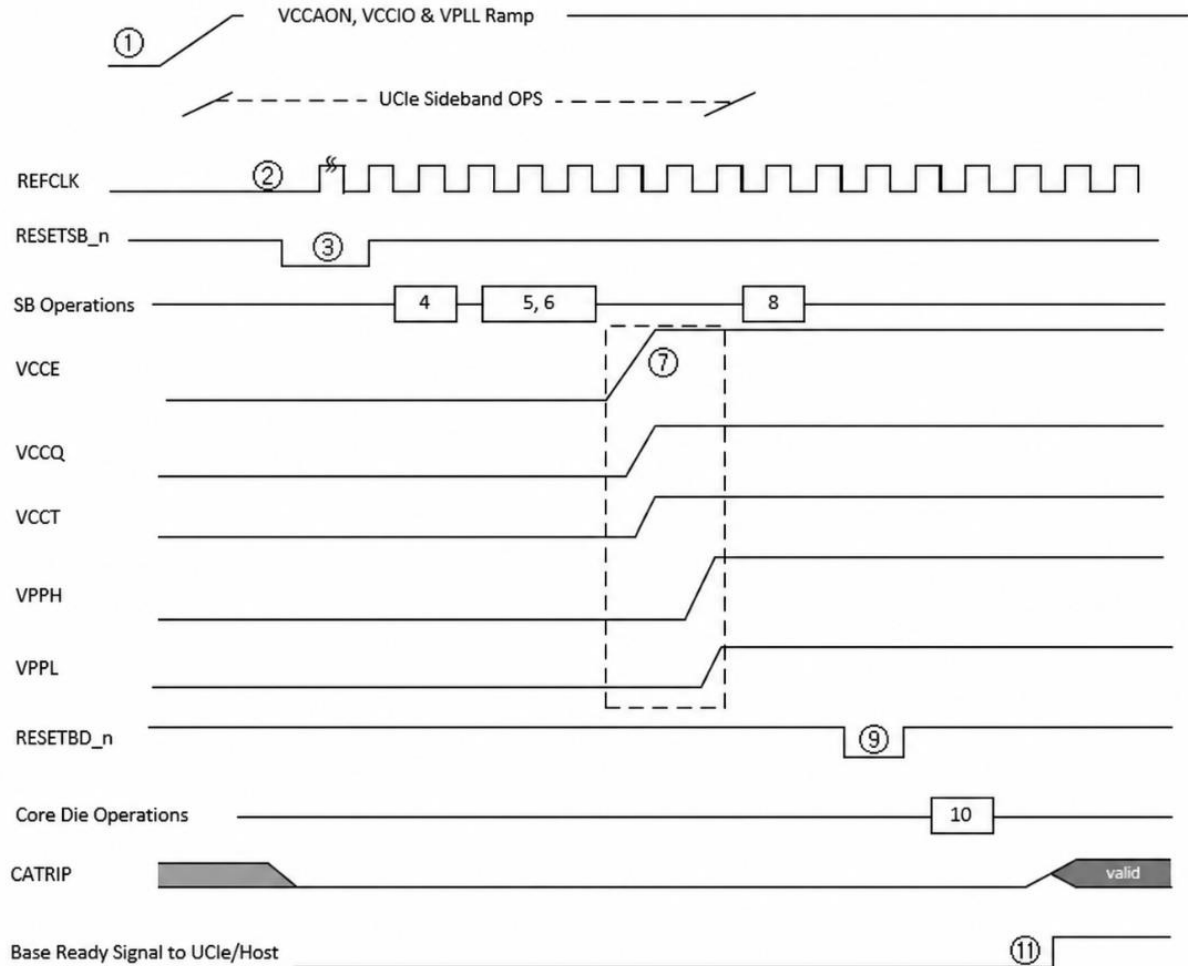


Figure 31: HBF Power Up Sequence

Notes:

- Power ramp time starts when any supply is at 300mV
- Base die requirement: VCCE is fully up before VCCQ starts to ramp
- Ramp time shall be less than 200ms
- Training encapsulates HBF IO training and UCle link up
- The following HBM requirements are noted here, because HBF is expected to be placed near HBMs. The HBF spec intent is to keep the voltage rail requirement deviations from HBM be minimum.
- HBM requirement: VCC shall be greater than VCCQ, VCCT
- HBM requirement: VCCQ and VCCT should ramp at the same time (and same time or earlier than VCCQL)

- HBM requirement: VCCQL and VCCT shall be greater than VCCQL-200mV during the power up sequence

The power ramp sequence is as follows:

1. Host drives VCCAON, VCCIO, and VPLL
2. Host drives 100MHz clock
3. Apply base reset to the chip for UCle sideband reset
4. Sideband PLL produces 800MHz UCle SB clock from 100MHz reference to activate sideband.
5. Base die and Host die sideband handshake to discover each other
6. Host reads vendor-specific registers to get voltages
7. Read the Voltage Pointer register 0x200 that points to a vendor-specific address where the voltage set is defined. The vendor-specific address contains the voltage start address register and the immediate next address contains the voltage end address register. The voltage start address and end address registers are 32-bit elements. The list of voltage rail specifications for that vendor is present between the start and end address registers. These registers are in the UCle D2D Adapter vendor-specific area. All the voltages in the list are also 32-bit elements, and the voltage format is defined in the next section
8. Host drives the set of voltages in increasing order of the address
9. Host writes to Base die vendor-specific register to indicate voltages are ready. A 2nd reset signal is asserted by the host to the Base die to reset and boot the Base die
10. Base/Core die are reset
11. HBF asserts READY to host to indicate that the Base die is ready to receive commands.

Note: CATTRIP data shall stay LOW from the end of reset de-assertion to the end the Core die operations.

7.5. Voltage Register Format

The list of voltages defined between the voltage start and voltage end address registers format is defined as follows. Each is a 32-bit element. The decimal point is placed between bit 15 and bit 16.

Table 24: Reset timing parameters

Bit positions	bit 31 – bit 24	bit 23 – bit 16	Decimal Point (.)	bit 15 – bit 8	bit 7 – bit 0
Place value	Tens	Ones	.	Tenths	Hundredths
Example 0.05	0000	0000	.	0000	0101
Example 0.10	0000	0000	.	0001	0000
Example 0.75	0000	0000	.	0111	0101
Example 3.9	0000	0011	.	1001	0000
Example 11.0	0001	0001	.	0000	0000

7.6. Power Down Sequence

Figure 32 shows the HBF power down sequence and is followed by the Base die.

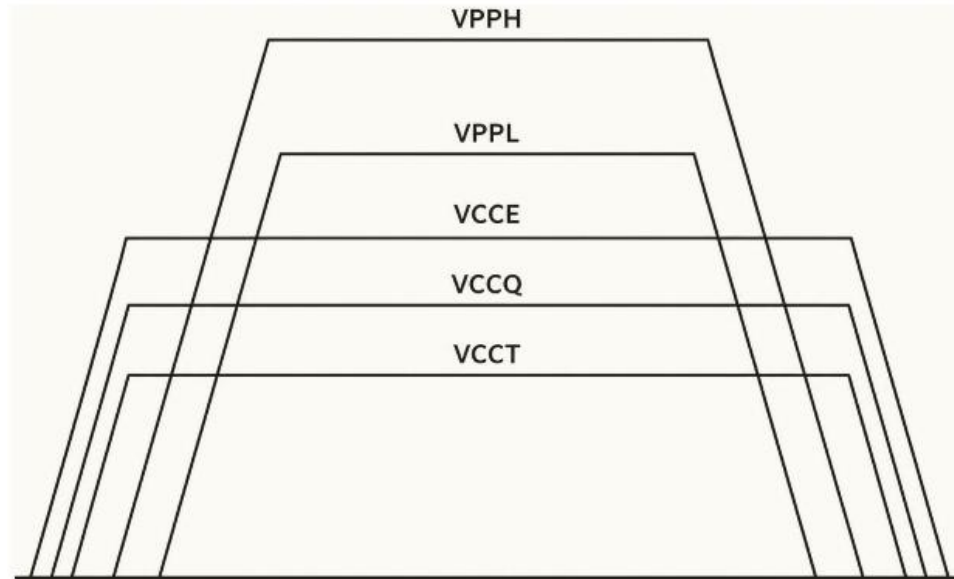


Figure 32: HBF Power Down Sequence

8. Test & Debug Interface

The HBF device provides three distinct interfaces to support comprehensive testing, debugging, and diagnostics throughout the entire product lifecycle, from SiP manufacturing to system-level operation.

8.1. Test & Debug Overview

These interfaces are strategically routed to different physical connection points to optimize for specific usage scenarios:

- DA Port: Dedicated to SiP-level manufacturing and Core die's deep physical analysis.
- IEEE1149.1 (JTAG) Port: Exposed via module pins for standalone hardware verification and firmware debugging.
- IEEE 1500 Port: Connected internally to the xPU for system-level diagnostics and lifecycle management including repair.

Table 25 describes the expected usage of the three Test & Debug interfaces. These may vary depending on the system environment; please refer to the vendor datasheet for details.

Table 25: Purpose of HBF Test & Debug Interfaces

	IEEE 1149.1(JTAG)	IEEE 1500	Direct Access
Purpose	FW Debug	Manufacturing Test, Maintenance	Core die Debug
Target Connection in SiP Package ¹	External Debugger	xPU	External Tester/Debugger
Target connection in Module Level ¹	External Debugger	No Connection	No Connection

NOTE ¹: The module and SiP interfaces may be customized by the customer and the manufacturer. Please refer to the vendor specific HBF product specification on any customizations.

8.2. Concept of HBF Manufacturing Test

The HBF will be manufactured as illustrated in Figure 33. While all defects originating from components are screened and identified using vendor-side test processes, additional defects may arise due to thermal stress incurred during the assembly process. Therefore, the customer shall perform final functional validation steps in accordance with the HBF product's official test guidelines via IEEE 1500 instructions to ensure reliability and compliance with operational specifications.

When a failure occurs, root cause analysis is efficiently performed using the IEEE 1149.1 (JTAG) interface, the Direct Access Port and IEEE 1500 interface.

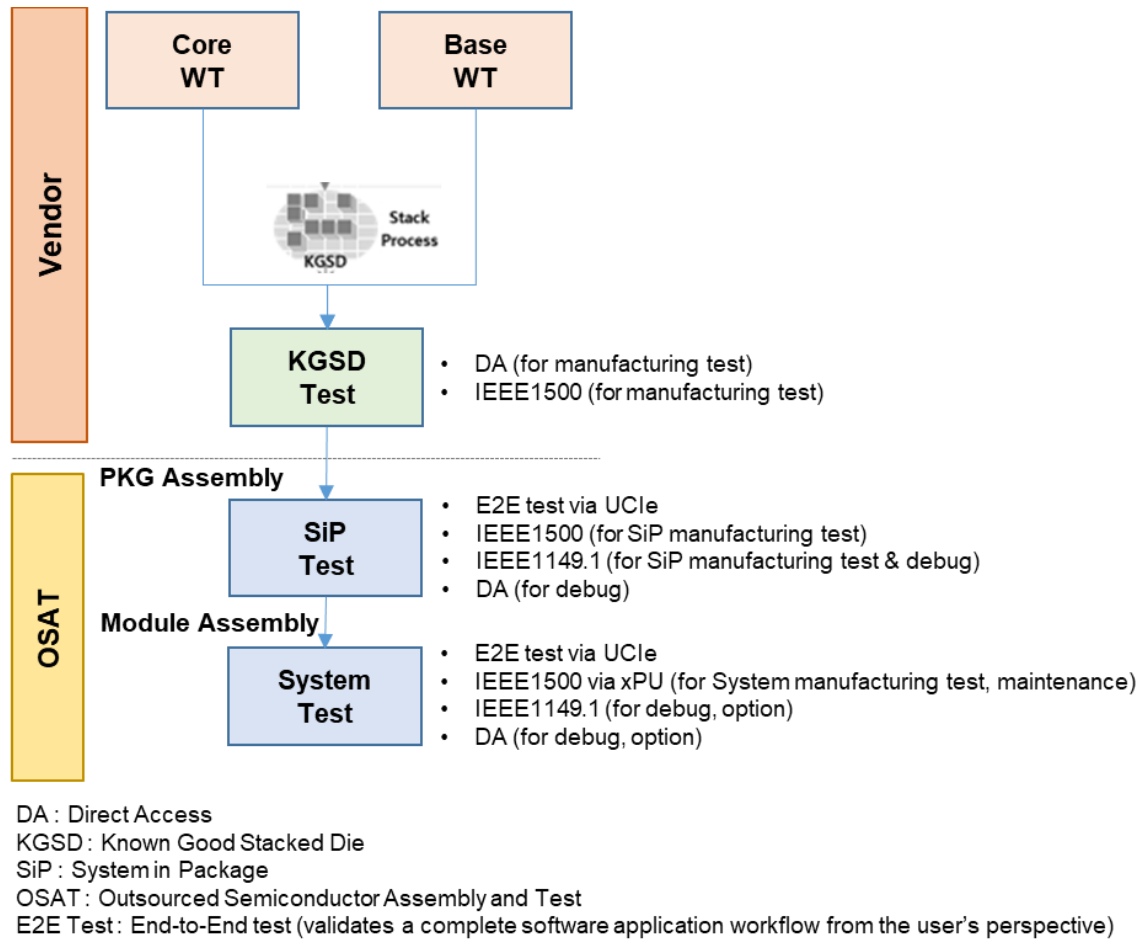


Figure 33: Example of HBF manufacturing test process

8.3. Direct Access(DA) Test Port

8.3.1. Overview

The Direct Access (DA) Test Port provides low-level, parallel access directly to the NAND Core Dies and PHY, explicitly bypassing the Base Die Controller and FW algorithm. This interface enables engineers to read/write internal NAND Core Die's registers and cells and perform targeted fault isolation. This interface is designed for:

- Deep Diagnostic & Debug: Direct characterization of NAND cell properties (e.g., threshold voltage, read disturb) without algorithmic interference.
- Fail Isolation: Distinguishing between Base Die Controller defects and NAND Core/PHY physical failures.
- If DA tests pass but standard interfaces (UCle/IEEE 1500) fail, the defect is isolated to the Base Die Controller. If DA tests fail, the defect is confirmed in the NAND Core or PHY.

- Vendor-Specific Testing: Execution of proprietary test patterns and repair algorithms not accessible via standard protocols.

8.3.2. Pin Configuration

A Direct Access (DA) test port is available via DA[39:0]. Support for this port may vary by product; refer to the specific product data sheet for availability.

- Point-to-Point (8 pins): DA[39:32] (or vendor-defined subset) are designated for dedicated connection to each HBF device for unique addressing.
- Multi-Drop Bus (32 pins): DA[39:20], DA[11:0] are configured for shared bus operation, allowing multiple devices (e.g., HBF and HBM in a SiP package) to connect to a single tester interface.
- Pin Function: The specific mapping of data, address, and control signals to the DA[39:0] pins are vendor-specific and defined in the separate Test Integration Guide.
- Note: Support for the DA port and the specific pinout configuration may vary by product grade. Refer to the specific product data sheet for availability.

Table 26: Direct Access(DA) pin allocation

Pin Group	DA Pin List	Pin Count
Point to Point	DA[19:12]	8
Multi Drop	DA[39:20],DA[11:0]	32

8.3.3. DA Mode Control

Access to the DA test port is strictly controlled by the DA[12] pin and LOCKOUT MMIO register.

- Default State (In-Active): The DA[12] pin includes an internal pull-down resistor, ensuring the port remains inactive (Logic LOW) even if the pin is left floating.
- In-Activate : When DA[12] = LOW or LOCKOUT=ENABLE
- All DA data/address drivers (DA[39:13], DA[11:0]) are placed in High-Impedance (Hi-Z) state.
 - Input receivers are disabled to prevent noise coupling.
 - The device operates in normal mode (or IEEE 1500/JTAG mode if selected by other signals).
- Activate : When DA[12] = HIGH and LOCKOUT=DISABLE
 - The DA drivers and receivers are enabled.
 - All on-the-fly commands queued in the Base Die Controller, or UCIe layer are abandoned and cleared.
 - The device enters Direct Access Mode, allowing vendor-specific test features.
 - All UCIe PHY interfaces are taken out of service.
- Data stored in HBF may be lost, depending on the ongoing operation.

8.3.4. Operational Constraints & Warnings

Access to the DA test port is controlled by the LOCKOUT register at MMIO address 0x0140. To disable the DA port and prevent unauthorized or unintended access, the host system or firmware shall write 0x01 to Byte 0 of this register. Once set, the DA test port remains inactive until the HBF is powered off. The lockout state shall be retained across all reset conditions, including RESET_n assertion and HBF_RESET via IEEE 1500 instructions. When operating in DA mode, the debugging entity shall be responsible for managing collisions and contention as appropriate.

8.4. HBF IEEE 1149.1 Debug Port

8.4.1. Overview

The HBF device incorporates a standard IEEE 1149.1 (JTAG) Test Access Port (TAP) designed for system-level boundary scan, firmware debugging, and silicon bring-up.

8.4.2. Pin Configuration

The JTAG port consists of 5 pins, as defined in Table 27.

Table 27: IEEE1149.1 Debug Port Signal List and Description

Symbol	Type	Description
TCK	Input	Primary clock signal for the IEEE 1149.1 TAP controller.
TMS	Input	Controls the state of the TAP Controller State Machine.
TDI	Input	Serial data input for the Instruction Register (IR) and Data Registers (DR).
TDO	Output	Serial data output from the Instruction Register or Data Registers.
TRST_n	Input	Asynchronous reset signal for the TAP Controller (Vendor Specific).

This specification does not restrict the physical connection topology between the xPU, external test probes, and target devices. The interconnection scheme (e.g., point-to-point, daisy-chain, or star topology) shall be determined based on specific application requirements and the selected xPU variant.

8.4.3. Operational Constraints & Warnings

IEEE 1149.1 is designated for firmware debugging and remains operational during UCle Main Band/Side Band and IEEE 1500 activities. While concurrent operation with UCle Main Band/Side Band is supported, simultaneous operation of IEEE 1500 and IEEE 1149.1 is not guaranteed and may be restricted depending on the internal implementation and system topology.

Upon customer request for security reasons, IEEE 1149.1 pins may be configured as floating inputs via fuse programming during the manufacturing phase.

8.5. HBF IEEE 1500 Test Port

8.5.1. Overview

The HBF implements an IEEE 1500-compliant test access port, providing a direct test interface between the host(xPU) and the HBF, as well as between external test equipment and the HBF.

- **Standard Compliance & Channel Architecture:** The HBF utilizes a 16-channel host interface via UCle PHY. The HBF maintains full compatibility with the IEEE 1500 WIR decoding rule, where channel selection is explicitly handled within the WIR. This ensures seamless integration with existing host-based test infrastructure.
- **Controller and Firmware Integration:** The HBF architecture uniquely integrates a dedicated Controller and Firmware (FW) engine that interfaces directly with the IEEE 1500 wrapper logic. This integration extends standard boundary-scan capabilities, enabling advanced diagnostic operations such as Firmware Download, Software-Based BIST(Built-In Self-Test), and complex NVM(Non-Volatile Memory) management tasks. Any test operation not explicitly defined in this specification strictly conform to the IEEE 1500 standard.

8.5.2. Pin Configuration

The test port of IEEE 1500 consists of dedicated control inputs and a 16-bit serial output that enables instruction selection, data capture, shifting, and result update operations on the target wrapper registers. The Table 28 lists all HBF IEEE 1500 test port signals, their type, and functional description.

Table 28: HBF IEEE1500 Test Port Signal List and Description

Symbol Type Description	Type	Description
WRCK	Input	Dedicated clock used to operate IEEE 1500 functions.
WRST_n	Input	When pulled LOW, WRST_n asynchronously puts the IEEE 1500 test port into its normal system mode. No WRCK clocks are required when WRST_n is LOW. See WDR Reset State
WSI	Input	IEEE 1500 test port serial input
SelectWIR	Input	SelectWIR determines whether the instruction register (WIR) or a wrapper data register is being accessed.
CaptureWR	Input	Controls a capture operation in the selected wrapper register (WR)
ShiftWR	Input	Controls a shift operation in the selected wrapper register (WR)
UpdateWR	Input	Controls an update operation in the selected wrapper register (WR)
WSO[15:0]	Output	IEEE 1500 test port serial output, which has a bit per a channel.

8.5.3. HBF IEEE 1500 Mode Control

The device supports three primary operational modes controlled by the WRST_n pin, DA[12] pin, and LOCKOUT register. Mode transitions are strictly gated to prevent bus contention, ensure test integrity, and maintain compatibility with existing host-based test infrastructure.

8.5.4. Operational Constraints & Warning

HBF IEEE 1500 operations are executed independently of the UCle PHY status. The interface remains fully functional during normal memory operation, low power mode without requiring an active UCle clock or link state.

- **DA[12] Pin Default State & Mode Selection:** The DA[12] pin serves as the primary hardware selector between the Direct Access (DA) test port and the IEEE 1500 interface. The pin is equipped with an internal pull-down resistor, ensuring a default Logic LOW state when left floating. A Logic LOW state on DA[12] selects the IEEE 1500 mode (or normal Mission Mode when combined with other control signals). Conversely, driving DA[12] HIGH activates the DA port, which automatically disables the IEEE 1500 port and UCle Mainband/Sideband to prevent bus contention. If DA mode is required, a proper external pull-up network shall be implemented to guarantee a stable HIGH state.

Failure to maintain correct pull-up/pull-down configuration may result in unintended mode transitions or floating pin noise coupling.

- **WRST_n Signal Behavior & Interface Gating:** WRST_n is the asynchronous reset and mode control signal for the IEEE 1500 wrapper. When pulled LOW, WRST_n forces the wrapper into its normal system mode, placing all wrapper registers in the BYPASS instruction state and disabling WRCK-dependent operations. During DA Mode, WRST_n is designated as “Don’t Care” as the DA port operates independently of the wrapper logic. However, during normal Mission Mode operation, it is strongly recommended to hold WRST_n LOW. This prevents accidental assertion of the IEEE 1500 interface, ensures BYPASS remains the active instruction, and avoids unintended capture/shift/update events that could disrupt memory controller operations or firmware execution.
- **WSO Signal Integrity & Data Interpretation:** The WSO[15:0] parallel output lines continuously reflect the current state of the Wrapper Data Register (WDR) shift stage, governed by the active WIR instruction and WRCK timing. Even when the IEEE 1500 interface is not actively shifting test data, WSO[15:0] remains electrically valid and driven by the wrapper logic. System designers shall ensure that the host controller or external probes do not misinterpret these signals as active test data or valid memory responses during idle or non-test periods. Proper signal masking, WIR channel selection, and explicit BYPASS instruction usage are required to isolate WSO outputs and prevent data misinterpretation or bus arbitration conflicts.
- **Additional Hardware & Timing Constraints**
 - **Clock Synchronization:** WRCK shall remain stable and toggle continuously during CaptureWR, ShiftWR, and UpdateWR phases. Asynchronous deassertion of WRCK during active wrapper operations may corrupt WDR shift states.
 - **Unselected Channel Isolation:** Channels not selected via WIR[12:8] are gated off and will not respond to Capture, Shift, or Update events. This prevents cross-channel data corruption but requires explicit WIR decoding for multi-die/channel testing.
 - **Reset Behavior:** HBF_RESET instruction (or equivalent) does not reset the Wrapper Serial Port (WSP) control logic or WDRs. The WSP remains active, and the BYPASS instruction state is preserved unless explicitly overwritten.

8.5.5. HBF IEEE 1500 Timing Diagram

The Detailed Operation Description illustrates the minimum clock cycle sequence for operating the HBF IEEE 1500 port. This sequence demonstrates how to load a Wrapper Instruction (e.g., FW_DOWNLOAD or SW_BIST), capture status data from the internal Base die controller and updates the command registers.

The operation relies on the synchronization of the WRCK (Wrapper Clock) with control signals (SelectWIR, CaptureWR, ShiftWR, UpdateWR).

Detailed Operation Description:

- Instruction Register Selection (SelectWIR = High)
 - At clock edge, the SelectWIR signal is asserted High.
 - This connects the Wrapper Instruction Register (WIR) between WSI (Input) and WSO[0] (Output).
 - Action: The host begins shifting the 14-bit (or 32-bit) instruction opcode into the WIR via WSI.
 - Constraint: SelectWIR shall remain stable (High) throughout the entire instruction loading phase until the sequence completes at T4. Other control signals (CaptureWR, ShiftWR, UpdateWR) shall remain inactive (Low) during this transition to prevent unintended register operations.
- Data Capture (CaptureWR = High)
 - At clock edge, the CaptureWR signal is pulsed High.
 - Action: This triggers the selected Wrapper Data Register (WDR) to capture parallel data from the HBF internal logic.
 - HBF Context: This typically captures status flags or error codes from the Controller or NVM array into the WDR shift stage. For example, if the previous instruction was GET_RESULT, this step captures the Pass/Fail result.
- Data Shift (ShiftWR = High)
 - At clock edge, the ShiftWR signal is pulsed High.
 - Action: The data loaded in the WDR shift stage is shifted out serially via WSO. Simultaneously, new test data is shifted in via WSI if the register supports Read/Write (R/W) mode.
 - HBF Context: Since HBF supports 16-channel parallel output (WSO[15:0]), this step allows the host to read status from multiple channels simultaneously, significantly reducing test time compared to single-bit serial output.
- Data Update (UpdateWR = High)
 - At clock edge, the UpdateWR signal is pulsed High.
 - Action: This transfers the data currently in the WDR shift stage to the update stage (latch).
 - HBF Context: This is the critical execution trigger. For commands like FW_DOWNLOAD, the actual data is committed to the Controller or Fuse Array only after this UpdateWR pulse. Until this point, the data remains in the temporary shift buffer and does not affect device operation.
- Sequence Completion & Stabilization
 - At clock edge, the SelectWIR signal is de-asserted (returned to Low) to switch back to Data Register mode or kept High for another instruction load.
 - Action: The port returns to an idle or ready state, awaiting the next command sequence.

- Note: The WRCK shall continue toggling until all control signals are stable to ensure synchronous operation.

8.5.6. HBF IEEE 1500 Logical Architecture

The internal logic of the HBF IEEE 1500 port is organized hierarchically to support both standard register access and controller-mediated operations. The HBF IEEE 1500 Logical Architecture comprises the following key components.

- Wrapper Instruction Register (WIR): A 14-bit shift register that decodes incoming instructions. When SELECTWIR is asserted, the WIR captures the opcode and channel selection bits.
- Wrapper Data Registers (WDR): Includes the Wrapper Boundary Register (WBR), Wrapper Bypass Register (WBY), and extended Data Registers connected to the internal Base die controller. These registers support Shift (S), Capture (C), and Update (U) stages as required by the instruction.
- Controller & FW Engine: The HBF-specific block that interprets WDR data to execute high-level commands (e.g., FW_DOWNLOAD, SW_BIST). It acts as a bridge between the serial test interface and the parallel NVM array.
- Channel Multiplexer: Routes control signals (CAPTURE, SHIFT, UPDATE) and data to specific channels (CH_SELECT[0:15]) based on the WIR decoding, enabling parallel operation across 16 channels.

8.5.7. HBF IEEE 1500 WIR Encoding

The HBF device supports flexible test targeting through its 14-bit Wrapper Instruction Register (WIR). This architecture enables efficient testing strategies ranging from single channel to massive parallel operations across the entire device.

When SELECTWIR is asserted HIGH, the WIR is connected between WSI and WSO[0] for instruction loading. The WIR bit allocation is defined as follows:

- WIR[7:0] (Opcode): Exclusively defines the test operation (e.g., BYPASS, FW_DOWNLOAD, SW_BIST).
- WIR[13:8] (Mode & Channel Select): Determines the scope of the operation by selecting specific channels or groups of channels within the 16-channel HBF architecture.

This dual-function encoding allows the host controller to execute a single instruction that simultaneously targets multiple channels, leveraging the parallel WSO[15:0] output structure to maximize test throughput.

The mapping of WIR[13:8] bits to channel selection modes is summarized in Table 29. The upper bit (WIR[13]) act as the Mode Selector, while the lower 5 bits (WIR[12:8]) define the specific Channel Group or Individual Channel.

Table 29: WIR Channel Mode Definition

MODE	ModeSelector	Channel Selection		OP Code
	WIR[13]	WIR[12]	WIR[11:8]	WIR[7:0]
Single Channel Selection	0x0	Don't care	0x0 - Channel0 0x1 - Channel1 0x2 - Channel2 0x3 - Channel3 0x4 - Channel4 0x5 - Channel5 0x6 - Channel6 0x7 - Channel7 0x8 - Channel8 0x9 - Channel9 0xA - Channel10 0xB - Channel11 0xC - Channel12 0xD - Channel13 0xE - Channel14 0xF - Channel15	Instruction OP Code
Multi Channels Selection	0x1	0x1	0x8 - 8Channels - 1 st Group. 0x9 - 8Channels - 2 nd Group 0xA - 4Channels - 1 st Group 0xB - 4Channels - 2 nd Group 0xC - 4Channels - 3 rd Group 0xD - 4Channels - 4 th Group The other values are vendor specific	Instruction OP Code
All Channels Selection	0x1	0x1	0xF - All Channels	Instruction OP Code

- Single Channel Selection (WIR[13]=0):
 - Targets exactly one specific channel (0–15) defined by WIR[11:8].
 - Use Case: Ideal for detailed failure analysis, characterizing specific weak bits, or performing repair operations on a single die/channel without disturbing others.
 - Output: Only the corresponding bit in WSO[15:0] will return valid data; other bits may not be valid.
- Multi Channel Selection (WIR[13:12]=11b, WIR[11:8] = 0x8~0xD):
 - Targets a predefined group of channels simultaneously.
 - Group Definitions:
 - 0x8: Channels 0–7 (1st Half)
 - 0x9: Channels 8–15 (2nd Half)
 - 0xA–0xD: Quadrant groups (e.g., 0–3, 4–7, etc. – *Exact mapping is Vendor Specific*).
 - Use Case: Balances test time and power consumption. Useful for block-level operations like erasing or programming specific regions of the NVM array.

- All Channels Selection (WIR[13:8]=0x3F):
 - Broadcasts the instruction to all 16 channels simultaneously.
 - Use Case: Maximum throughput operations such as global BYPASS, full-device SW_BIST execution, or bulk status checking.
 - Output: All 16 bits of WSO[15:0] will output data in parallel, allowing the host to read the status of the entire device in a single shift cycle.

By utilizing the All Channels or Multi Channel modes, the HBF leverages its 16-bit parallel WSO interface to significantly reduce test time compared to traditional serial scan architectures. For example, during a SW_BIST operation:

- The host loads the BIST_START opcode with WIR[13:8] = 0x3F (All Channels).
- The internal Base die controller initiates BIST on all 16 channels concurrently.
- Upon completion, a single CaptureWR followed by ShiftWR retrieves 16 bits of status data (one per channel) simultaneously.

8.5.8. HBF IEEE 1500 Wrapper Data Register Types

- Read only(R) Wrapper Data Registers:

Read-only WDR bit fields latch data into the shift stage register upon a CaptureWR event. These registers maintain their current state during an UpdateWR event and do not incorporate an update stage. During a ShiftWR event, the stored data is shifted out via WSO. Any data shifted into WSI during this phase is disregarded.

- Write only(W) Wrapper Data Registers:

Write-only WDR bit fields transfer all data bits to the update stage register upon an UpdateWR event. If a write-only WDR is routed between WSI and WSO, CaptureWR events do not influence its operation. The register contents are shifted out during a ShiftWR event.

- Read and Write (R/W) Wrapper Data Registers:

R/W WDRs combine the operational characteristics of both read-only and write-only registers. Data bits are captured into the shift stage register during a CaptureWR event. Subsequently, at the UpdateWR event, these bits are simultaneously transferred from the shift stage to the update stage.

- WDR Reset State:

Asserting WRST_n to LOW asynchronously asserts these states on the HBF's IEEE 1500 test port logic:

- All WDRs place their update and/or shift stages (where applicable) into a state that ensures that the HBF returns to mission mode operation and all test modes are disabled
- Any ongoing BIST operation will be terminated.

8.5.9. HBF IEEE 1500 Test Port AC Timing Parameters**Table 30: IEEE1500 Test Port AC timings**

Group	Timing Parameter	Symbol	Value		Unit	Note
			Min	Max		
IEEE1500 Port I/O Timings	WRCK clock period	tCKTP	10		ns	
	WRCK clock high pulse width	tCKTPH	0.45		tCKTP	
	WRCK clock low pulse width	tCKTPL	0.45		tCKTP	
	WRST_n pulse width low	tWRSTL	100		ns	
	IEEE1500 port operation after WRST_n deassertion	tWINIT1			tCKTP	
	Rising WRST_n edge to WRCK setup time	tSWRST			ns	
	WSI input setup time to WRCK rising edge	tSR			ns	
	WSI input hold time from WRCK rising edge	tHR			ns	
	WSI input setup time to WRCK falling edge	tSF			ns	
	WSI input hold time from WRCK falling edge	tHF			ns	
	WSO output valid time from WRCK falling edge	tOVWSO			ns	
Channel_Disable	Channel disable to input and out buffer disable delay	tCHDIS			us	
Data Output	Channel output delay after setting level	tCHCHN			us	
Channel_ID	Output high time from WRCK falling edge	tOVCHN			ns	
	Output return to default state delay	tOZCHN			ns	
MMIO Register	Transfer delay from WDR to MMIO Register	tUPMMIO			ns	
	Transfer delay from MMIO register to WDR	tDWMMIO			ns	
Configuration Register	Transfer delay from WDR to CONFIGURATION Register	tUPCONFIG			ns	
	Transfer delay from CONFIGURATION Register to WDR	tDWCONFIG			ns	
FW Download	FW commit operation latency	tFWCOM			s	
	FW commit result latency	tFWRST			us	
SW BIST	BIST Execution latency	tBIST			s	
	BIST Result latency	tBISTRST			us	
Status	Update status latency	tSTA			us	
Reset	Hard Reset latency	tHRES			ms	
	Soft Reset latency	tSRES			ms	

Data Transfer Delay	Transfer delay from WDR to Buffer	tUPBUF			us	
	Transfer delay from Buffer to WDR	tDWBUF			us	

8.5.10. HBF IEEE 1500 Advanced Functionality

The HBF extends the standard IEEE 1500 usage by leveraging the internal Base Die controller and Firmware (FW) for complex test scenarios.

8.6. Mission Mode and Test Mode Interaction and Control Signals

The HBF supports normal operation (Mission Mode) alongside manufacturing and debugging test modes (IEEE 1500, IEEE 1149.1, and DA Mode). Control signals and clock states are explicitly defined to prevent interface conflicts between modes and to ensure deterministic state transitions. Table 31 summarizes the recommended required Control Signals and Clock Statuses for Mission Mode and the three Test Modes, providing a reference baseline for system designers and testers to accurately configure signal gating and interface activation states during mode transitions.

Table 31: Interface Control Signals and Clock State Definitions Per Mode

Control Signals & Clock	Mission Mode (UCle Mainband/Sideband)	Test Mode IEEE 1500	Test Mode IEEE 1149.1	Test Mode DA Mode
WRST_N	LOW	HIGH	HIGH	LOW
DA[12]	LOW	LOW	LOW	HIGH
LOCKOUT Register	ENABLE	ENABLE	ENABLE	DISABLE
UCle Clock	ACTIVE	Vendor Specific	ACTIVE	Vendor Specific
IEEE 1149.1 Clock	Don't Care (Not Required)	Vendor Specific	ACTIVE	Vendor Specific
IEEE 1500 Clock	Don't Care Not Required	ACTIVE	Vendor Specific	Vendor Specific

- Operational Recommendations: Maintain Mission Mode Safety: During normal operation, enforce LOCKOUT=ENABLE, WRST_N=LOW, and DA[12]=LOW to prevent unintended test mode activation.

- **Mode Transition Protocol:** Prior to entering IEEE 1500 or DA Mode, disable the UCle link and stop the UCle Clock to eliminate bus contention risks. Avoid switching UCle operating modes while IEEE 1500 memory accesses are in progress.
- **Concurrency Restrictions:** While Mission Mode and Test Modes are physically isolated, simultaneous operation is technically feasible but strongly discouraged due to potential memory access interruptions and host task cancellations. Concurrent operation across IEEE 1500, IEEE 1149.1, and DA is unnecessary; hardware gating via DA[12] and WRST_N ensures mutual exclusivity.
- **Vendor-Specific Configuration:** UCle Clock and IEEE 1149.1 Clock states marked as “Don't Care” or “Vendor Specific” depend on SiP routing and product tier. Refer to the vendor datasheet for exact timing and electrical requirements.

8.7. HBF IEEE 1500 Instructions

HBF supports IEEE 1500 instructions as specified in Table 32 which shows the IEEE 1500 instruction set. These instructions are designed to align with the HBM standard to enable sharing of the IEEE 1500 test bus. For detailed definitions and usage guidance, please refer to the product data sheet and test guide note.

Table 32: HBF IEEE 1500 instruction set

MODE	WIR [7:0] OPCODE	Instruction	Description	Reg Type	WDR Len
X	0x00	BYPASS	Bypass	R/W	1
S/A	0x01	EXTEST_RX	Micro bump boundary scan Rx (Open/Short)	R	TBD
S/A	0x02	EXTEST_TX	Micro bump boundary scan Tx (Open/Short)	W	TBD
	0x03	RFU (Reserved for Future Use)			
	0x04	RFU (Reserved for Future Use)			
X	0x05	HBF_RESET	Functional Reset	W	1
	0x06-0x0D	RFU (Reserved for Future Use)			
S/A	0x0E	DEVICE_ID	Returns the HBF's unique identification code.	R	TBD
S/A	0x0F	TEMPERATURE	Returns binary temperature code.	R	9
	0x10-0x11	RFU (Reserved for Future Use)			
S	0x12	SOFT_LANE_REPAIR	Soft Lane Remapping -WSO RM Pin	R/W	TBD
S	0x13	HARD_LANE_REPAIR	Hard Lane Remapping -WSO RM Pin	R/W	TBD
S/A	0x14	CHANNEL_DISABLE	Disables a channel (All-channel disable is optional)	W	1

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S/A	0x15	CHANNEL_TEMPERATURE	Returns binary channel temperature code of each Cores and Base die.(9bit binary)	R	144
	0x16-0x17	RFU (Reserved for Future Use)		W	1
S/A	0x18	ECS ERROR LOG	Error Check and Scrub(ECS) Error Log Information	R	TBD
S/A	0x19	HS_REP_CAP	Returns whether core's TSV have repair sources or not	R	TBD
S/A	0x1A	SELF_REP	Self-repair the internal TSV	R/W	TBD
S/A	0x1B	SELF_REP_RESULTS	Self-repair results after SELF_REP	R	TBD
-	0x1C-0x1D	RFU (Reserved for Future Use)			
S/A	0x20	SET_CONFIG_REGISTER	Set Config Register	R/W	Vendor Specific
S/A	0x21	GET_CONFIG_REGISTER	Returns Configuration Registers	R/W	Vendor Specific
S/A	0x22	SET_MMIO_REGISTER	Set MMIO Register	R/W	Vendor Specific
S/A	0x23	GET_MMIO_REGISTER	Returns MMIO Registers	R/W	
S/A	0x24	GET_DEVICE_INFORMATION	Returns the device information. Ex). smart log, error information	R/W	Vendor Specific
S/A	0x30	FW_DOWNLOAD_START	FW download - Target Address	R/W	36
S/A	0x31	FW_DOWNLOAD_DATA	FW download - Payload	W	MPS (Maximum payload size)
S/A	0x32	FW_DOWNLOAD_COMMIT	FW download - FW Commit - GET_STATUS, GET_RESULT activated.	R/W	36
S/A	0x33	SW_BIST_EXECUTE	Execute FW - GET_STATUS, GET_RESULT activated.	W	Vendor Specific
S/A	0x35	GET_RESULT	Returns the test result for the instruction issued lastly.	R/W	Instruction Specific
S/A	0x36	GET_STATUS	Returns status for the instruction issued lastly. - Ready/Busy	R	8

			- Buffer Valid/Not Valid		
S/A	0x37	INT_TRAINING	Training internal channels and return eye of measurement data. - GET_STATUS, GET_RESULT activated.	R/W	Vendor Specific
S/A	0x38	AUTHENTICATION	Verifying the identity of a host, it ensures that the entity requesting access is who or what it claims to be	R/W	Vendor Specific
	0x39-0x3F	RFU (Reserved for Future Use)			
Vendor Specific	0x40-0xFF	Vendor Specific			

NOTE 1 : S : Single Channel Selected, A : Broadcast. All Channel Selected, M : Multi Channels Selected.

NOTE 2 : Unsupported instruction codes will default to the BYPASS instruction when the WIR is updated with the unsupported encoding.

NOTE 3 : Channels that are not selected by WIR do not respond to the instruction and ignore any Update, Capture and Shift events.

NOTE 4 : WDRs shift out the least significant bit on the WSO port at the first WRCK of the shift sequence.

timing and valid data window are defined in Table .

NOTE 5 : All Channel Selected instruction (ex) WIR[13:8] =0x3F) drives the same data on the WSO outputs of all active channels during ShiftWR events. Inactive channels (channels that are marked as “not present / not working” in the DEVICE_ID WDR and channels that have been disabled using the CHANNEL_DISABLE instruction) drive a static LOW on their WSO outputs.

8.8. Firmware Download Sequence

8.8.1. Overview

This specification defines the firmware download procedure using the IEEE 1500 standard interface. The primary purpose of this protocol is to enable firmware updates during the manufacturing process or in-field operations. The Vendor shall provide a firmware image aligned to the ****MPS****(Maximum Payload Size), which corresponds to the IEEE 1500 WDR (Wrapper Data Register) size. The image shall include option parameters for the download process, along with a digital signature and CRC code to ensure data security and data integrity. The Host segments the image into MPS-sized chunks and transmits them via the FW_DOWNLOAD_DATA command. Upon completion, the Host issues a FW_COMMIT command, triggering the device to internally verify the header, signature, and CRC before writing to Non-Volatile Memory (NVM). The device reports the result to the Host. Finally, a RESET command is issued to activate the new firmware.

8.8.2. Firmware Binary Recommendation

- Alignment: The firmware binary shall be padded to align with the ****MPS****(WDR Size).
- Header Structure: The image header shall contain:
 - Download option flags (e.g., target channel, boot mode).

- Total image size and version information.
- Security Block: Digital Signature (e.g., ECDSA/RSA) for authentication.
- Integrity Block: CRC-16/32 code for data error detection.
- Source: Provided by the Vendor as a secure binary file.

8.8.3. Firmware Download Sequence

Step 1: Initialization

The Host issues the HBF_RESET command (Hard Reset) to clear any previous states and prepare the device for firmware download.

Step 2: Start FW Download

The Host issues the FW_DOWNLOAD_START command to initiate the download sequence with the target address.

Step 3: Data Transmission (Segmented Write)

The Vendor shall provide a firmware image aligned to the Maximum Payload Size (MPS), which corresponds to the IEEE 1500 Wrapper Data Register (WDR) bit-width. The firmware binary shall be padded to the nearest MPS boundary if the actual image size is not a multiple of the WDR size. The aligned image shall be segmented into MPS-sized chunks for sequential transmission via the IEEE 1500 WDR.

Step 4: FW Download Commit

After all data segments are transmitted, the Host issues the FW_DOWNLOAD_COMMIT instruction. Upon the UpdateWR event, the Device might perform the following internal operations:

- Header Validation: Verifies the image format, target options, and version parameters.
- Security Check: Authenticates the digital signature (e.g., ECDSA/RSA) to prevent unauthorized firmware.
- Integrity Check: Calculates and compares the CRC-16/32 code to detect data corruption.

If all checks pass, the validated data is atomically written to the NVM. If any check fails, the write operation is aborted, the transmitted buffer is discarded, and the previous valid firmware is retained. No corrupted or unverified image will overwrite existing NVM contents.

Following the completion of internal verification, the Device enters a non-deterministic latency period (tFWCOM). The Host shall monitor the download completion by issuing the GET_STATUS instruction and reading the status field via the WDR. During the NVM write phase, the Host shall maintain a stable power supply and shall not issue any reset or power-down commands.

Step 5: Activation

Upon receiving a "Success" status, the Host issues the HBF_RESET command to reboot the device. The device boots up using the newly updated firmware. The type of HBF_RESET is Vendor specific.

8.8.4. Error Handling

- **Verification Failure:** If the signature or CRC check fails during the COMMIT phase, the device returns a specific error code. The Host shall retry the entire sequence from Step 1.
- **Timeout:** If the device does not respond within tFWCOM, tFWRST, the Host shall abort the process and flag a hardware error.
- **Data Retain:** Under no circumstances will a corrupted or unverified image overwrite the existing valid firmware in NVM. The system remains in its pre-update state. Regarding the detailed error handling, please refer to vendor datasheet.

8.8.5. Summary of FW Download Sequence

Figure 34 illustrates the FW Download Sequence via IEEE 1500.

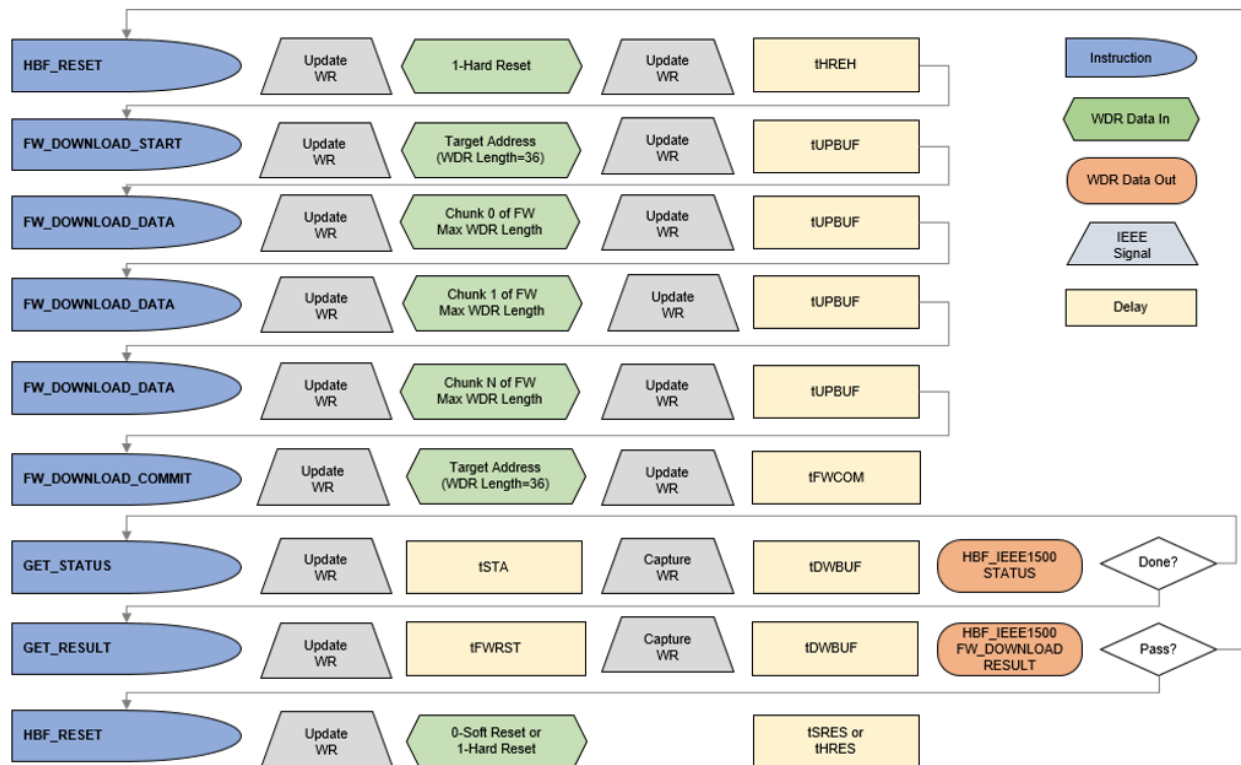


Figure 34: FW Download Sequence via IEEE1500

NOTE: HBF_RESET's type is vendor specific.

8.9. SW Based BIST Test

8.9.1. Overview

This section defines the software-based Built-In Self-Test (BIST) framework accessible via the IEEE 1500 test port. Unlike fixed hardware test logic, the SW-based BIST architecture delivers exceptional flexibility by allowing test vectors, patterns, and diagnostic routines to be updated dynamically without hardware redesign. It supports vendor-specific instruction encoding and WDR configuration, enabling tailored testing for novel failure modes or process variations. Furthermore, it provides robust in-field failure response capabilities; unexpected defects detected during operation or manufacturing is addressed through secure firmware updates and soft-repair routines, ensuring continuous device reliability and extended lifecycle management.

8.9.2. SW Based BIST Sequence

The SW Based BIST sequence operates through the standard IEEE 1500 TAP controller and Wrapper registers. The procedure follows a deterministic, multi-stage sequence:

Step1. SW BIST FW Download: The target SW BIST firmware is securely deployed to NVM via the FW_DOWNLOAD_START, FW_DOWNLOAD_DATA, and FW_DOWNLOAD_COMMIT commands. Data segments are aligned to the MPS/WDR size, and integrity is verified via digital signature and CRC before executing SW Based BIST testing. It's same as the firmware download sequence.

Step2. BIST Execution: The host loads the SWBIST_EXECUTE instruction into the WIR. Configuration parameters (e.g., target channels via WIR[13:8], test patterns, timeout settings) are written to the WDR using ShiftWR and committed via UpdateWR.

Step3. BIST Polling: The internal Controller initiates BIST on the selected channels. The host monitors progress by polling the GET_STATUS instruction.

Step4. BIST Result Retrieval: Upon completion, the GET_RESULT instruction captures pass/fail flags and error codes into the WDR shift stage. The data set is vendor specific.

8.9.3. Error Handling

- Operational Failure: If the BIST execution fails due to many failures and critical issues internally while testing, the device returns a specific error code via the WDR. The Host shall capture the SWBIST_RESULT register and associated diagnostic logs and consult with the device manufacturer for failure analysis and corrective action.
- Timeout: If the device does not respond within the defined timeout period (e.g., tBIST_MAX) due to an internal hang, the Host shall abort the process and perform a hardware reset or power cycle.

- Data Retain: Upon reboot by HBF_RESET or Power Off/On, the HBF device will initialize using the Mission Mode FW and resume normal system operation regardless of the result and any errors.
- The vendor data sheet shall contain the detailed error handling requirements.

8.9.4. Summary of SW Based BIST Test Sequence

Figure 35 illustrates the FW Download Sequence via IEEE 1500.

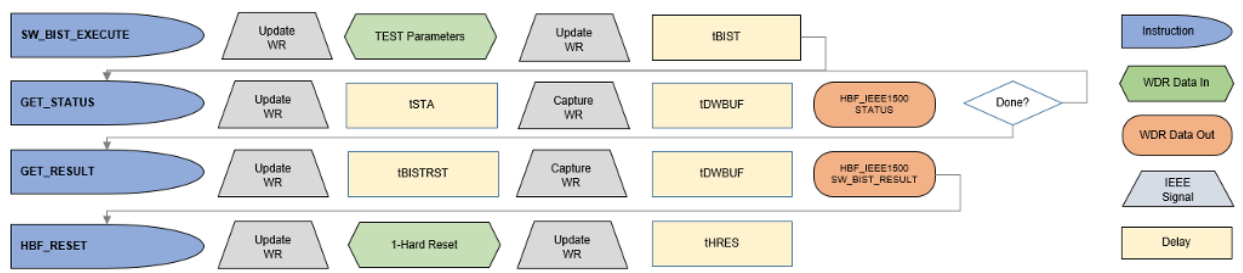


Figure 35: SW Based BIST Test Sequence

9. Reliability Specifications

The reliability characteristics of HBF are like those of non-volatile memory (NVM) technologies such as SSD (Solid State Drive) and UFS (Universal Flash Storage). Unlike HBM (High Bandwidth Memory), HBF is subject to endurance limitations. The endurance index may vary depending on the customer's workload and specific product configuration. Power On data retention is guaranteed for 24 hours at 85°C, meaning that the Core dies can retain stored data for the specific duration. In Power Off, HBF may not retain data after the specific duration, equivalent to HBM. For permanent data storage, persistent storage devices such as SSDs are recommended. Table 33 describes the reliability specifications of HBF.

Table 33: Reliability Specifications.

Life Span	Endurance	Read Disturb	Power On Data Retention @85°C [hour]	UBER	MTBF [hour]/AFR
10years ¹	Product Specific ²	Product Specific ³	24 Hours	10 ⁻¹⁶	20M/ 0.04%

NOTE1: The device is rated for a lifespan of 10 years or until the endurance indicator reaches 100%, whichever occurs first. Actual lifespan is directly influenced by host workload, environmental conditions, and the wear leveling support.

NOTE 2: The HBF device provides critical endurance parameters via Protocol-Specific MMIO Registers to enable host-managed lifecycle control: Offset 0x144 (MAXPEC): The maximum PEC(Program/Erase Cycle) value(design specification), Offset 0x148 (AVGPEC): The current average PEC counts across all NAND blocks. These values are product specific.

The host system shall utilize these provided values to actively monitor and manage device endurance. The host is responsible for determining the remaining lifespan by comparing the dynamic AVGPEC against the static MAXPEC limit and shall take appropriate action when the reported usage approaches the defined maximum.

NOTE 3: Refer to product specifications.

9.1. Operating Temperature

The HBF operates within an operating junction temperature range of 0°C to 105°C. The host shall monitor the HBF operating temperature using the IEEE 1500 instructions. This temperature is the junction temperature of HBF.

9.2. Temperature Throttling and Shutdown

HBF addresses thermal throttling and shutdown features to prevent overheating and permanent damage. This process is managed by a thermal sensor that monitors the HBF's temperature. The HBF operates in four distinct modes based on the HBF's temperature thresholds, as summarized in Table 34.

Table 34: Thermal Throttling Mode

Thermal Throttling Mode	Performance	CATTRIP Signal	Required Action
Normal	Full Performance	Low	-
Light Throttling ¹	Low Performance	Low	-
Severe Throttling	Limited service ²	High	Warning Register ³

Shutdown	-	High	Check the cooling system.
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¹ when entering 'Light Throttling', the device set to 1 on 0x0018 in the MMIO registers

² The HBF halts IO operations. Only maintenance commands shall be supported.

³ Please refer to HBF's Register Table Protocol-Specific MMIO map registers.

The address of CATTRIP register is 0x0018 in the MMIO registers and it is sticky

Normal Mode:

The HBF maintains maximum performance with no thermal restrictions. All components operate at normal clock speeds.

Light Throttling:

When the temperature reaches a thermal throttling threshold while operating, the device uses some mechanism (Ex: like dynamically reducing clock frequency) to lower power dissipation automatically. Normal operation resumes automatically when the temperature returns to a safe operating range. The read and write responses may be slow, but the operation is normal.

Severe Throttling:

When the temperature remains elevated despite using some thermal reduction mechanism (Ex: reducing clock frequency), the HBF asserts the CATTRIP signal high to issue a thermal alert to the host system. The HBF then halts all operations by deasserting the Link Layer AXI Ready signal from the HBF device side. This will backpressure the host commands until CATTRIP signal goes low. Before deasserting the AXI Ready signal, all the commands in flight shall be completed with regular responses or error responses(command status error code 0x9). In severe throttling mode, the credits to the host shall not be given until the system comes out of severe throttling mode.

Shutdown:

When the temperature continues to rise despite the host system receiving a thermal alert via the CATTRIP signal, HBF shall automatically enter shutdown. While entering the shutdown, the host will see the link error, and this is an indication that the HBF device is going to shut down.

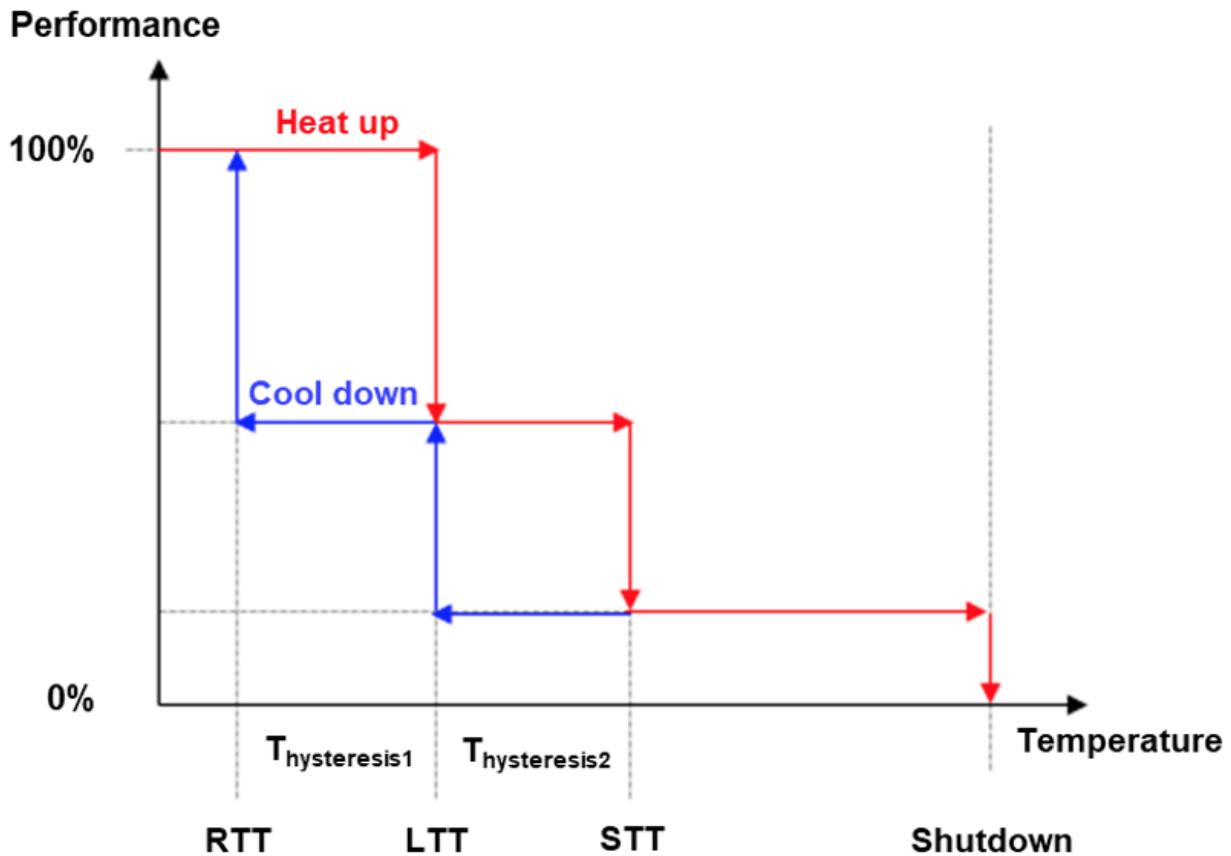


Figure 36: Thermal Throttling Hysteresis and temperatures

Figure 36 illustrates the performance levels during temperature heat-up (red arrows) and cool-down (blue arrows), defined by RTT, LTT, and STT registers

- RTT (Released Throttling Temperature): The temperature threshold at which the system exits the light throttling mode and restores full (100%) performance. (Leftmost point on the graph)
- LTT (Light Throttling Temperature): The temperature threshold at which the system transitions from Normal Mode to Light Throttling Mode (first stage of performance limitation) during the heat-up phase.
- STT (Severe Throttling Temperature): The temperature threshold at which the system transitions from Light Throttling Mode to Severe Throttling Mode (aggressive performance limitation) during the heat-up phase.

The three thermal throttling thresholds and enabling/disabling throttling mode are defined at MMIO Address 0x150.

9.3. ESD Requirements

All bumps in the HBF device are protected by on-chip ESD circuits, except for the power bumps, as described in HBF Power Rails. ESD requirements are defined in Table 35. For D2D connection uBumps, high ESD specifications are not required. For C2C connections, μ Bumps shall meet HBM and CDM ESD requirements as described in Table 35. Power bumps require external TSV diodes mounted on the PCB or interposer, as detailed in the product specification document.

Table 35: ESD Requirements

Category	Connection	Parameter	Min	Max	Unit
μ Bump	C2C	HBM		1000	V
		CDM		250	V
uBump	D2D	CDM		30	V

NOTE1: C2C: Chip-to-Chip, D2D: Die-to-Die, HBM: High Bandwidth Memory, CDM: Charged Device Model

NOTE2: Power bumps will be protected via external TVS diodes on either interposer or PCB

NOTE3: ESD requirements may be changed on the product specification.

9.4. Storage Temperature Conditions

The storage temperature range of the device varies depending on its form factor and packaging state. The storage temperature of KGSD complies with the JEDEC Solid State Technology Association, JESD49B.01: Procurement Standard for Semiconductor Die Products Including Known Good Die (KGD), Publication Date: October 2023.

10. Packaging Specification

10.1. HBF Bump Map

- The HBF uses a 45um pitch in the UCle region. It specifically uses Figure 5-23 in the Rev3p0 UCle spec, 10-column x64 Advanced Package Bump Map for ≤ 32 GT/s.
- In the non-UCle region, the HBF uses a 65um pitch.
- The HBF Bump Map shown in Figure 37 needs to be refined.

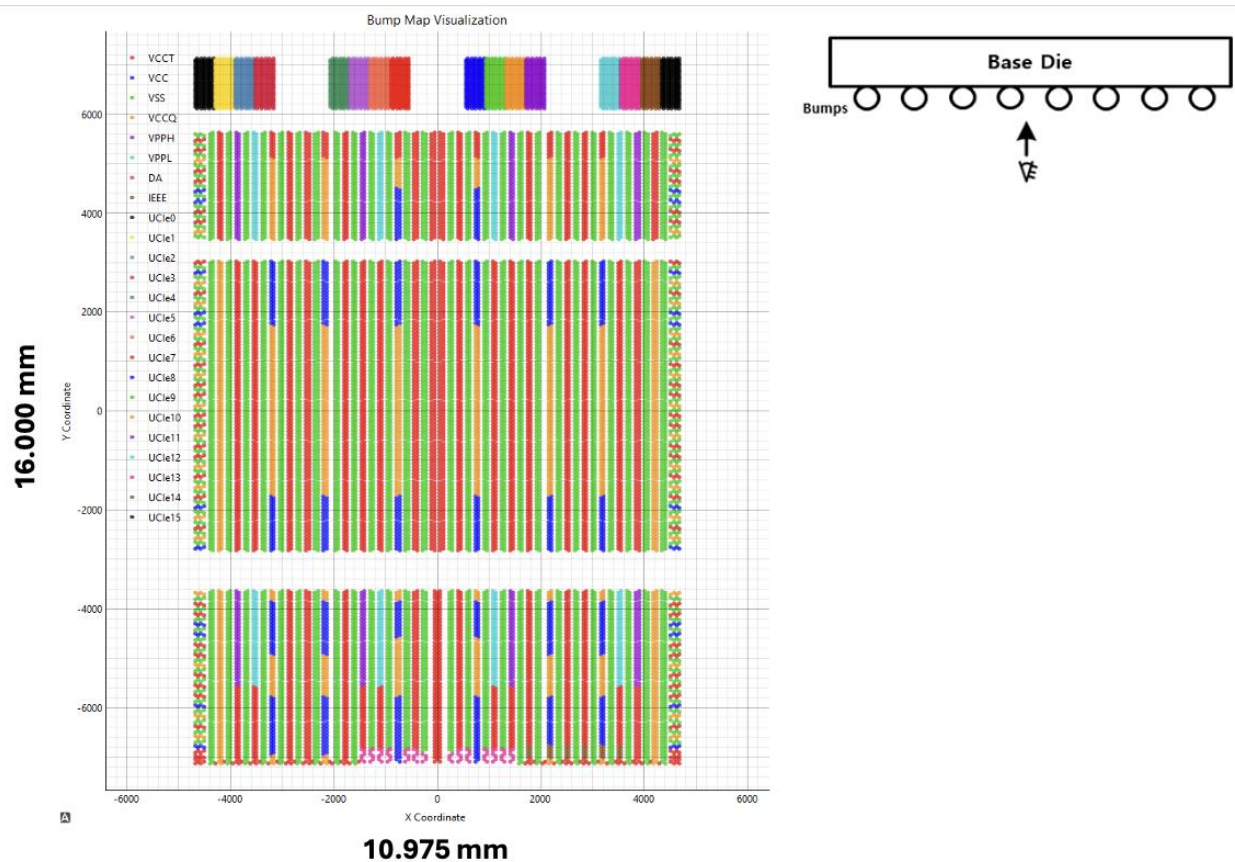


Figure 37: Initial HBF Bump Map (Dead Bug View)

10.2. Microbump Positions

The uBump pattern is shown in Figure 38. There are two uBump patterns, one representing the UCle region and the other representing the non-UCle region.

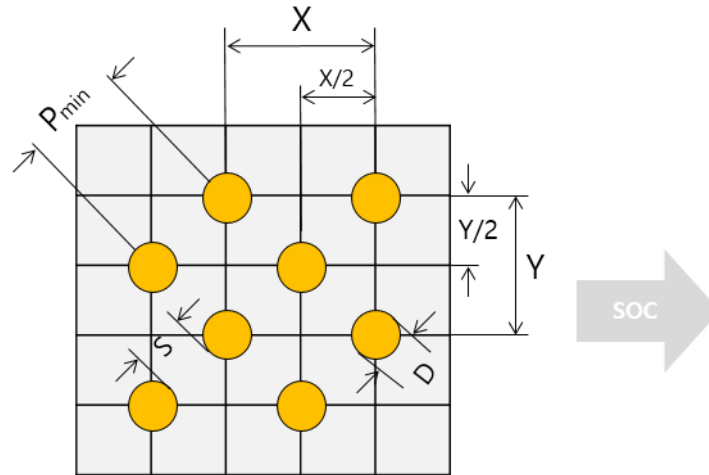


Figure 38: Staggered Microbump Pattern

Table 36: Geometric Parameters of the Staggered Microbump Pattern

Label	Non-UCle Region Nominal Value	UCle Region Nominal Value	Description
X	70um	45um	Hor. Pitch of two adjacent uBump
Y	110um	78um	Hor. Pitch of two adjacent uBump
Pmin	65um	45um	Minimum pitch of the bump field
D	25um	25um	uBump diameter
S	40um	20um	Bump-to-Bump air gap; S=Pmin-D

The initial HBF bump map is shown in Figure 37. It is a preliminary version and will be updated with more voltage rail bumps in the next HBF specification revision.

10.3. HBF Device Dimensions

Parameter	Symbol	Minimum	Nominal	Maximum	Unit	Notes
Width	X	TBD	16	TBD	mm	1,3
Length	Y	TBD	10.975	TBD	mm	1,3
Height	Z	TBD	775	TBD	μm	2,3

NOTE 1: The width and length noted in this table are approximations.

NOTE 2: The height noted in this table applies to 4, 8, 12, and 16-High configurations.

NOTE 3: TBD will be completed in v1.0 HBF Specification.

11. Software Userguide

11.1. HBF Memory Mapping

11.1.1. Memory System Architecture

Using all UCle channels of the HBF in parallel can achieve maximum performance. However, due to the structural design of the HBF, data is distributed to the channels in units of NAND page size (e.g., 4 KiB), one at a time.

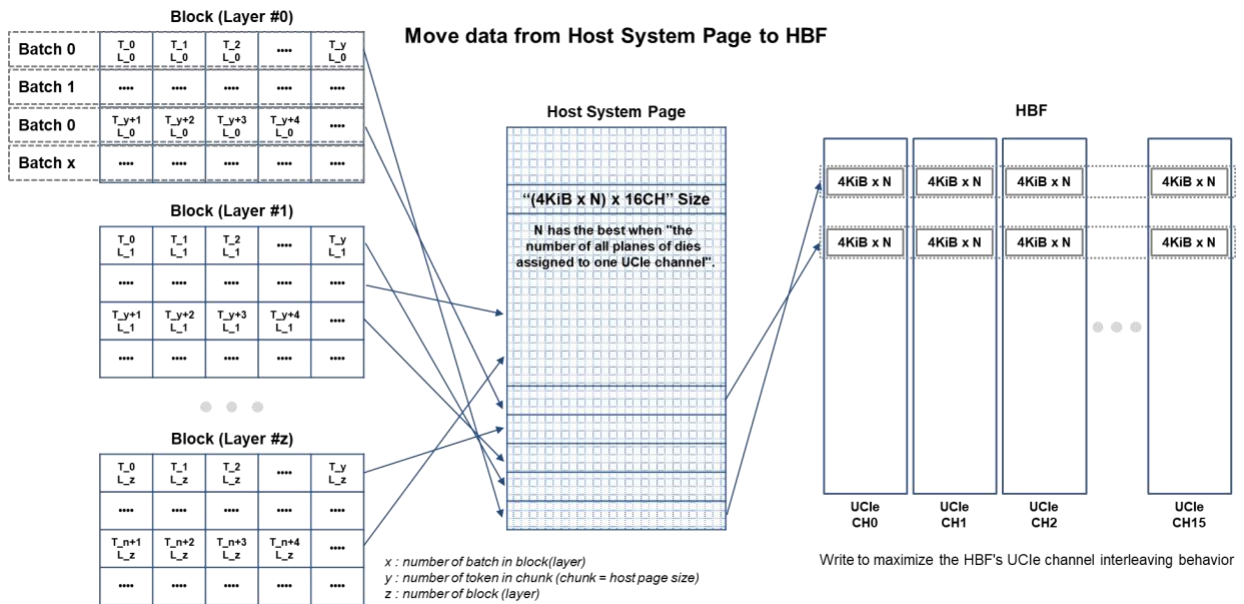


Figure 39: Data Transfer from Host System Page to HBF

The maximum performance of HBF is achieved when the host system page buffer is managed to transfer "(4KiB * N) * all UCle channels in HBF" at once, where N is the number of banks or planes in a die. And N is maximized when N equals the "the number of planes in die * the number of dies per UCle CH".

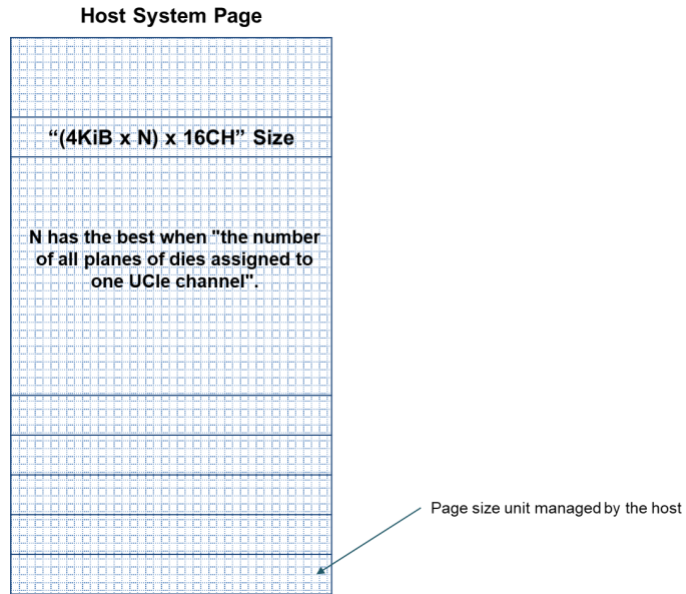


Figure 40: Host System Page

11.1.2. Address Space & Layout

Figure 41 shows an example of local address allocation when two 512GiB HBF units and four 36 GiB HBM units are used together. Due to the large difference in memory capacity for HBF and HBM, memory management shall be performed separately if used together. HBF has 16 UCle channels, each with its own linear local address memory space.

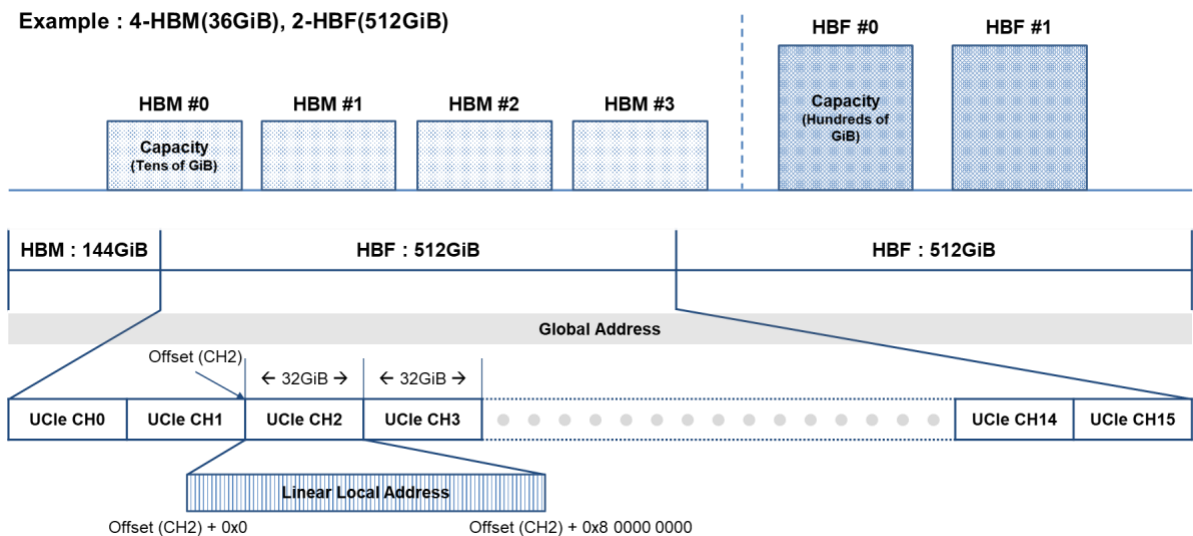


Figure 41: HBM, HBF – Memory Size Issue, Separate Memory Management

The number of 4KiB units in the buffer of the xPU does not necessarily have to be matched to all UCle 16 channels. When multiple 4KiB units in the xPU Buffer are written to the HBF, all UCle channels in the HBF are interleaved. If the split size is 64B, the 64 global addresses (GA) that make up 4KiB match the local addresses (LA) as shown in Figure 42.

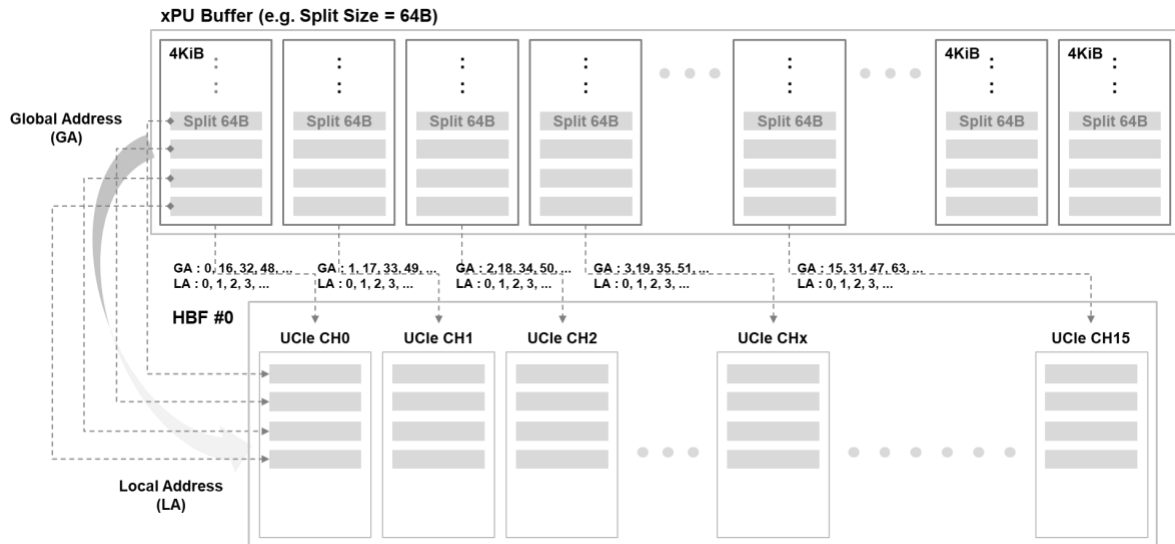


Figure 42: 4KiB Interleaving each UCle channel of HBF

Global addresses in units of 64 bytes within 4KiB need not be continuous, but all global addresses shall be linear when the addresses are collected. For example, if you collect 64 global addresses of 64B split, there should be no missing or duplicated addresses within 4KiB.

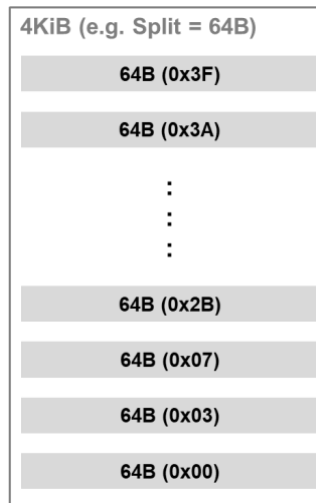


Figure 43: Linear Address

11.2. Programming Model

11.2.1. Read Operation Behavior

The read requests between channels do not maintain order, and the maximum performance is achieved when all possible UCle channels are read simultaneously.

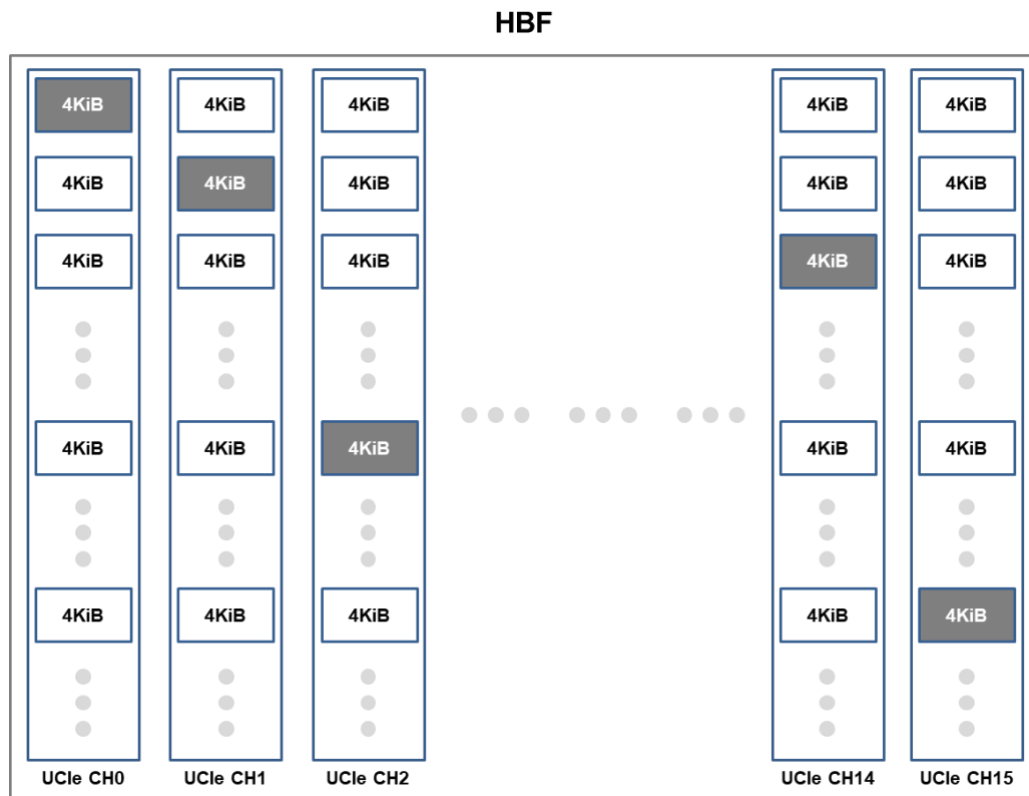


Figure 44: Read Operation Example

11.2.2. Write Optimization

The best write performance is achieved when the host writes to all banks in all the host channels. All banks within the same NAND die shall have the same page number for maximum performance, while different NAND die of the same host channel or different host channels may have different page numbers.

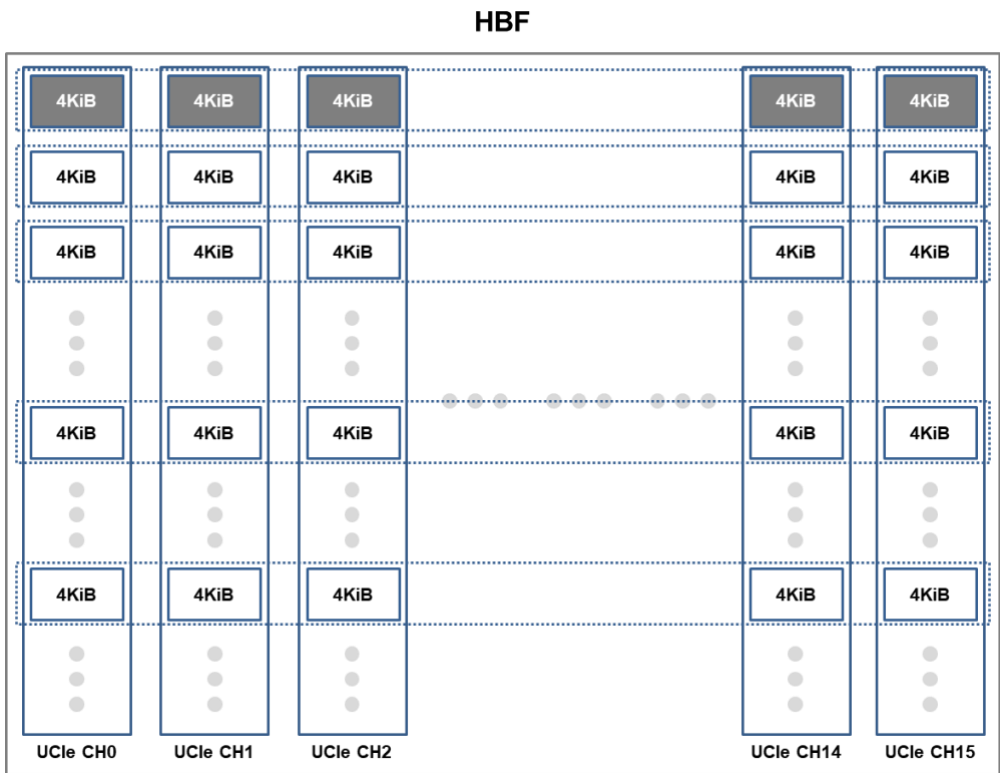


Figure 45: Write Operation Example

11.3. Scratchpad SRAM Usage Model

The host can read and write temporary data to the SRAM in the HBF, which does not hold data after power cycle. The scratchpad size of the SRAM region is obtained by the configuration register reads, SRAM start address, and size as defined in the MMIO register space. The scratchpad region is fully maintained by the host software and the HBF just supports reads and writes at full bandwidth of the UCle channel. All read and write requests to scratchpad memory are high-priority requests and serviced immediately as soon as possible in the HBF. Each UCle channel has individual SRAM space, and the total space is used by the host software as it sees fit to improve the LLM or any other workload that it is running.

There are many uses while executing the LLM or other workloads in the xPU. One such use case is intermediate activations and layer to layer buffers are held in the SRAM.

Note: The scratchpad SRAM support is optional and is product specific. The HBF product datasheet provides additional information.

11.4. HBF Wear Leveling and GC

HBF vendors provide host/xPU software that allows selection of HBF Base-die-driven wear leveling (product vendor-specific) or host-controlled wear leveling with zone remapping. It is recommended to enable either host-controlled wear-leveling or Base die controlled wear-leveling.

HBF does not support any GC (garbage collection) or the transfer of active data from one physical location to another. HBF only supports remapping the UCIe channel local address from one physical location to another, as a part of wear leveling, and further writes to that UCIe local channel address with wearing out new assign physical location.

11.4.1. Zone Remapping

HBF supports host/xPU controlled zone remapping, where the host can divide each UCIe channel capacity into N zones, and each zone size/capacity should be the same. HBF Base die supports the host to track/read each zone's PEC (program-erase cycles) from CSR registers. Host software can also maintain a record of each zone's PEC if required. Based on the assignment of each zone and current usages or future usages, the host software can decide to swap the Hot Zone (High PEC) with the Cold Zone (Low PEC). Before swapping, the host software needs to ensure that the data is invalid and no further IO read requests are issued to these zones before rewriting.

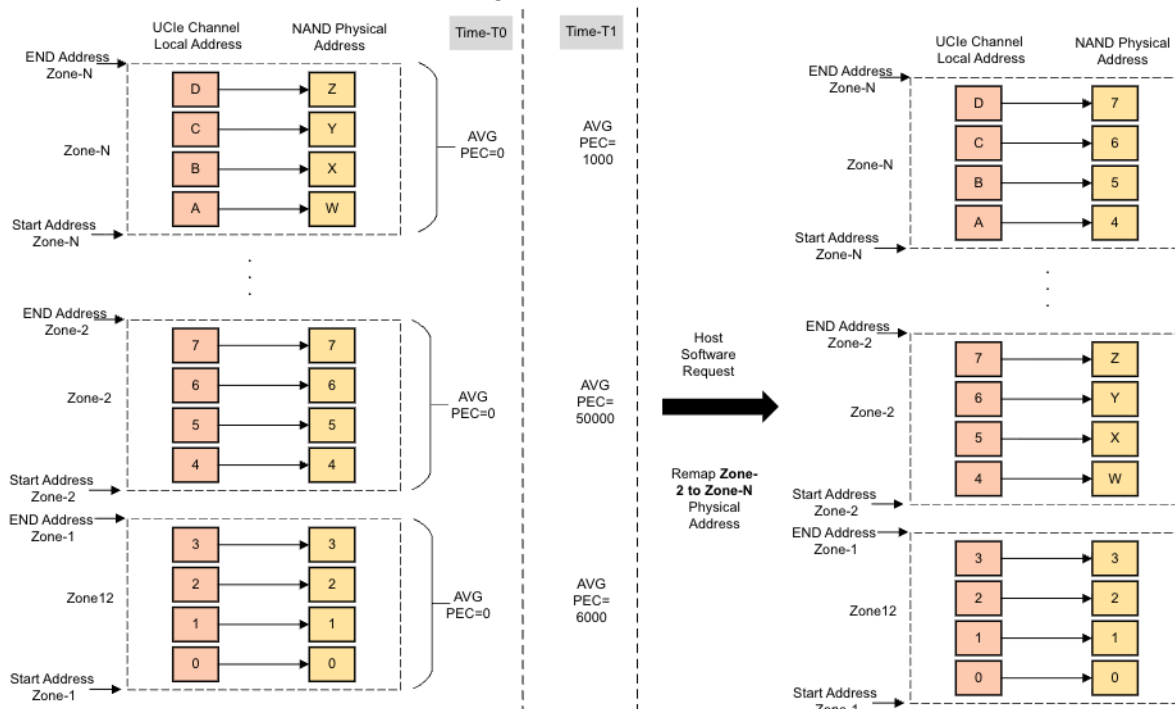


Figure 46: Example of Zone Remapping

Limitations:

- The host shall ensure that the zone size is a multiple of the Core die NAND block size.
- The product may impose a limit - the average PEC difference between zones cannot exceed a certain threshold.

11.5. HBF to Host Error Status and Actions Needed by Host

To enable seamless, low-latency data access for the Host/xPU as defined in the product specification, periodic data refresh is required at intervals specified in the reliability specification (typically every 24–48 hours). It is recommended that the host software schedule these refresh operations during maintenance periods or interleave them alongside ongoing operations in a distributed manner.

If the host chooses to interleave refresh with active workloads, it shall ensure that refresh commands and read commands are not issued to the same die at the same time. Otherwise, some read operations may experience increased latency.

Data refresh conditions

- Time-duration (Data retention time): This is HBF product specific and it shall publish the guaranteed data retention of NAND block during power-on and power-off scenarios. Host software needs to ensure that data is refreshed at this interval.
- Excess reads from any physical NAND block: Due to NAND characteristics, with each access, the NAND block data health deteriorates. It is the host's responsibility to refresh the NAND block's data after reading NAND block data X (product-specific) times. The host can periodically check each NAND block's read count using the CSR command; however, when Base die observes that a specific physical NAND block is at risk of data loss, it will indicate this in the command status.

Due to NAND characteristics in specific rare scenarios, it may be possible that the health of some NAND blocks deteriorates earlier than the Read Disturb/Data Retention specification, or the Base die encounters an error due to NAND characteristics, an invalid field in the command, or the current Base die status. In these cases, the Base die will report an error in the command status, and a specific follow-up action is required from the host as described in the following subsections.

11.5.1. Read Errors

11.5.1.1. Invalid Address

The error is reported if a host-issued read command is out of the UCle channel capacity or to a region of the UCle channel capacity that has already retired. The host needs to ensure the address in the command is correct, and if the already-retired capacity has not been retired to retire it.

Block capacity is retired at the granularity of the NAND block size, and the maximum retired capacity can reach the UCIe channel capacity.

Host Action: At this error, the host software needs to retire capacity for the NAND block, in case the host software misses it when HBF initially reports the retired capacity, while in the original request, the exact size of the retired capacity is reported.

11.5.1.2. Temporarily Restricted Read

The error is reported if Base die temporarily restricts a specific address range and still observes read requests for the following address range. The reasons for restriction and duration of restriction are HBF product-specific and will be described in the product specification.

Host needs to wait for the duration (event-based or time-based) to complete before accessing the restricted address range.

11.5.1.3. Invalid User Field in Command

The error is reported if Base die observes an Invalid field in the command field that is not supported by HBF. The host needs to recheck all command fields and reissue the command in the HBF specification command format.

11.5.1.4. UECC (Block Data Refresh required)

The error is reported if the data at the following address has gone bad and couldn't be corrected by Base die's error-correcting engine. At the same time, the user [8:6] describes the root cause of these errors.

It is recommended to use the host software to refresh this NAND block data for seamless operation or for the time being till the active workload utilizes HBF's full bandwidth. Access the following data from the backup copy and refresh it later.

As data refresh happens at physical NAND block granularity, and UECC reported for this read physical location, it may be possible that Base die can correct some other physical location of this NAND block data. Still, those read latencies will be high. So, HBF recommends that once UECC reports, do not access this NAND block data before refresh if the host software has a backup copy.

11.5.1.5. CECC (Block Data Refresh required)

The error is reported if the data in the following NAND block has a potential risk of data loss. Base die is still able to correct it. user [8:6] indicated that Base die is experiencing it because the NAND block read count exceeds the product threshold.

Host software can decide when to refresh this NAND block data, based on the current active workload and the type of data in this NAND block. However, HBF recommends refreshing this NAND block data ASAP, as NAND block health is deteriorating before continuing to access this NAND block for a long time.

11.5.1.6. UECC - Read Retry

The error is reported if the data at the following address has gone bad and couldn't be corrected by Base die's error-correcting engine at this try but retry this read again with the opcode provided in user [8:6], as it may require recalibration in Base die rather than data refresh in the NAND block.

Following error is reported only if the host chooses to host driven-read-retry for transparent latency.

Host software needs to reissue the same read command with copying Ruser[8:6] in ARuser[8:6].

11.5.1.7. Read from Erase Page

The error is reported if the host attempts to read a physical location that has not yet been written with valid data. The host needs to reconfirm the read command address. The host needs

11.5.1.8. Mark UCI channel Local Capacity-unusable

In specific rare scenarios, part of the UCI channel capacity is not recoverable due to hard failures. Base die will report the error with granularity of capacity mark bad in Ruser[8:6].

Host software needs to calculate the reported capacity address space and mark it bad from the host software usage point of view. After a power cycle, the following information could be retrieved from the CSR registers by using CSR Command(opcode=0x0A). The 1024 CSR which is 1024-bitmap will contain the unusable logical capacity information per NAND block. The 1024 CSRs support 1MiB, 2MiB, or 4MiB per NAND block capacity. Each bit in the CSR represents 1 NAND block of HBF. If set to 1, it is unusable, and if it is set to 0, it is usable.

11.5.1.9. Die temporary block for recovery

In a specific recovery scenario, Base die may block all access to Die and complete the command with this error code, so the host is aware of it rather than wait for recovery to finish for processing the command.

Host software can read the Die status from the CSR register; however, it shall wait until the recovery operation completes before issuing a read/write request to a die. The host can poll the CSR status register to check the Die status.

11.5.2. Write Errors

11.5.2.1. Invalid Address

The error is reported if a host-issued write command is out of the UCIe channel capacity or to a region of the UCIe channel capacity that has already retired. The host needs to ensure the address in the command is correct, and if the already retired capacity has not been retired, please retire it.

Block capacity is retired at the granularity of the NAND block size, and the maximum retired capacity can reach the UCIe channel capacity.

At this error, the host software needs to retire capacity for the NAND block, in case the host software misses it when HBF initially reports the retired capacity, while in the original request, the exact size of the retired capacity is reported.

11.5.2.2. Overlap Address

The error is reported if a given UCIe channel address command is already pending or has already been written to NAND, as HBF doesn't support random write within the NAND block.

To rewrite the NAND block, the host software needs to start writing from NAND block page 0, wordline 0, and string 0.

11.5.2.3. Invalid User Field in Command

The error is reported if Base die observes an Invalid field in the command field that is not supported by HBF. The host needs to recheck all command fields and reissue the command in the HBF specification command format.

11.5.2.4. Maximum Pending Device Logical Unit (4KiB) Limit Reached

HBF shall publish the maximum pending device logical unit write supported in the Base die (product-specific). The host needs to ensure that it does not write more than the

limited device logical units, while the host can continue issuing writes to send complete 4KiB data to the pending device logical units.

If the host software is willing to issue more device logical unit writes, wait for the completion of any pending device logical unit writes.

11.5.2.5. Complete Device Logical Unit (4KiB) Accumulation Timeout

The host can configure the maximum wait time for sending complete 4KiB data of device logical units. If Base die didn't receive complete 4KiB data within the set time limit, Base die will send a completion of all issued writes for this device logical unit with this error code. The host software can resend the entire 4KiB data write request.

11.5.2.6. Write Order Violation

HBF requires the host to write sequentially within a NAND block, while the host can write multiple random NAND blocks simultaneously. If the Base die detects the host did not follow the write order within the NAND block or the next page within the NAND block write request initiated earlier than the prior page, the Base die will complete the write command with this error opcode response. Host software needs to ensure all prior page write has been sent before reissuing this write.

In a particular scenario, the following error shall indicate that Base die is expecting a write to page N, where as the host sent the write to page N+1 with additional info 0x1 as part of the error code.

In another scenario, if the write to page N is a NAND program error, then all the pending writes behind the write to page N shall receive a write order violation error with additional info 0x2 as part of the error code.

The host software can follow the recommended action for page N write.

11.5.2.7. Program Fail, Replay NAND block Data

The error is indicated when Base die encounters a program failure on a specific page of a NAND block. All follow-up writes to this NAND block will be completed with an error type write order violation.

The host software needs to calculate the UCIe local channel address for this NAND block and start rewriting this NAND block from Page-0 (wordline-0, String-0) to the failed page number. After completing this host software, continue with the follow-up write to the next page of this NAND block, however it is advisable if host has local copy of this NAND block data use the local copy in-place of reading from HBF.

11.5.2.8. Mark UCle Channel Local Capacity as Unusable

In specific rare scenarios, part of the UCle channel capacity is not recoverable due to hard failures. Base die will report the error code with granularity of capacity mark bad in Ruser[8:6]. The following error may occur if the bank runs out of spare NAND blocks and the requested write-new NAND block couldn't be allocated.

Host software needs to calculate the reported capacity address space and mark it bad from the host software usage point of view. After a power cycle, the following information could be retrieved from the CSR registers.

11.5.2.9. Die Temporary Block for Recovery

In a specific recovery scenario, Base die may block all access to Die and complete the command with this error code, so the host is aware of it, rather than waiting for recovery to finish for processing the command.

Host software can read the Die status from the CSR register; however, it shall wait until the recovery operation completes before issuing a read/write request to a die. The host can poll the CSR status register to check the Die status.

12.Security

The HBF security features are optional and product specific. The HBF is connected to the xPU in the same package. The HBF NAND security during power on is protected by the same level of security to access the xPU, like HBM. While the power is turned off, there may be some additional level of protection needed to secure the contents in the HBF, though it is extremely hard to remove the NAND flash from the xPU package and remove the Base die controller and access the NAND contents by connecting the TSVs to external controller.

The data-at-rest security refers to protecting the data stored in NAND flash memory when the device is powered off. There are many known SSD techniques available to protect the data-at-rest in NAND. These SSD techniques may not be directly applicable or used in HBF as it operates at very high bandwidth, and any inline encryption/decryption are a very high-power consuming operation and affecting the performance. The security mechanisms enabled to support HBF data-at-rest are optional and any feedback on how the HBF data is accessed at rest would be helpful to add the security features. There are couple of low-cost features that are implemented to protect the HBF data-at-rest without affecting the high bandwidth operation and consumes less power.

All the security features in this chapter are optional and may not be implemented. Refer to the product specification to check the list of security features like Sanitize (NAND block erase or overwrite) and others may be implemented.

12.1. Power-on Erase

Power-on erase works by obfuscating the data in the HBF every time the HBF is powered on makes sure the data cannot be read by the xPU. And any data read by the xPU before writing/erasing the HBF will return ECC errors to the xPU for the full capacity of all pages in the HBF. The data needs to be written before use by the xPU after power-on.

12.2. HBF Sanitize

The HBF sanitize is a command used to securely erase HBF data, ensuring that all data, including cached and buffered information, is completely removed. The secure erase command is described in section 5.2.4.3.

12.3. Physical Tamper Proof

A physical theft of the cards that contain the xPU and HBF package from datacenters is possible. A metal clamp strapped to heat sink can prevent the physical theft of the cards when it passes through a metal detector. An optional embedded RFID could also provide a low-cost solution.

13.HBF Applications

13.1. Single LLM Serving Using HBF

13.1.1. With Host Channel (UCle) Interleaving

The following Figure 47 describes the read operation when interleaving all host channels in single LLM Serving. At the time of loading the model to the HBF, the data is written at $4\text{KiB} * N * 16$ sequentially across all channels, where N is the number of parallel banks/planes across all dies in each channel. When the host reads the pages at the time of inferencing, all $4\text{KiB} * N * 16$ pages are accessed in parallel in one HBF read latency time, which could be hundreds of KiB of data. The QKV Projection Weight is split into pages. The size of the split pages is $4\text{KiB} * N * 16$ Channels. In addition, Output Projection, Gate Up Projection, and Down Projection also have the same size and structure. The reason for taking the page size to $4\text{KiB} * N * 16$ Channels is to keep the read bandwidth to the maximum through interleaving of all channels.

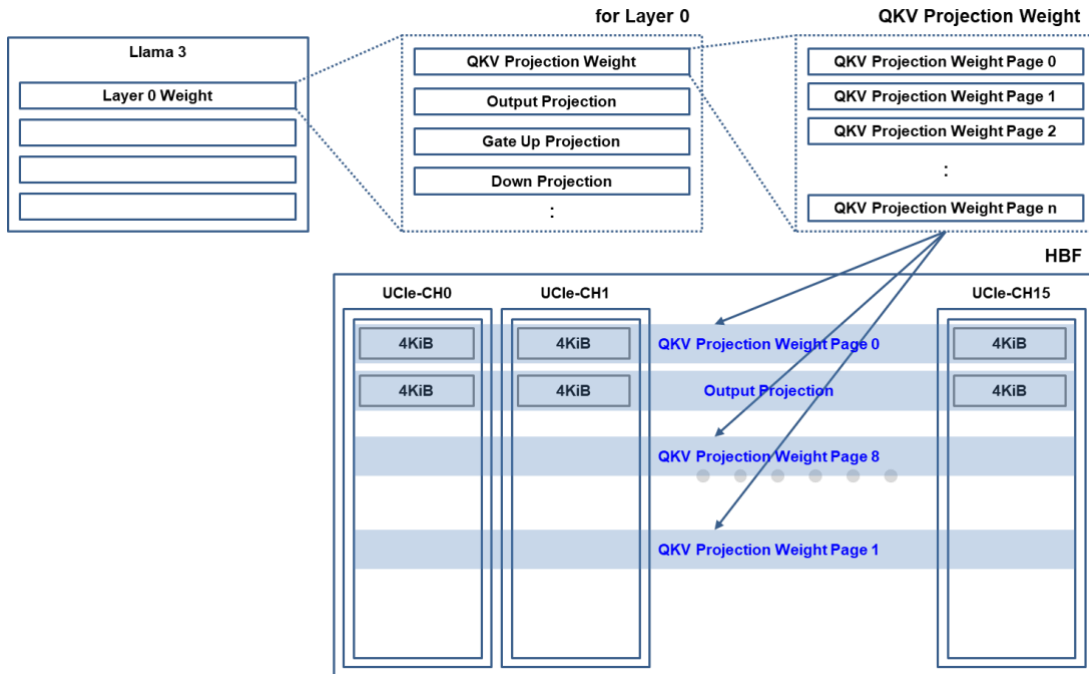


Figure 47: UCle Channel Interleaving – Single Model Case

13.2. Multiple LLM Serving Using HBF

13.2.1. With Host Channel Interleaving

In multiple LLM services, when interleaving of all channels on the host is activated, it can always be read with full bandwidth, as shown in Figure 48. This case is suitable for loading

various weight models and accessing only active models. In traditional GPUs, new weights had to be loaded from SSDs to GPU's memory when changing the model, but in this case, the model load has already been completed and is accessed immediately by changing the model without any delay. Multiple LLM Serving will be easy to understand as an extended version of Single LLM Serving.

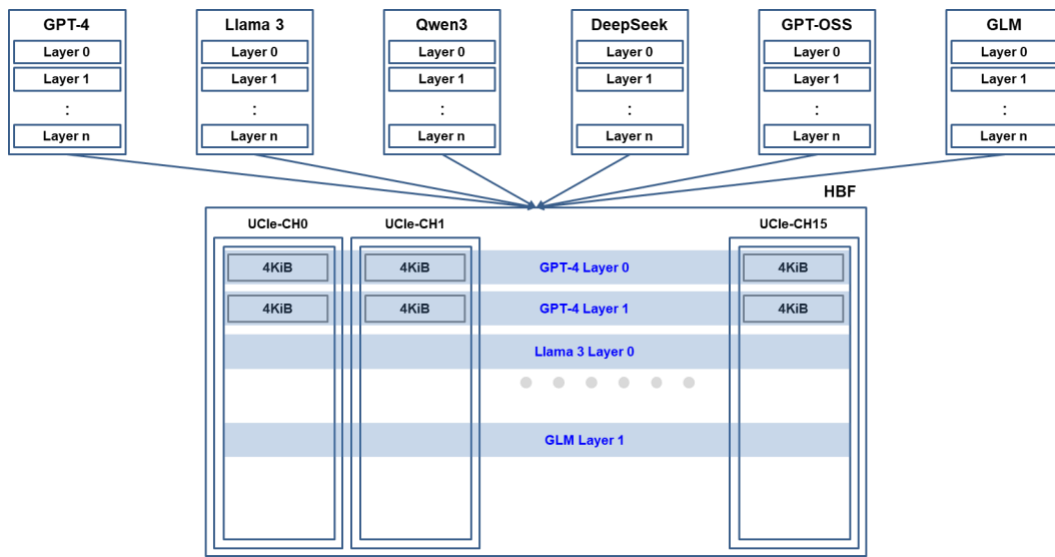


Figure 48: UCle Channel Interleaving – Multi-Model Case

13.2.2. Without Host Channel Interleaving

Figure 49 describes the case of weighting each area of each channel without channel interleaving in multiple LLM services. In this case, in the absence of channel interleaving, the host should know the address of each channel and design a model to write to each channel. This case demonstrates its strength in a scenario where GPU cores can simultaneously access the models by separating them. As there is no host channel interleaving, the total page size for each model is of 4KiB * N size. As there is no collision between models accessing the weights from the same channel, this method may be advantageous in certain model serving cases.

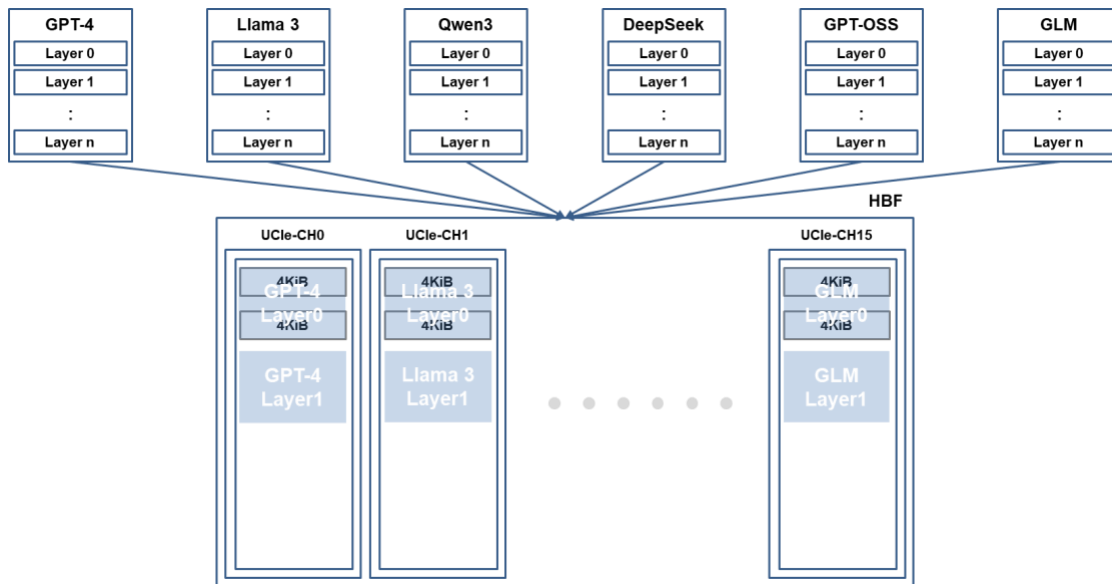


Figure 49: Dedicated UCle Channel – Multi Model Case

13.2.3. MoE Example

Multimodal LLMs Using HBF, MoE (Mixture of Experts), Agentic Workloads:

Models such as Multiple LLMs, MoE (Mixture of Experiments), and Agentic Workloads will also not be much different from the cases of "With Host Channel Interleaving" and "Without Host Channel Interleaving" described in Multiple LLM Serving. Depending on its purpose (performance or capacity utilization), you can choose whether to write or store the model in only a few channels.

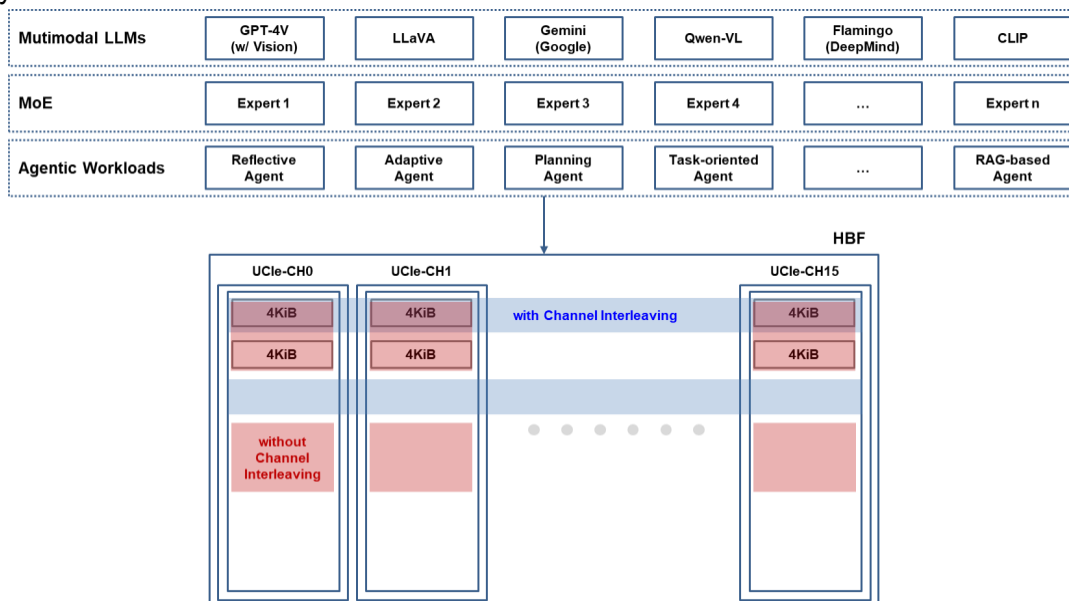


Figure 50: Example – Multimodal LLMs using HBF, MoE and Agentic Workload

13.3. HBF Usage Examples

13.3.1. AI Parameter Store & Loading

Due to the nature of AI applications, various parameters need to be loaded at every token generation. Since AI parameters can range from several GiB to tens or hundreds of GiB, the bandwidth of the entire channel should be utilized for the full performance at parameter loading when generating tokens. Therefore, to make the most of the bandwidth of the entire channel, the parameter is written parallel using channel interleaving, while reads shall also be interleaved in the same way.

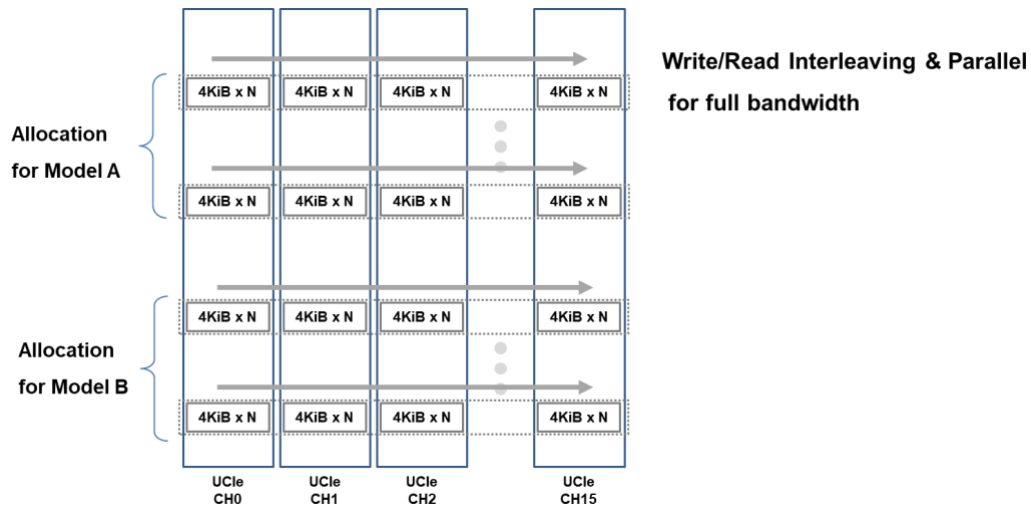


Figure 51: Example – Write/Read LLM Weight

13.3.2. KV Cache data write/read guidelines

During prefill, the KV Cache is generated for each layer and the decode uses these KV cache for next token generation. A new KV Cache is generated by using all previous layers' KV Cache of the same layer of token. Therefore, write and read performance may vary depending on the KV Cache memory layout by the host system. It is expected that the host knows the LLM or other AI/ML architecture well to optimize the data placement for the best write and read performance. The goal of the HBF is to provide all the hooks necessary from the hardware perspective as described in the earlier sections to utilize the HBF in the best possible way to improve the user experience while various AI/ML applications and LLM serving.

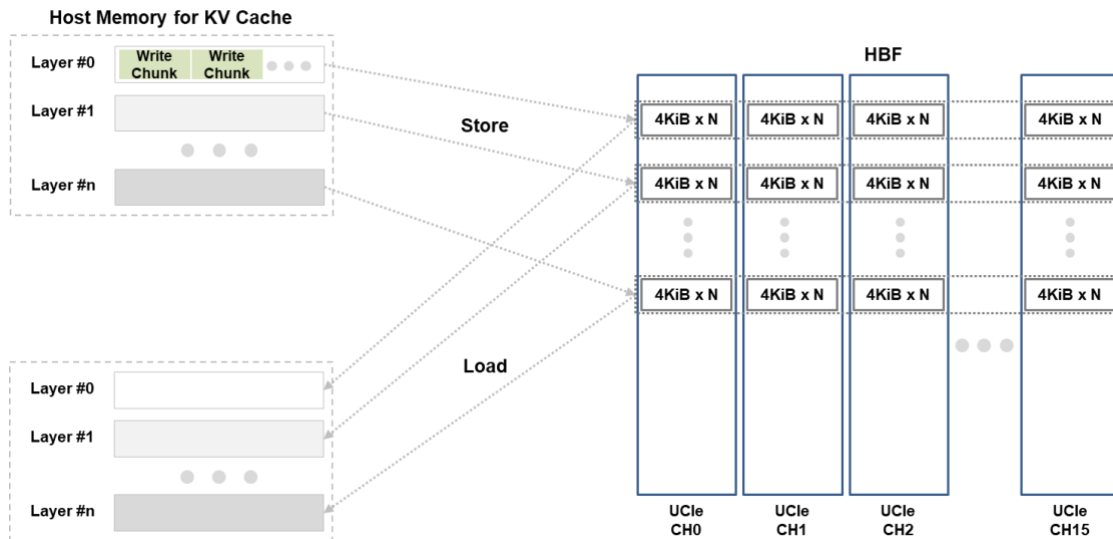


Figure 52: Example – Write/Read KV Cache Data

13.3.3. Partitioning Weights and KV Cache Writes in Different Channels

If KV Cache and model weight with different read/write patterns are mixed in one zone, it may be disadvantageous to overall endurance and capacity utilization. Therefore, it is possible to choose how to partition the total capacity in units of channels and write to KV Cache and weight for each partition. Channel partitioning can select one of the following depending on the purpose.

Scenario 1: Endurance Perspective (Uniform Channel Partitioning)

The entire channel is divided into two to write weights and KV Cache separately.

Scenario: Channels are divided into equal sized groups (Group 0 and Group 1). During Weight write operations or rewrite due to retention, weights are migrated to the group with lower PE cycle.

Expectation: Simplified host-driven wear-leveling algorithm

Scenario 2: Maximize Capacity Utilization (Non-uniform Channel Partitioning)

The entire channel is divided arbitrarily to write weight and KV Cache.

Scenario: Determine the number of channels based on the weight size of the model to be used.

Expectation: Maximize capacity utilization

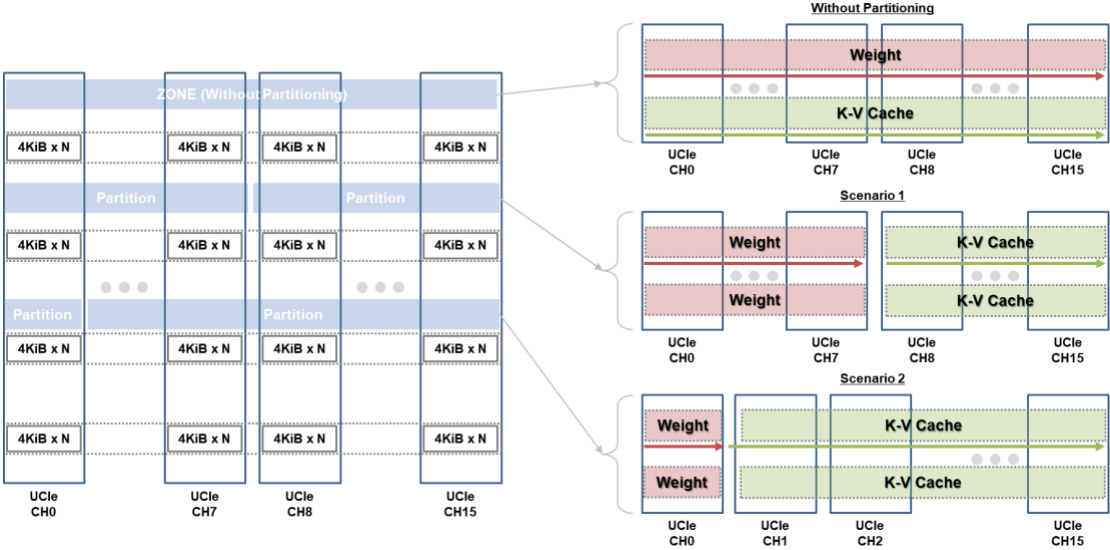


Figure 53: Example – Partition of LLM Weights and KV Cache Data